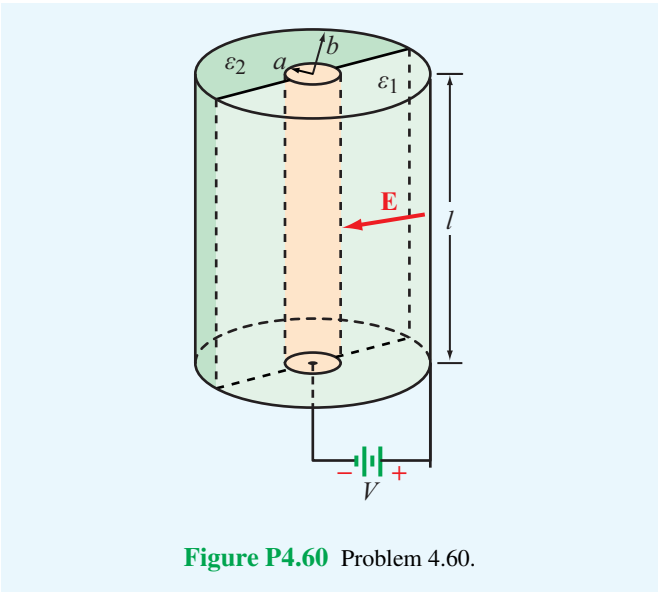


4.60 A coaxial capacitor consists of two concentric, conducting, cylindrical surfaces: one of radius a and another of radius b , as shown in **Fig. P4.60**. The insulating layer separating the two conducting surfaces is divided equally into two semi-cylindrical sections: one filled with dielectric ϵ_1 and the other filled with dielectric ϵ_2 .

(a) Develop an expression for C in terms of the length l and the given quantities.

*** (b)** Evaluate the value of C for $a = 2$ mm, $b = 6$ mm, $\epsilon_{r1} = 2$, $\epsilon_{r2} = 4$, and $l = 4$ cm.



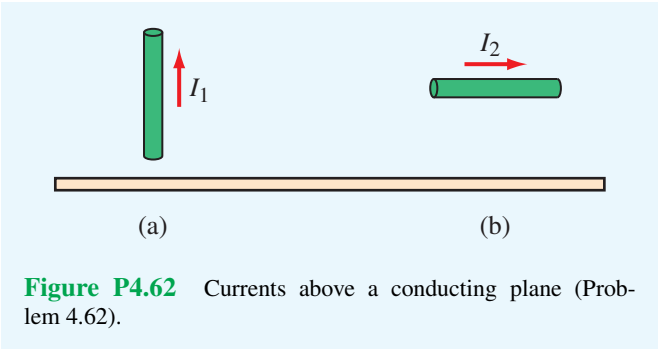
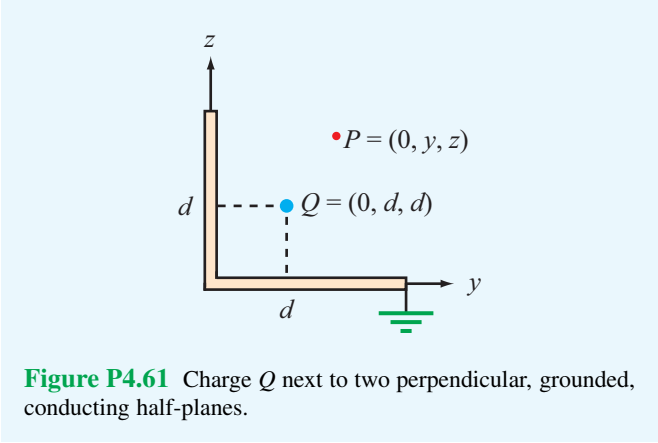
Section 4-12: Image Method

4.61 With reference to **Fig. P4.61**, charge Q is located at a distance d above a grounded half-plane located in the x - y plane and at a distance d from another grounded half-plane in the x - z plane. Use the image method to

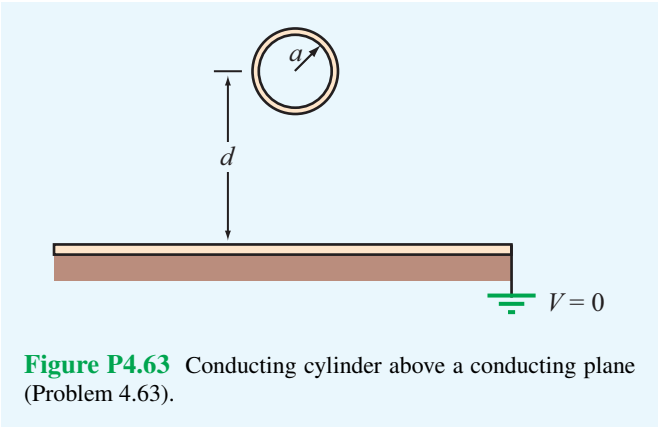
(a) Establish the magnitudes, polarities, and locations of the images of charge Q with respect to each of the two ground planes (as if each is infinite in extent).

(b) Find the electric potential and electric field at an arbitrary point $P = (0, y, z)$.

4.62 Conducting wires above a conducting plane carry currents I_1 and I_2 in the directions shown in **Fig. P4.62**. Keeping in mind that the direction of a current is defined in terms of the movement of positive charges, what are the directions of the image currents corresponding to I_1 and I_2 ?



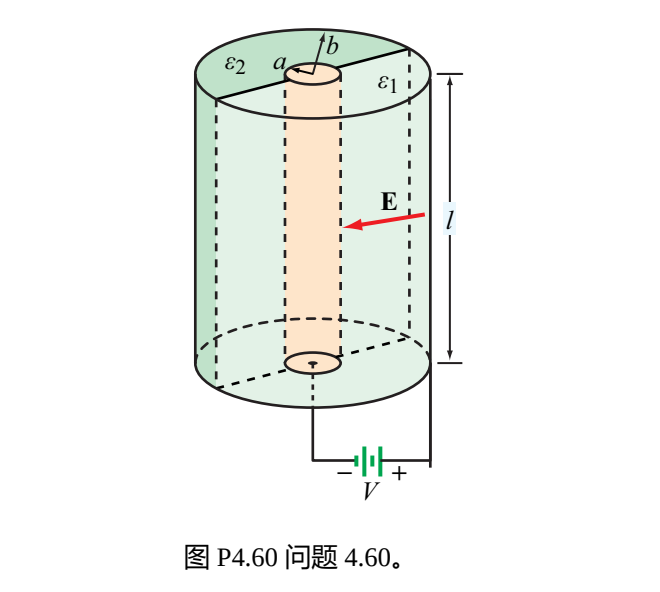
*** 4.63** Use the image method to find the capacitance per unit length of an infinitely long conducting cylinder of radius a situated at a distance d from a parallel conducting plane, as shown in **Fig. P4.63**.



4.60 同轴电容器由两个同心的导电圆柱形表面组成：一个半径为 a ，另一个半径为 b ，如图 P4.60 所示。分隔两个导电表面的绝缘层被等分为两个半圆柱形部分：一个填充有电介质 ϵ_1 ，另一个填充有电介质 ϵ_2 。

(a) 根据长度 l 和给定数量建立 C 的表达式。

*** (b)** 计算 $a = 2$ mm、 $b = 6$ mm、 $\epsilon_{r1} = 2$ 、 $\epsilon_{r2} = 4$ 和 $l = 4$ cm 的 C 值。

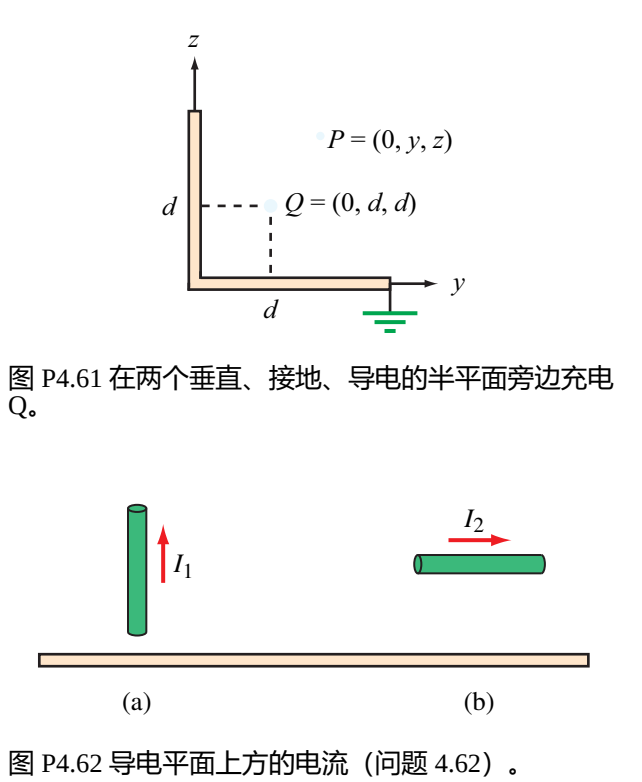


第 4-12 节：镜像法

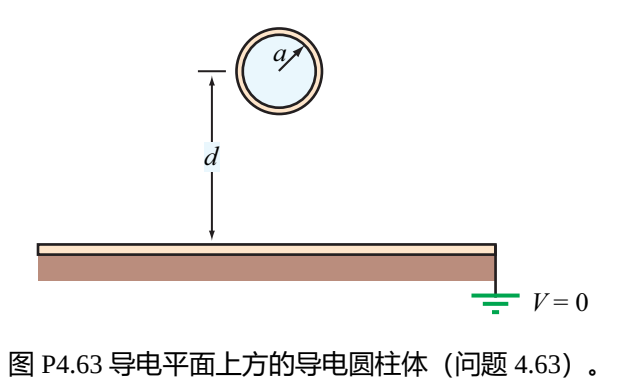
4.61 参考图 P4.61，电荷 Q 位于 x - y 平面中接地半平面上方距离 d 处，并且与 x - z 平面中另一个接地半平面距离 d 处。使用图像方法

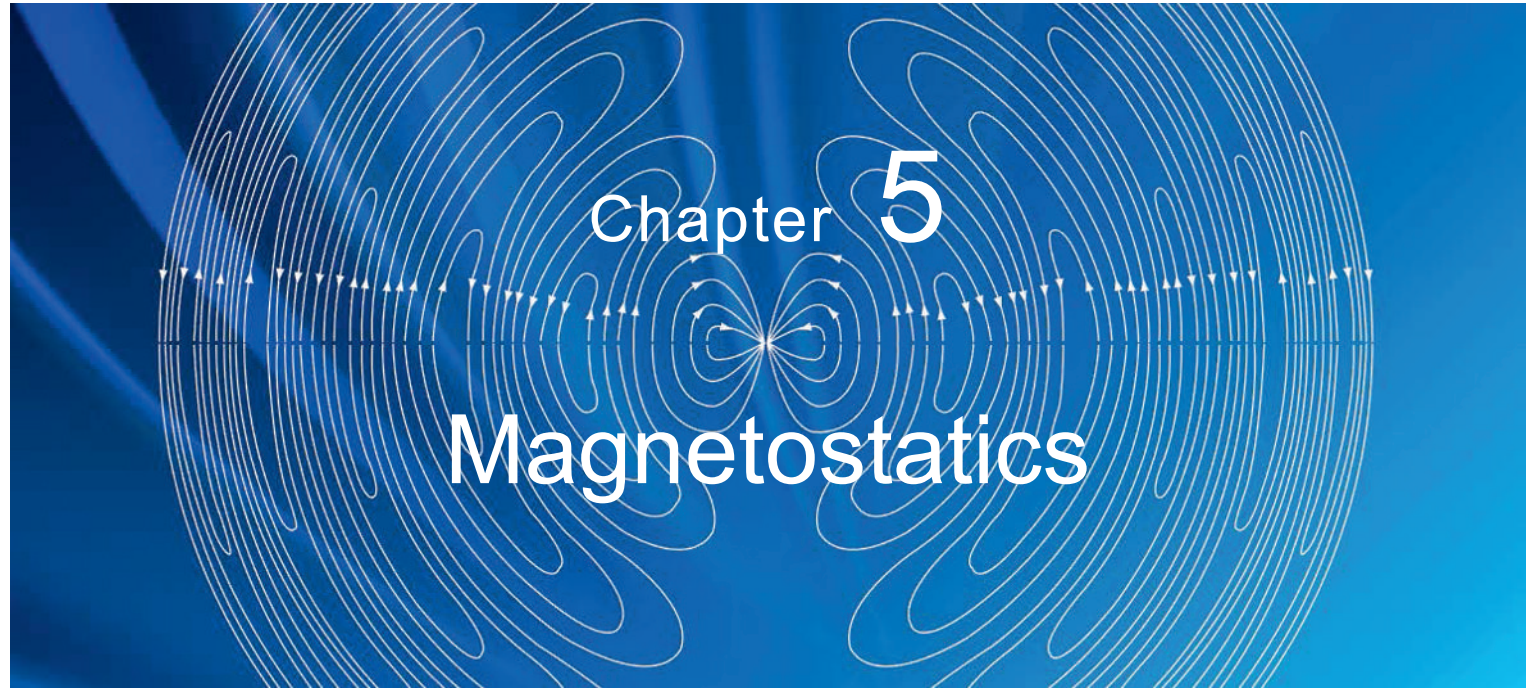
(a) 确定电荷 Q 的图像相对于两个接地平面的大小、极性和位置（就好像每个接地平面的范围都是无限的）。(b) 求任意点 $P = (0, y, z)$ 处的电势和电场。

4.62 导电平面上方的导线沿图 P4.62 所示方向承载电流 I_1 和 I_2 。请记住，电流的方向是根据正电荷的移动来定义的，对应于 I_1 和 I_2 的镜像电流的方向是什么？



*** 4.63** 用镜像法求半径为 a 的无限长导电圆柱体与平行导电面距离 d 处的单位长度电容，如图 P4.63 所示。





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Objectives

- Upon learning the material presented in this chapter, you should be able to:
1. Calculate the magnetic force on a current-carrying wire placed in a magnetic field and the torque exerted on a current loop.
 2. Apply the Biot–Savart law to calculate the magnetic field due to current distribution.
 3. Apply Ampère’s law to configurations with appropriate symmetry.
 4. Explain magnetic hysteresis in ferromagnetic materials.
 5. Calculate the inductance of a solenoid, a coaxial transmission line, or other configurations.
 6. Relate the magnetic energy stored in a region to the magnetic field distribution in that region.



章节内容

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目标

学习本章介绍的材料后，您应该能够：

1. 计算放置在磁场中的载流导线上的磁力以及施加在电流环上的扭矩。
2. 应用毕奥-萨伐尔定律计算电流分布产生的磁场。
3. 将安培环路定律应用于具有适当对称性的配置。
4. 解释铁磁材料中的磁滞现象。
5. 计算螺线管、同轴传输线或其他配置的电感。
6. 将某个区域中存储的磁能与该区域的磁场分布联系起来。

Overview

This chapter on magnetostatics parallels the preceding one on electrostatics. Stationary charges produce static electric fields, and steady (i.e., non–time-varying) currents produce static magnetic fields. When $\partial/\partial t = 0$, the magnetic fields in a medium with magnetic permeability μ are governed by the second pair of Maxwell’s equations (Eqs. (4.3a,b)):

$$\nabla \cdot \mathbf{B} = 0,$$
(5.1a)

$$\nabla \times \mathbf{H} = \mathbf{J},$$
(5.1b)

where $\mathbf{J}$ is the current density. The **magnetic flux density** $\mathbf{B}$ and the **magnetic field intensity** $\mathbf{H}$ are related by

$$\mathbf{B} = \mu \mathbf{H}.$$
(5.2)

When examining electric fields in a dielectric medium in Chapter 4, we noted that the relation $\mathbf{D} = \epsilon \mathbf{E}$ is valid only when the medium is linear and isotropic. These properties, which hold true for most materials, allow us to treat the permittivity ϵ as a constant, scalar quantity that is independent of both the magnitude and the direction of $\mathbf{E}$. A similar statement applies to the relation given by Eq. (5.2). With the exception of ferromagnetic materials, for which the relationship between $\mathbf{B}$ and $\mathbf{H}$ is nonlinear, most materials are characterized by constant permeabilities.

► Furthermore, $\mu = \mu_0$ for most dielectrics and metals (excluding ferromagnetic materials). ◀

The objectives of this chapter are to develop an understanding of the relationship between steady currents and the magnetic flux $\mathbf{B}$ and field $\mathbf{H}$ due to various types of current distributions and in various types of media and to introduce a number of related quantities, such as the magnetic vector potential $\mathbf{A}$, the magnetic energy density w_m , and the inductance of a conducting structure, L . The parallelism that exists between these magnetostatic quantities and their electrostatic counterparts is elucidated in **Table 5-1**.

5-1 Magnetic Forces and Torques

The electric field $\mathbf{E}$ at a point in space has been defined as the electric force $\mathbf{F}_e$ per unit charge acting on a charged test particle placed at that point. We now define the **magnetic flux density** $\mathbf{B}$ at a point in space in terms of the **magnetic force** $\mathbf{F}_m$ that acts on a charged test particle moving with

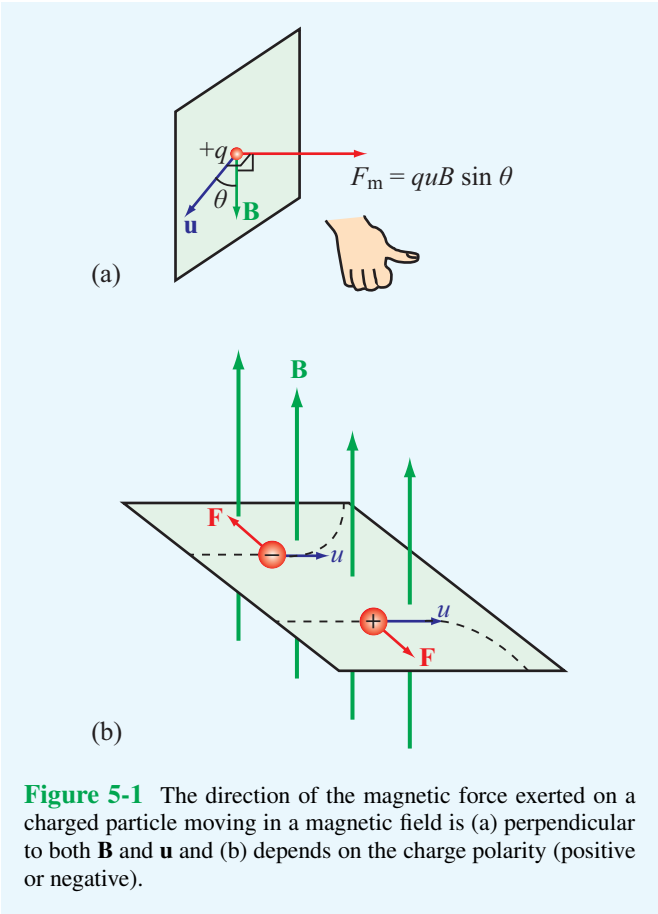


Figure 5-1 The direction of the magnetic force exerted on a charged particle moving in a magnetic field is (a) perpendicular to both $\mathbf{B}$ and $\mathbf{u}$ and (b) depends on the charge polarity (positive or negative).

velocity $\mathbf{u}$ through that point. Experiments revealed that a particle of charge q moving with velocity $\mathbf{u}$ in a magnetic field experiences a magnetic force $\mathbf{F}_m$ given by

$$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B} \quad (\text{N}).$$
(5.3)

Accordingly, the strength of $\mathbf{B}$ is measured in newtons/(C·m/s), which is also called the tesla (T). For a positively charged particle, the direction of $\mathbf{F}_m$ is that of the cross product $\mathbf{u} \times \mathbf{B}$, which is perpendicular to the plane containing $\mathbf{u}$ and $\mathbf{B}$ and governed by the right-hand rule. If q is negative, the direction of $\mathbf{F}_m$ is reversed (**Fig. 5-1**). The magnitude of $\mathbf{F}_m$ is given by

$$F_m = quB \sin \theta,$$
(5.4)

where θ is the angle between $\mathbf{u}$ and $\mathbf{B}$.

► We note that F_m is maximum when $\mathbf{u}$ is perpendicular to $\mathbf{B}$ ($\theta = 90^\circ$) and zero when $\mathbf{u}$ is parallel to $\mathbf{B}$ ($\theta = 0$ or 180°). ◀

概述

本章关于静磁学的内容与前一章关于静电学的内容类似。静电荷产生静电场，稳定（即非时变）电流产生静磁场。当 $\partial/\partial t = 0$ 时，磁导率为 μ 的介质中的磁场由第二对麦克斯韦方程组控制（方程 (4.3a,b)）：

$$\nabla \cdot \mathbf{B} = 0,$$
(5.1a)

$$\nabla \times \mathbf{H} = \mathbf{J},$$
(5.1b)

其中 $\mathbf{J}$ 是电流密度。磁通密度 $\mathbf{B}$ 和磁场强度 $\mathbf{H}$ 的关系为

$$\mathbf{B} = \mu \mathbf{H}.$$
(5.2)

在第 4 章研究介电介质中的电场时，我们注意到关系式 $\mathbf{D} = \epsilon \mathbf{E}$ 仅当介质是线性且各向同性时才有效。这些对于大多数材料都适用的属性使我们能够将介电常数 ϵ 视为一个恒定的标量，它与 $\mathbf{E}$ 的大小和方向无关。类似的陈述适用于方程 1 给出的关系。(5.2)。除了铁磁材料（ $\mathbf{B}$ 和 $\mathbf{H}$ 之间的关系是非线性的）之外，大多数材料都具有恒定磁导率的特征。

► 此外， $\mu = \mu_0$ 对于大多数电介质和金属（不包括铁磁材料）。◀

本章的目标是加深对由于各种类型的电流分布和各种介质中的稳定电流与磁通量 $\mathbf{B}$ 和磁场 $\mathbf{H}$ 之间的关系的理解，并引入许多相关量，例如磁矢量势 $\mathbf{A}$ 、磁能密度 w_m 和导电结构的电感 L 。表 5-1 阐明了这些静磁量与其静电对应量之间存在的并行性。

5-1 磁力和扭矩

空间中某一点的电场 $\mathbf{E}$ 被定义为作用于放置在该点的带电测试粒子上的每单位电荷的电力 $\mathbf{F}_e$ 。现在，我们用作用在带电测试粒子上的磁力 $\mathbf{F}_m$ 来定义空间中某一点的磁通密度 $\mathbf{B}$

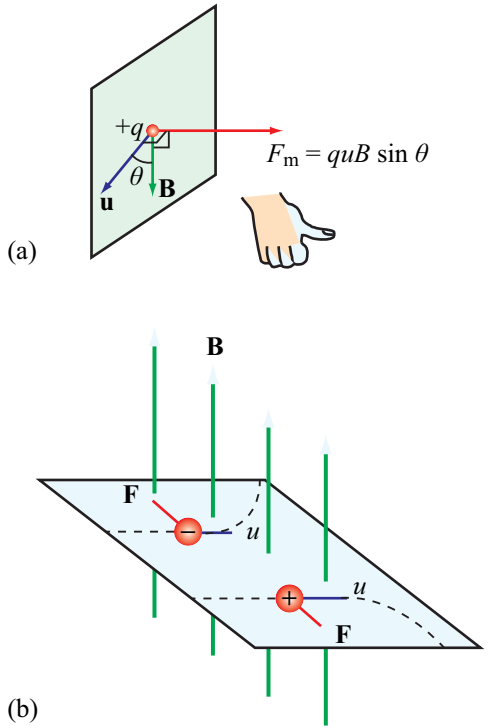


图 5-1 施加在磁场中运动的带电粒子上的磁力方向 (a) 垂直于 $\mathbf{B}$ 和 $\mathbf{u}$, (b) 取决于电荷极性 (正或负)。

通过该点的速度 $\mathbf{u}$ 。实验表明，在磁场中以速度 $\mathbf{u}$ 运动的电荷 q 粒子会受到由下式给出的磁力 $\mathbf{F}_m$

$$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B} \quad (\text{N}).$$
(5.3)

因此， $\mathbf{B}$ 的强度以牛顿/(C·m/s) 为单位，也称为特斯拉 (T)。对于带正电的粒子， $\mathbf{F}_m$ 的方向是叉积 $\mathbf{u} \times \mathbf{B}$ 的方向，垂直于包含 $\mathbf{u}$ 和 $\mathbf{B}$ 的平面并受右手定则约束。如果 q 为负，则 $\mathbf{F}_m$ 的方向相反（图 5-1）。 $\mathbf{F}_m$ 的大小由下式给出

$$F_m = quB \sin \theta,$$
(5.4)

其中 θ 是 $\mathbf{u}$ 和 $\mathbf{B}$ 之间的角度。

► 我们注意到 F_m 当 $\mathbf{u}$ 垂直于 $\mathbf{B}$ 时最大 ($\theta = 90^\circ$) 且当 $\mathbf{u}$ 平行于 $\mathbf{B}$ ($\theta = 0$ 或 180°)。◀

Table 5-1 Attributes of electrostatics and magnetostatics.

Attribute	Electrostatics	Magnetostatics
Sources	Stationary charges ρ_v	Steady currents $\mathbf{J}$
Fields and Fluxes	$\mathbf{E}$ and $\mathbf{D}$	$\mathbf{H}$ and $\mathbf{B}$
Constitutive parameter(s)	ϵ and σ	μ
Governing equations - Differential form - Integral form	$\nabla \cdot \mathbf{D} = \rho_v$ $\nabla \times \mathbf{E} = 0$ $\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$ $\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$	$\nabla \cdot \mathbf{B} = 0$ $\nabla \times \mathbf{H} = \mathbf{J}$ $\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$ $\oint_C \mathbf{H} \cdot d\mathbf{l} = I$
Potential	Scalar V , with $\mathbf{E} = -\nabla V$	Vector $\mathbf{A}$, with $\mathbf{B} = \nabla \times \mathbf{A}$
Energy density	$w_e = \frac{1}{2} \epsilon E^2$	$w_m = \frac{1}{2} \mu H^2$
Force on charge q	$\mathbf{F}_e = q\mathbf{E}$	$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B}$
Circuit element(s)	C and R	L

If a charged particle resides in the presence of both an electric field $\mathbf{E}$ and a magnetic field $\mathbf{B}$, then the total *electromagnetic force* acting on it is

$$\mathbf{F} = \mathbf{F}_e + \mathbf{F}_m = q\mathbf{E} + q\mathbf{u} \times \mathbf{B} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}). \quad (5.5)$$

The force expressed by Eq. (5.5) also is known as the *Lorentz force*. Electric and magnetic forces exhibit a number of important differences:

- Whereas the electric force is always in the direction of the electric field, the magnetic force is always perpendicular to the magnetic field.
- Whereas the electric force acts on a charged particle whether or not it is moving, the magnetic force acts on it only when it is in motion.
- Whereas the electric force expends energy in displacing a charged particle, the magnetic force does no work when a particle is displaced.

This last statement requires further elaboration. Because the magnetic force $\mathbf{F}_m$ is always perpendicular to $\mathbf{u}$, $\mathbf{F}_m \cdot \mathbf{u} = 0$.

Hence, the work performed when a particle is displaced by a differential distance $d\mathbf{l} = \mathbf{u} \, dt$ is

$$dW = \mathbf{F}_m \cdot d\mathbf{l} = (\mathbf{F}_m \cdot \mathbf{u}) \, dt = 0. \quad (5.6)$$

► Since no work is done, a magnetic field cannot change the kinetic energy of a charged particle; the magnetic field can change the direction of motion of a charged particle, but not its speed. ◀

Exercise 5-1: An electron moving in the positive x direction perpendicular to a magnetic field is deflected in the negative z direction. What is the direction of the magnetic field?

Answer: Positive y direction. (See )

Exercise 5-2: A proton moving with a speed of 2×10^6 m/s through a magnetic field with magnetic flux density of 2.5 T experiences a magnetic force of magnitude 4×10^{-13} N. What is the angle between the magnetic field and the proton's velocity?

表 5-1 静电学和静磁学的属性。

Attribute	Electrostatics	Magnetostatics
Sources	Stationary charges ρ_v	Steady currents $\mathbf{J}$
Fields and Fluxes	$\mathbf{E}$ and $\mathbf{D}$	$\mathbf{H}$ and $\mathbf{B}$
Constitutive parameter(s)	ϵ and σ	μ
Governing equations - Differential form - Integral form	$\nabla \cdot \mathbf{D} = \rho_v$ $\nabla \times \mathbf{E} = 0$ $\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$ $\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$	$\nabla \cdot \mathbf{B} = 0$ $\nabla \times \mathbf{H} = \mathbf{J}$ $\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$ $\oint_C \mathbf{H} \cdot d\mathbf{l} = I$
Potential	Scalar V , with $\mathbf{E} = -\nabla V$	Vector $\mathbf{A}$, with $\mathbf{B} = \nabla \times \mathbf{A}$
Energy density	$w_e = \frac{1}{2} \epsilon E^2$	$w_m = \frac{1}{2} \mu H^2$
Force on charge q	$\mathbf{F}_e = q\mathbf{E}$	$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B}$
Circuit element(s)	C and R	L

如果带电粒子同时存在电场 $\mathbf{E}$ 和磁场 $\mathbf{B}$ ，则作用在其上的总电磁力为

$$\mathbf{F} = \mathbf{F}_e + \mathbf{F}_m = q\mathbf{E} + q\mathbf{u} \times \mathbf{B} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}). \quad (5.5)$$

力由方程式表示。(5.5) 也称为洛伦兹力。电力和磁力表现出许多重要的差异：

- 电力始终沿电场方向，而磁力始终垂直于磁场方向。
- 无论带电粒子是否运动，电力都会作用在带电粒子上，而磁力仅在带电粒子运动时才作用在带电粒子上。
- 电力在移动带电粒子时会消耗能量，而磁力在粒子移动时不起作用。

最后的陈述需要进一步阐述。因为磁力 $\mathbf{F}_m$ 始终垂直于 $\mathbf{u}$ ，所以 $\mathbf{F}_m \cdot \mathbf{u} = 0$ 。

因此，当粒子位移差分距离 $d\mathbf{l} = \mathbf{u} \, dt$ 时所做的功为

$$dW = \mathbf{F}_m \cdot d\mathbf{l} = (\mathbf{F}_m \cdot \mathbf{u}) \, dt = 0. \quad (5.6)$$

► 由于没有做功，磁场不能改变带电粒子的动能；磁场可以改变带电粒子的运动方向，但不能改变其速度。 ◀

练习 5-1：在垂直于磁场的正 x 方向上移动的电子会在负 z 方向上偏转。磁场的方向是什么？

答案：y 轴正方向。（参见^{EM}。）

练习 5-2：质子以 2×10^6 m/s 的速度穿过磁通密度为 2.5 T 的磁场时，会受到强度为 4×10^{-13} N 的磁力。磁场与质子速度之间的角度是多少？

Answer: $\theta = 30^\circ$ or 150° . (See [EM](#).)

Exercise 5-3: A charged particle with velocity $\mathbf{u}$ is moving in a medium with uniform fields $\mathbf{E} = \hat{\mathbf{x}}E$ and $\mathbf{B} = \hat{\mathbf{y}}B$. What should $\mathbf{u}$ be so that the particle experiences no net force?

Answer: $\mathbf{u} = \hat{\mathbf{z}}E/B$. ($\mathbf{u}$ may also have an arbitrary y component u_y .) (See [EM](#).)

5-1.1 Magnetic Force on a Current-Carrying Conductor

A current flowing through a conducting wire consists of charged particles *drifting* through the material of the wire. Consequently, when a current-carrying wire is placed in a magnetic field, it experiences a force equal to the sum of the magnetic forces acting on the charged particles moving within it. Consider, for example, the arrangement shown in [Fig. 5-2](#) in which a vertical wire oriented along the z direction is placed in a magnetic field $\mathbf{B}$ (produced by a magnet) oriented along the $-\hat{\mathbf{x}}$ direction (into the page). With no current flowing in the wire, $\mathbf{F}_m = 0$ and the wire maintains its vertical orientation ([Fig. 5-2\(a\)](#)), but when a current is introduced in the wire, the wire deflects to the left ($-\hat{\mathbf{y}}$ direction) if the current direction is upward ($+\hat{\mathbf{z}}$ direction) and to the right ($+\hat{\mathbf{y}}$ direction) if the current direction is downward ($-\hat{\mathbf{z}}$ direction). The directions of these deflections are in accordance with the cross product given by Eq. (5.3).

To quantify the relationship between $\mathbf{F}_m$ and the current I flowing in a wire, let us consider a small segment of the wire of cross-sectional area A and differential length $d\mathbf{l}$ with the direction of $d\mathbf{l}$ denoting the direction of the current. Without loss of generality, we assume that the charge carriers constituting the current I are exclusively electrons, which is always a valid assumption for a good conductor. If the wire contains a free-electron charge density $\rho_{ve} = -N_e e$, where N_e is the number of moving electrons per unit volume, then the total amount of moving charge contained in an elemental volume of the wire is

$$dQ = \rho_{ve} A dl = -N_e e A dl, \tag{5.7}$$

and the corresponding magnetic force acting on dQ in the presence of a magnetic field $\mathbf{B}$ is

$$d\mathbf{F}_m = dQ \mathbf{u}_e \times \mathbf{B} = -N_e e A dl \mathbf{u}_e \times \mathbf{B}, \tag{5.8a}$$

where $\mathbf{u}_e$ is the drift velocity of the electrons. Since the direction of a current is defined as the direction of flow of positive charges, the electron drift velocity $\mathbf{u}_e$ is parallel to $d\mathbf{l}$,

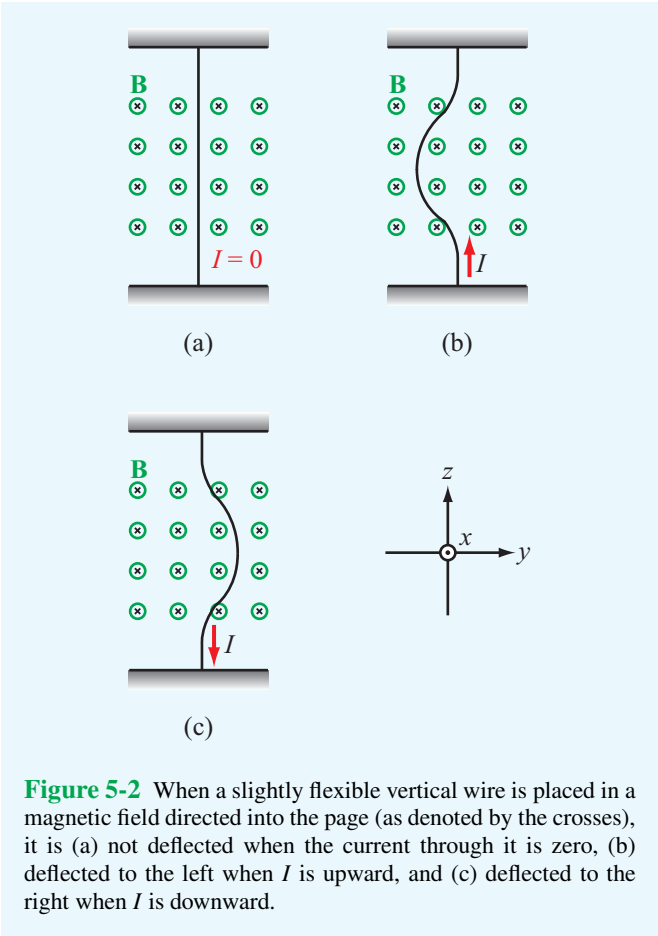


Figure 5-2 When a slightly flexible vertical wire is placed in a magnetic field directed into the page (as denoted by the crosses), it is (a) not deflected when the current through it is zero, (b) deflected to the left when I is upward, and (c) deflected to the right when I is downward.

but opposite in direction. Thus, $d\mathbf{l} \mathbf{u}_e = -d\mathbf{l} u_e$ and Eq. (5.8a) becomes

$$d\mathbf{F}_m = N_e e A u_e d\mathbf{l} \times \mathbf{B}. \tag{5.8b}$$

From Eqs. (4.11) and (4.12), the current I flowing through a cross-sectional area A due to electrons with density $\rho_{ve} = -N_e e$ and moving with velocity $-u_e$ is

$$I = \rho_{ve}(-u_e)A = (-N_e e)(-u_e)A = N_e e A u_e.$$

Hence, Eq. (5.8b) may be written in the compact form

$$d\mathbf{F}_m = I d\mathbf{l} \times \mathbf{B} \quad (\text{N}). \tag{5.9}$$

For a closed circuit of contour C carrying a current I , the total magnetic force is

$$\mathbf{F}_m = I \oint_C d\mathbf{l} \times \mathbf{B} \quad (\text{N}). \tag{5.10}$$

答案: $\theta = 30^\circ$ 或 150° 。(参见 [EM](#)。)

练习 5-3: 速度为 u 的带电粒子在均匀场 $E = \hat{x}E$ 和 $B = \hat{y}B$ 的介质中运动。你应该怎样才能使粒子不受净力的影响?

答案: $u = \hat{z}E/B$ 。(u 也可以有任意 y 分量 u_y 。) (参见 [EM](#)。)

5-1.1 载流导体上的磁力

流经导线的电流由漂移穿过导线材料的带电粒子组成。因此，当载流导线放置在磁场中时，它所受到的力等于作用在其中移动的带电粒子上的磁力之和。例如，考虑图 5-2 中所示的布置，其中沿 z 方向定向的垂直导线放置在沿 $-\hat{x}$ 方向（进入页面）定向的磁场 $\mathbf{B}$ （由磁铁产生）中。当导线中没有电流流动时， $\mathbf{F}_m = 0$ 且导线保持垂直方向（图 5-2 (a) ），但当导线中引入电流时，如果电流方向向上（ $+\hat{z}$ 方向），则导线向左偏转（ $-\hat{y}$ 方向）；如果电流方向向下（ $-\hat{z}$ 方向），导线向右偏转（ $+\hat{y}$ 方向）。这些偏转的方向与方程式给出的叉积一致。（5.3）。

为了量化 $\mathbf{F}_m$ 与导线中流动的电流 I 之间的关系，让我们考虑一小段横截面积为 A 且长度微分为 $d\mathbf{l}$ 的导线，其中 $d\mathbf{l}$ 的方向表示电流方向。不失一般性，我们假设构成电流 I 的电荷载体完全是电子，这对于良导体始终是有有效的假设。如果导线包含自由电子电荷密度 $\rho_{ve} = -N_e e$ ，其中 N_e 是每单位体积移动电子的数量，则导线元素体积中包含的移动电荷总量为

$$dQ = \rho_{ve} A dl = -N_e e A dl, \tag{5.7}$$

在磁场 $\mathbf{B}$ 存在的情况下，作用在 dQ 上的相应磁力为

$$d\mathbf{F}_m = dQ \mathbf{u}_e \times \mathbf{B} = -N_e e A dl \mathbf{u}_e \times \mathbf{B}, \tag{5.8a}$$

其中 u_e 是电子的漂移速度。由于电流方向定义为正电荷流动方向，因此电子漂移速度 u_e 平行于 $d\mathbf{l}$ ，

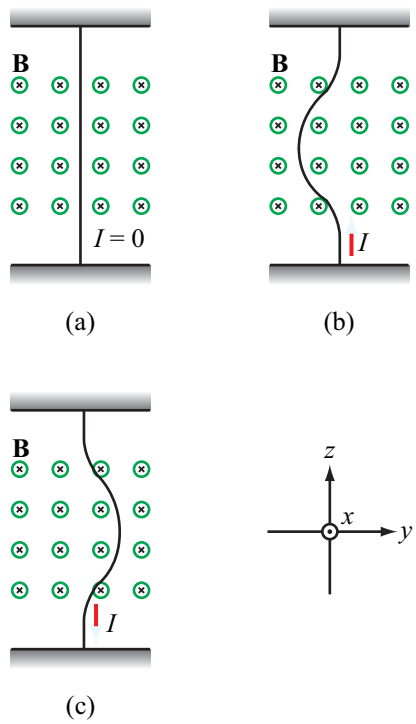


图 5-2 当一根稍微柔软的垂直导线放置在指向页面的磁场中（如十字所示）时，它 (a) 当通过它的电流为零时不偏转，(b) 当 I 向上时向左偏转，以及 (c) 当 I 向下时向右偏转。

但方向相反。因此， $d\mathbf{l} \mathbf{u}_e = -d\mathbf{l} u_e$ 和方程。(5.8a) 变为

$$d\mathbf{F}_m = N_e e A u_e d\mathbf{l} \times \mathbf{B}. \tag{5.8b}$$

从方程式。(4.11) 和 (4.12) 中，密度为 $\rho_{ve} = -N_e e$ 且以速度 $-u_e$ 运动的电子流过横截面积 A 的电流 I 为

$$I = \rho_{ve}(-u_e)A = (-N_e e)(-u_e)A = N_e e A u_e.$$

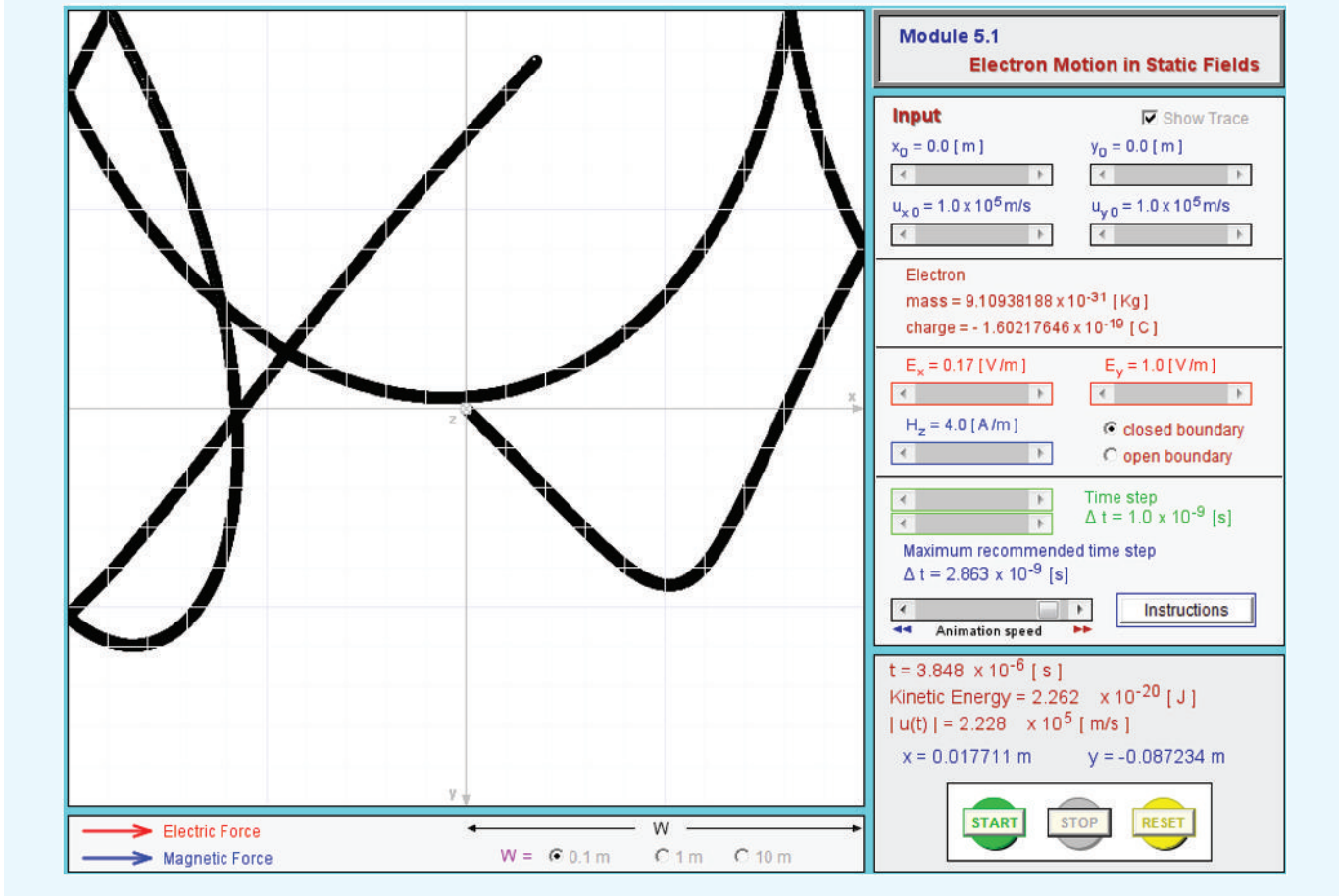
因此，等式。(5.8b) 可以写成紧凑形式

$$d\mathbf{F}_m = I d\mathbf{l} \times \mathbf{B} \quad (\text{N}). \tag{5.9}$$

对于承载电流 I 的轮廓 C 的闭合电路，总磁力为

$$\mathbf{F}_m = I \oint_C d\mathbf{l} \times \mathbf{B} \quad (\text{N}). \tag{5.10}$$

Module 5.1 Electron Motion in Static Fields This module demonstrates the Lorentz force on an electron moving under the influence of an electric field alone, a magnetic field alone, or both acting simultaneously.



If the closed wire shown in **Fig. 5-3(a)** resides in a uniform external magnetic field $\mathbf{B}$, then $\mathbf{B}$ can be taken outside the integral in Eq. (5.10), in which case

$$\mathbf{F}_m = I \left(\oint_C d\mathbf{l} \right) \times \mathbf{B} = 0. \quad (5.11)$$

► This result, which is a consequence of the fact that the vector sum of the infinitesimal vectors $d\mathbf{l}$ over a closed path equals zero, states that the total magnetic force on any closed current loop in a uniform magnetic field is zero. ◀

This does not mean that the force at every point along the wire is zero, but rather that the vector sum of all forces exerted on all parts of the closed wire adds up to zero.

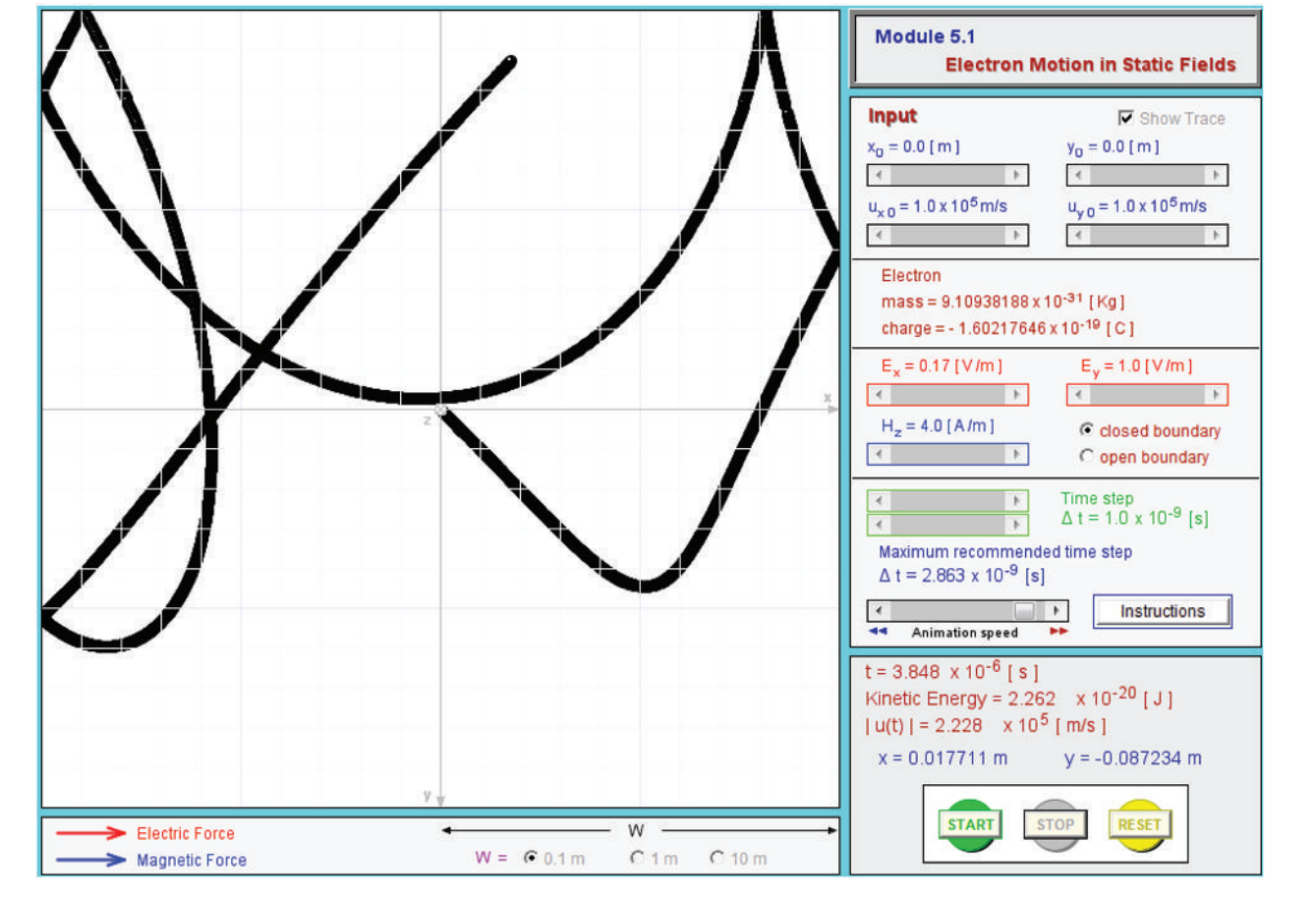
In the study of magnetostatics, all currents flow through closed paths. To understand why, consider the curved wire in **Fig. 5-3(b)** carrying a current I from point a to point b . In doing so, negative charges accumulate at a , and positive ones at b . The time-varying nature of these charges violates the static assumptions underlying Eqs. (5.1a,b).

If we are interested in the magnetic force exerted on a wire segment l (**Fig. 5-3(b)**) residing in a uniform magnetic field (while realizing that it is part of a closed current loop), we can integrate Eq. (5.9) to obtain

$$\mathbf{F}_m = I \left(\int_\ell d\mathbf{l} \right) \times \mathbf{B} = I\ell \times \mathbf{B}, \quad (5.12)$$

where ℓ is the vector directed from a to b (**Fig. 5-3(b)**). The integral of $d\mathbf{l}$ from a to b has the same value irrespective of the path taken between a and b . For a closed loop, points a and b

模块 5.1 静态场中的电子运动 该模块演示了在单独的电场、单独的磁场或两者同时作用的影响下运动的电子所受到的洛伦兹力。



如果图 5-3(a) 所示的闭合导线位于均匀的外部磁场 $\mathbf{B}$ 中，则 $\mathbf{B}$ 可以在方程 (1) 的积分之外取。 (5.10)，在这种情况下

$$\mathbf{F}_m = I \left(\oint_C d\mathbf{l} \right) \times \mathbf{B} = 0. \quad (5.11)$$

► 这个结果是闭合路径上无穷小矢量 $d\mathbf{l}$ 的矢量和等于零这一事实的结果，表明均匀磁场中任何闭合电流回路上的总磁力为零。 ◀

这并不意味着沿线的每一点的力都为零，而是施加在闭合线的所有部分上的所有力的矢量和加起来为零。

在静磁学的研究中，所有电流都流过闭合路径。要理解其中的原因，请考虑图 5-3(b) 中的弯曲导线，其从 a 点到 b 点承载电流 I 。在此过程中，负电荷在 a 处积聚，正电荷在 b 处积聚。这些电荷的随时间变化的性质违反了方程式的静态假设。 (5.1a, b)。

如果我们对施加在均匀磁场中的线段 l (图 5-3(b)) 上的磁力感兴趣 (同时意识到它是闭合电流环路的一部分)，我们可以对方程 1 进行积分。 (5.9) 得到

$$\mathbf{F}_m = I \left(\int_\ell d\mathbf{l} \right) \times \mathbf{B} = I\ell \times \mathbf{B}, \quad (5.12)$$

其中 ℓ 是从 a 到 b 的向量 (图 5-3 (b))。无论 a 和 b 之间采用的路径如何，从 a 到 b 的 $d\mathbf{l}$ 积分都具有相同的值。对于闭环，点 a 和 b

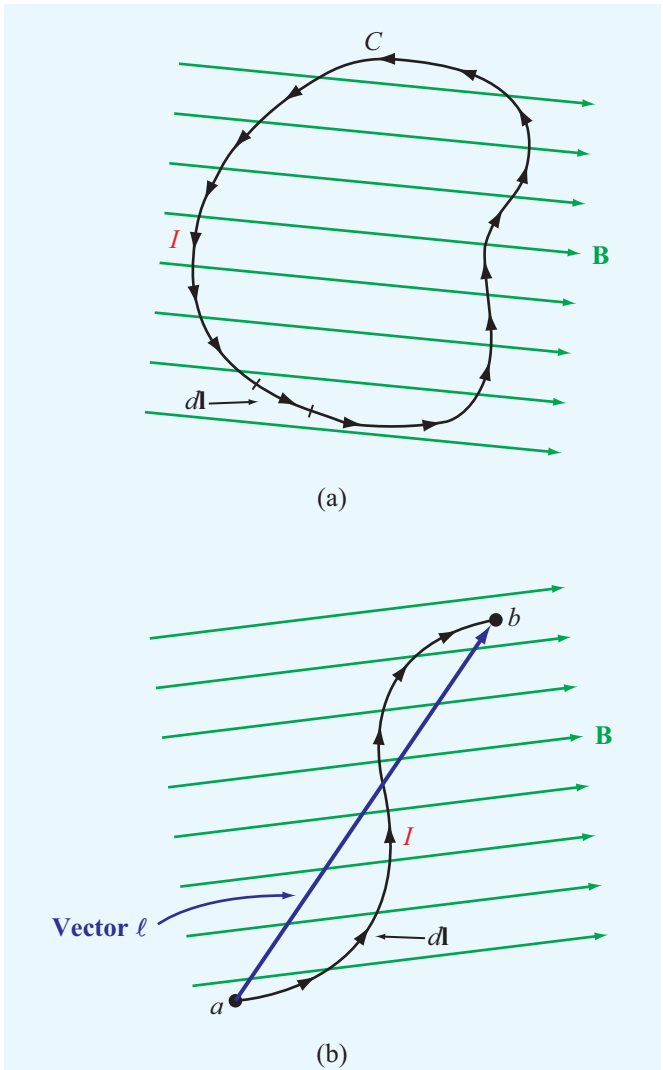


Figure 5-3 In a uniform magnetic field, (a) the net force on a closed current loop is zero because the integral of the displacement vector $d\mathbf{l}$ over a closed contour is zero and (b) the force on a line segment is proportional to the vector between the end point ($\mathbf{F}_m = I\boldsymbol{\ell} \times \mathbf{B}$).

become the same point, in which case $\ell = 0$ and $\mathbf{F}_m = 0$.

Example 5-1: Force on a Semicircular Conductor

The semicircular conductor shown in **Fig. 5-4** lies in the

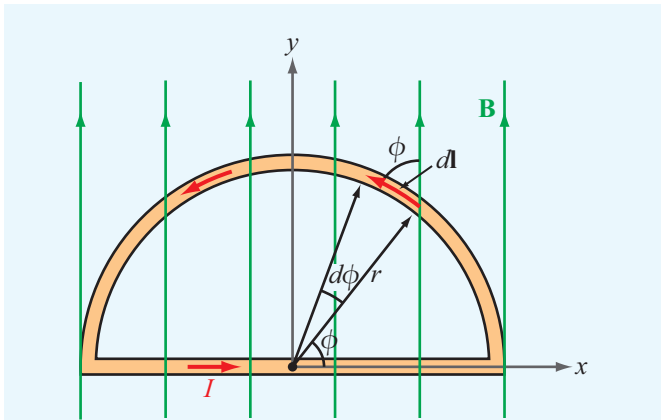


Figure 5-4 Semicircular conductor in a uniform field (Example 5-1).

x - y plane and carries a current I . The closed circuit is exposed to a uniform magnetic field $\mathbf{B} = \hat{y}B_0$. Determine (a) the magnetic force $\mathbf{F}_1$ on the straight section of the wire and (b) the force $\mathbf{F}_2$ on the curved section.

Solution: (a) To evaluate $\mathbf{F}_1$, consider that the straight section of the circuit is of length $2r$ and its current flows along the $+\hat{x}$ direction. Application of Eq. (5.12) with $\boldsymbol{\ell} = \hat{x}2r$ gives

$$\mathbf{F}_1 = \hat{x}(2Ir) \times \hat{y}B_0 = \hat{z}2IrB_0 \quad (\text{N}).$$

The $\hat{z}$ direction in **Fig. 5-4** is out of the page.

(b) To evaluate $\mathbf{F}_2$, consider a segment of differential length $d\mathbf{l}$ on the curved part of the circle. The direction of $d\mathbf{l}$ is chosen to coincide with the direction of the current. Since $d\mathbf{l}$ and $\mathbf{B}$ are both in the x - y plane, their cross product $d\mathbf{l} \times \mathbf{B}$ points in the negative z direction, and the magnitude of $d\mathbf{l} \times \mathbf{B}$ is proportional to $\sin \phi$, where ϕ is the angle between $d\mathbf{l}$ and $\mathbf{B}$. Moreover, the magnitude of $d\mathbf{l}$ is $d\mathbf{l} = r d\phi$. Hence,

$$\begin{aligned} \mathbf{F}_2 &= I \int_{\phi=0}^{\pi} d\mathbf{l} \times \mathbf{B} \\ &= -\hat{z}I \int_{\phi=0}^{\pi} rB_0 \sin \phi d\phi = -\hat{z}2IrB_0 \quad (\text{N}). \end{aligned}$$

The $-\hat{z}$ direction of the force acting on the curved part of the conductor is into the page. We note that $\mathbf{F}_2 = -\mathbf{F}_1$, implying that no net force acts on the closed loop, although opposing forces act on its two sections.

Concept Question 5-1: What are the major differences between the electric force $\mathbf{F}_e$ and the magnetic force $\mathbf{F}_m$?

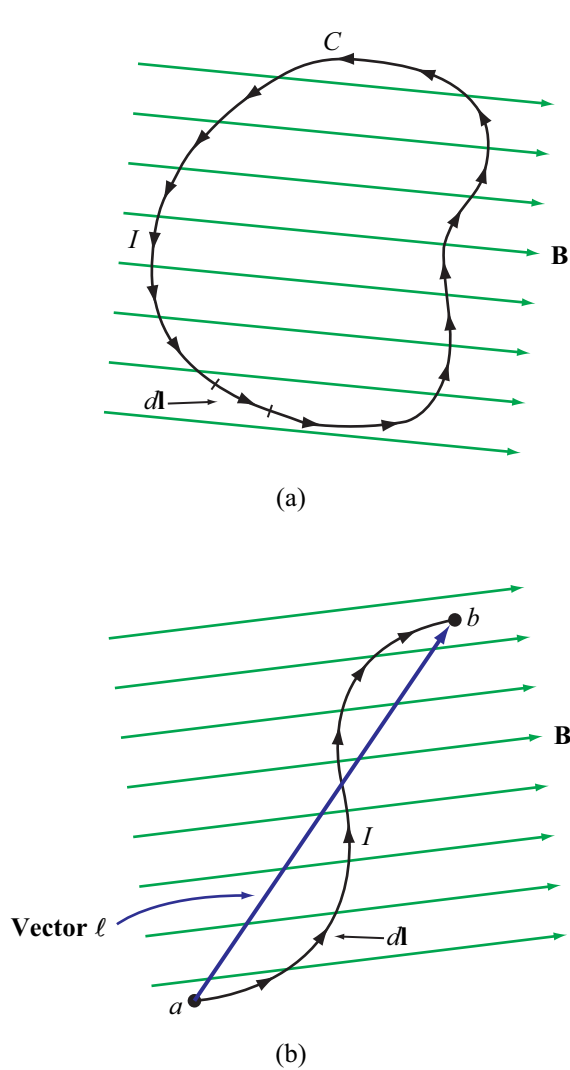


图 5-4 均匀场中的半圆形导体 (例 5-1)。

x - y 平面并承载电流 I 。闭合电路暴露在均匀磁场 $\mathbf{B} = \hat{y}B_0$ 中。确定 (a) 导线直线部分上的磁力 $\mathbf{F}_1$ 和 (b) 弯曲部分上的力 $\mathbf{F}_2$ 。

解：(a) 要计算 $\mathbf{F}_1$ ，请考虑电路的直线部分长度为 $2r$ ，并且其电流沿 $+\hat{x}$ 方向流动。方程的应用(5.12) 与 $\boldsymbol{\ell} = \hat{x}2r$ 给出

$$\mathbf{F}_1 = \hat{x}(2Ir) \times \hat{y}B_0 = \hat{z}2IrB_0 \quad (\text{N}).$$

图5-4中的 $\hat{z}$ 方向在页面之外。

(b) 为了计算 $\mathbf{F}_2$ ，请考虑圆的弯曲部分上的微分长度 $d\mathbf{l}$ 的线段。选择 $d\mathbf{l}$ 的方向与电流方向一致。由于 $d\mathbf{l}$ 和 $\mathbf{B}$ 都在 x - y 平面上，因此它们的叉积 $d\mathbf{l} \times \mathbf{B}$ 指向负 z 方向，并且 $d\mathbf{l} \times \mathbf{B}$ 的大小与 $\sin \phi$ 成正比，其中 ϕ 是 $d\mathbf{l}$ 和 $\mathbf{B}$ 之间的角度。此外， $d\mathbf{l}$ 的大小为 $d\mathbf{l} = r d\phi$ 。因此，

$$\begin{aligned} \mathbf{F}_2 &= I \int_{\phi=0}^{\pi} d\mathbf{l} \times \mathbf{B} \\ &= -\hat{z}I \int_{\phi=0}^{\pi} rB_0 \sin \phi d\phi = -\hat{z}2IrB_0 \quad (\text{N}). \end{aligned}$$

作用在导体弯曲部分的力的 $-\hat{z}$ 方向进入页面。我们注意到 $\mathbf{F}_2 = -\mathbf{F}_1$ ，这意味着没有净力作用在闭环上，尽管相反的力作用在其两个部分上。

概念问题 5-1：电力 $\mathbf{F}_e$ 和磁力 $\mathbf{F}_m$ 之间的主要区别是什么？

图5-4所示的半圆形导体位于

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Concept Question 5-2: The ends of a 10-cm long wire carrying a constant current I are anchored at two points on the x axis, namely $x = 0$ and $x = 6$ cm. If the wire lies in the x - y plane in a magnetic field $\mathbf{B} = \hat{y}B_0$, which of the following arrangements produces a greater magnetic force on the wire: (a) wire is V-shaped with corners at $(0,0)$, $(3,4)$, and $(6,0)$ or (b) wire is an open rectangle with corners at $(0,0)$, $(0,2)$, $(6,2)$, and $(6,0)$?

Exercise 5-4: A horizontal wire with a mass per unit length of 0.2 kg/m carries a current of 4 A in the $+x$ direction. If the wire is placed in a uniform magnetic flux density $\mathbf{B}$, what should the direction and minimum magnitude of $\mathbf{B}$ be in order to magnetically lift the wire vertically upward? (Hint: The acceleration due to gravity is $\mathbf{g} = -\hat{z}9.8 \text{ m/s}^2$.)

Answer: $\mathbf{B} = \hat{y}0.49 \text{ T}$. (See .)

5-1.2 Magnetic Torque on a Current-Carrying Loop

When a force is applied on a rigid body that can pivot about a fixed axis, the body will, in general, react by rotating about that axis. The angular acceleration depends on the cross product of the applied force vector $\mathbf{F}$ and the distance vector $\mathbf{d}$, which is measured from a point on the rotation axis (such that $\mathbf{d}$ is perpendicular to the axis) to the point of application of $\mathbf{F}$ (Fig. 5-5). The length of $\mathbf{d}$ is called the **moment arm**, and the cross product

$$\mathbf{T} = \mathbf{d} \times \mathbf{F} \quad (\text{N}\cdot\text{m}) \quad (5.13)$$

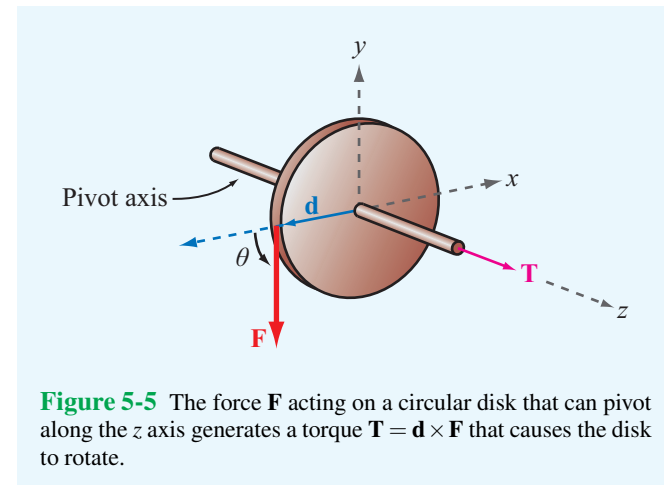


Figure 5-5 The force $\mathbf{F}$ acting on a circular disk that can pivot along the z axis generates a torque $\mathbf{T} = \mathbf{d} \times \mathbf{F}$ that causes the disk to rotate.

is called the **torque**. The unit for $\mathbf{T}$ is the same as that for work or energy, even though torque does not represent either. The force $\mathbf{F}$ applied on the disk shown in Fig. 5-5 lies in the x - y plane and makes an angle θ with $\mathbf{d}$. Hence,

$$\mathbf{T} = \hat{z}rF \sin \theta, \quad (5.14)$$

where $|\mathbf{d}| = r$, which is the radius of the disk, and $F = |\mathbf{F}|$. From Eq. (5.14), we observe that a torque along the positive z direction corresponds to a tendency for the cylinder to rotate counterclockwise and, conversely, a torque along the $-z$ direction corresponds to clockwise rotation.

► These directions are governed by the following **right-hand rule**: When the thumb of the right hand points along the direction of the torque, the four fingers indicate the direction that the torque tries to rotate the body. ◀

We now consider the **magnetic torque** exerted on a conducting loop under the influence of magnetic forces. We begin with the simple case where the magnetic field $\mathbf{B}$ is in the plane of the loop, and then we extend the analysis to the more general case where $\mathbf{B}$ makes an angle θ with the surface normal of the loop.

Magnetic Field in the Plane of the Loop

The rectangular conducting loop shown in Fig. 5-6(a) is constructed from rigid wire and carries a current I . The loop lies in the x - y plane and is allowed to pivot about the axis shown. Under the influence of an externally generated uniform magnetic field $\mathbf{B} = \hat{x}B_0$, arms 1 and 3 of the loop are subjected to forces $\mathbf{F}_1$ and $\mathbf{F}_3$ given by

$$\mathbf{F}_1 = I(-\hat{y}b) \times (\hat{x}B_0) = \hat{z}IbB_0, \quad (5.15a)$$

and

$$\mathbf{F}_3 = I(\hat{y}b) \times (\hat{x}B_0) = -\hat{z}IbB_0. \quad (5.15b)$$

These results are based on the application of Eq. (5.12). We note that the magnetic forces acting on arms 1 and 3 are in opposite directions, and no magnetic force is exerted on either arm 2 or 4 because $\mathbf{B}$ is parallel to the direction of the current flowing in those arms.

A bottom view of the loop, depicted in Fig. 5-6(b), reveals that forces $\mathbf{F}_1$ and $\mathbf{F}_3$ produce a torque about the origin O , causing the loop to rotate in a clockwise direction. The moment arm is $a/2$ for both forces, but $\mathbf{d}_1$ and $\mathbf{d}_3$ are in opposite directions, resulting in a total magnetic torque of

$$\begin{aligned} \mathbf{T} &= \mathbf{d}_1 \times \mathbf{F}_1 + \mathbf{d}_3 \times \mathbf{F}_3 \\ &= \left(-\hat{x} \frac{a}{2}\right) \times (\hat{z}IbB_0) + \left(\hat{x} \frac{a}{2}\right) \times (-\hat{z}IbB_0) \\ &= \hat{y}IabB_0 = \hat{y}IAB_0, \end{aligned} \quad (5.16)$$

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概念问题 5-2: 承载恒定电流 I 的 10 厘米长导线的末端固定在 x 轴上的两个点, 即 $x = 0$ 和 $x = 6$ 厘米。如果导线位于磁场 $\mathbf{B} = \hat{y}B_0$ 中的 x - y 平面上, 下列哪种布置会在导线上产生更大的磁力: (a) 导线为 V 形, 角点为 $(0,0)$ 、 $(3,4)$ 和 $(6,0)$ 或 (b) 导线为开放矩形, 角点为 $(0,0)$ 、 $(0,2)$ 、 $(6,2)$ 和 $(6,0)$?

练习 5-4: 单位长度质量为 0.2 kg/m 的水平导线在 $+x$ 方向承载 4 A 的电流。如果将导线放置在均匀磁通密度 $\mathbf{B}$ 中, 为了使导线垂直向上磁力提升, $\mathbf{B}$ 的方向和最小大小应为多少? (提示: 重力加速度为 $\mathbf{g} = -\hat{z}9.8 \text{ m/s}^2$.)

答案: $\mathbf{B} = \hat{y}0.49 \text{ T}$ 。(参见 .)

5-1.2 载流回路上的磁扭矩

当一个力施加在一个可以绕固定轴旋转的刚体上时, 该刚体通常会通过绕该轴旋转来做出反应。角加速度取决于施加的力矢量 $\mathbf{F}$ 和距离矢量 $\mathbf{d}$ 的叉积, 距离矢量 $\mathbf{d}$ 是从旋转轴上的一点 (使得 $\mathbf{d}$ 垂直于轴) 到 $\mathbf{F}$ 的施加点测量的 (图 5-5)。 $\mathbf{d}$ 的长度称为力臂, 叉积

$$\mathbf{T} = \mathbf{d} \times \mathbf{F} \quad (\text{N}\cdot\text{m}) \quad (5.13)$$

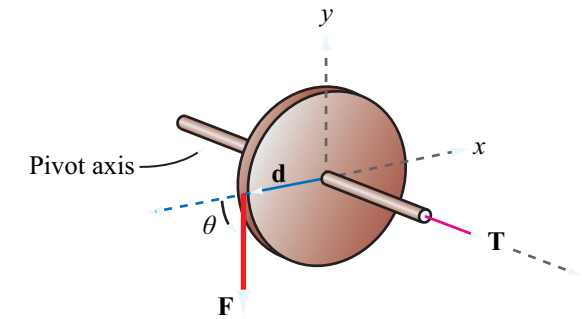


图 5-5 作用在可沿 z 轴枢转的圆盘上的力 $\mathbf{F}$ 会产生导致圆盘旋转的扭矩 $\mathbf{T} = \mathbf{d} \times \mathbf{F}$ 。

称为扭矩。T 的单位与功或能量的单位相同, 尽管扭矩并不代表这两者。如图 5-5 所示, 施加在圆盘上的力 $\mathbf{F}$ 位于 x - y 平面内, 并与 $\mathbf{d}$ 形成角度 θ 。因此,

$$\mathbf{T} = \hat{z}rF \sin \theta, \quad (5.14)$$

其中 $|\mathbf{d}| = r$ 是圆盘的半径, $F = |\mathbf{F}|$ 。从方程。根据方程 (5.14), 我们观察到沿正 z 方向的扭矩对应于圆柱体逆时针旋转的趋势, 相反, 沿 $-z$ 方向的扭矩对应于顺时针旋转。

► 这些方向受以下右手法则控制: 当右手的拇指指向扭矩的方向时, 四个手指指示扭矩试图旋转物体的方向。◀

我们现在考虑在磁力影响下施加在导电环上的磁扭矩。我们从磁场 $\mathbf{B}$ 位于环平面内的简单情况开始, 然后将分析扩展到更一般的情况, 其中 $\mathbf{B}$ 与环的表面法线形成角度 θ 。

环平面内的磁场

图 5-6(a) 所示的矩形导电环由刚性导线构成, 承载电流 I 。该环位于 x - y 平面内, 可以绕所示轴旋转。在外部产生的均匀磁场 $\mathbf{B} = \hat{x}B_0$ 的影响下, 环路的臂 1 和 3 受到由下式给出的力 $\mathbf{F}_1$ 和 $\mathbf{F}_3$

$$\mathbf{F}_1 = I(-\hat{y}b) \times (\hat{x}B_0) = \hat{z}IbB_0, \quad (5.15a)$$

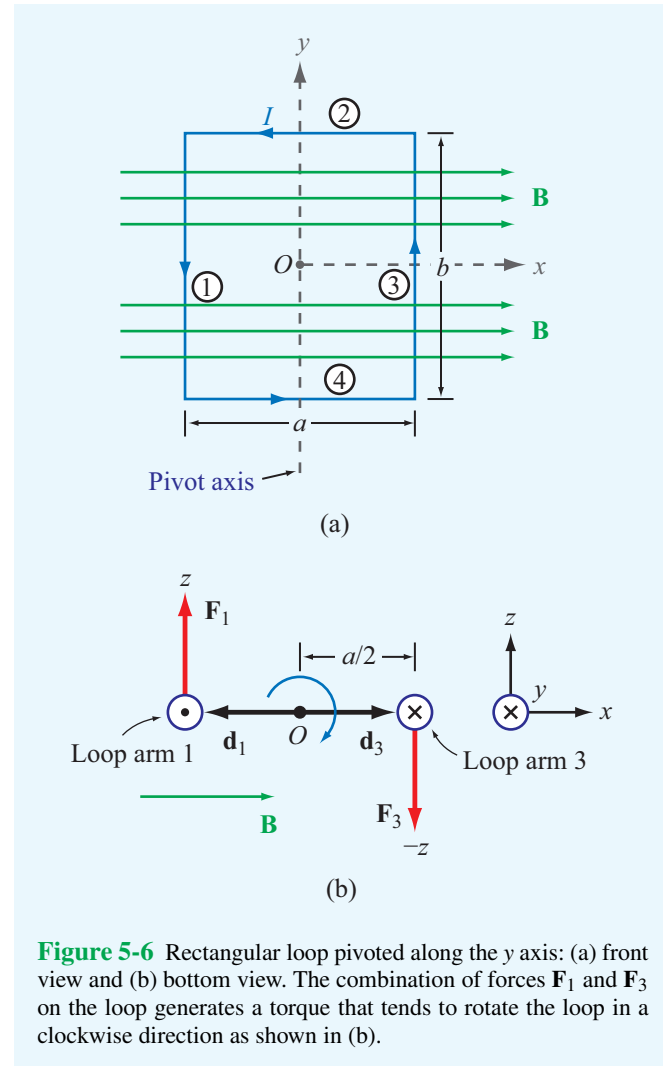
and

$$\mathbf{F}_3 = I(\hat{y}b) \times (\hat{x}B_0) = -\hat{z}IbB_0. \quad (5.15b)$$

这些结果基于方程的应用。(5.12)。我们注意到, 作用在臂 1 和 3 上的磁力方向相反, 并且没有磁力施加在臂 2 或 4 上, 因为 $\mathbf{B}$ 平行于这些臂中的电流流动方向。

如图 5-6(b) 所示, 环路的底视图显示, 力 $\mathbf{F}_1$ 和 $\mathbf{F}_3$ 产生绕原点 O 的扭矩, 导致环路沿顺时针方向旋转。两个力的力臂均为 $a/2$, 但 $\mathbf{d}_1$ 和 $\mathbf{d}_3$ 方向相反, 因此总磁矩为

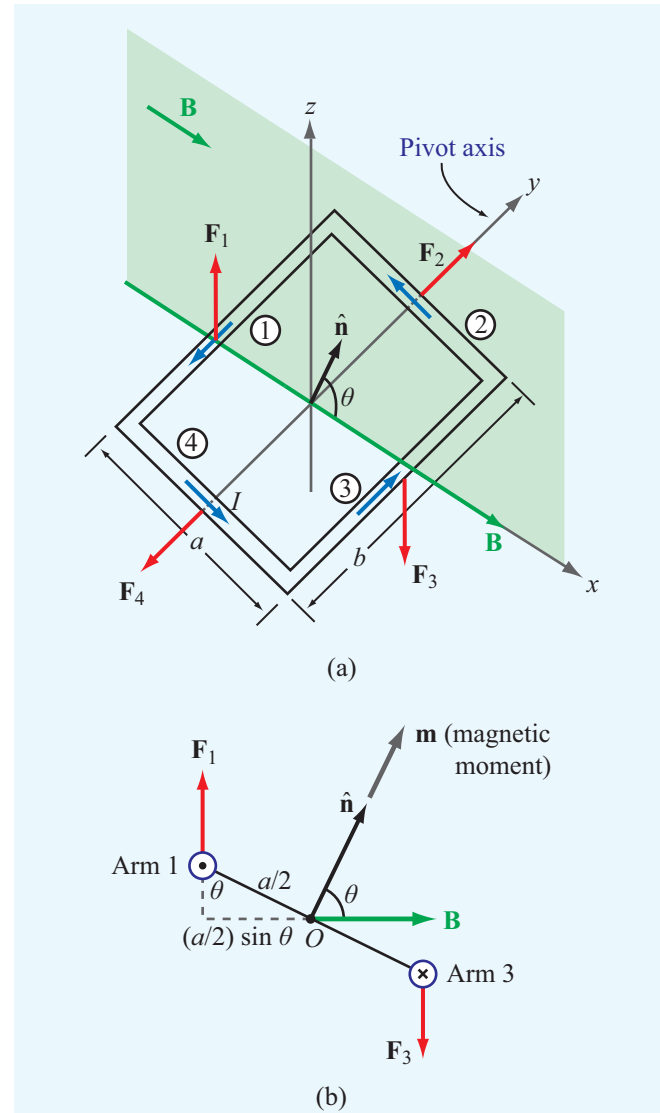
$$\begin{aligned} \mathbf{T} &= \mathbf{d}_1 \times \mathbf{F}_1 + \mathbf{d}_3 \times \mathbf{F}_3 \\ &= \left(-\hat{x} \frac{a}{2}\right) \times (\hat{z}IbB_0) + \left(\hat{x} \frac{a}{2}\right) \times (-\hat{z}IbB_0) \\ &= \hat{y}IabB_0 = \hat{y}IAB_0, \end{aligned} \quad (5.16)$$



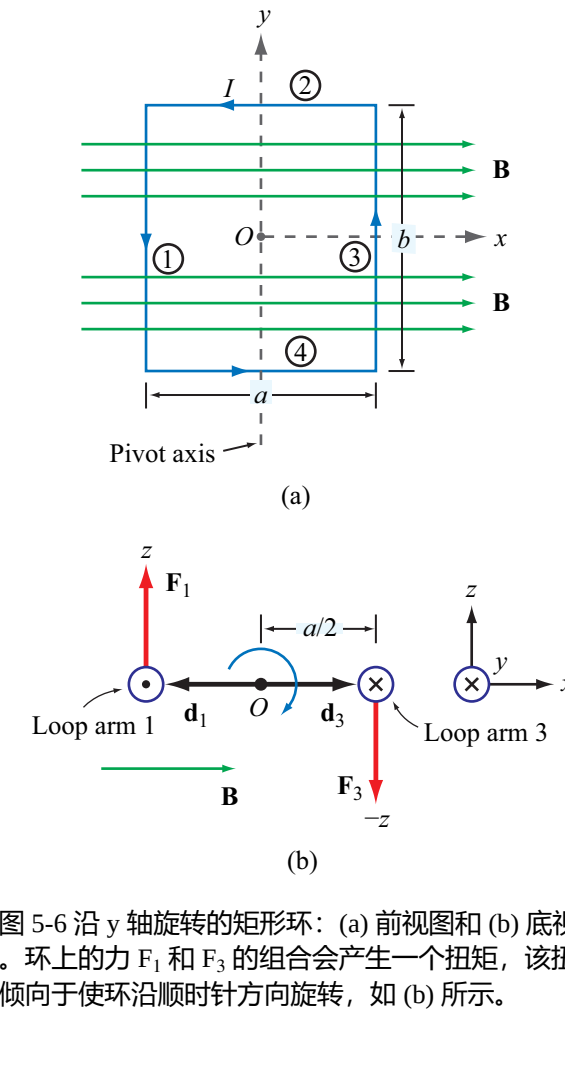
where $A = ab$ is the area of the loop. The right-hand rule tells us that the sense of rotation is clockwise. The result given by Eq. (5.16) is valid only when the magnetic field $\mathbf{B}$ is parallel to the plane of the loop. As soon as the loop starts to rotate, the torque $\mathbf{T}$ decreases, and at the end of one quarter of a complete rotation, the torque becomes zero, as discussed next.

Magnetic Field Perpendicular to the Axis of a Rectangular Loop

For the situation represented by Fig. 5-7, where $\mathbf{B} = \hat{x}B_0$, the field is still perpendicular to the loop's axis of rotation, but because its direction may be at any angle θ with respect to the loop's surface normal $\hat{n}$, we may now have nonzero forces on all four arms of the rectangular loop. However, forces $\mathbf{F}_2$



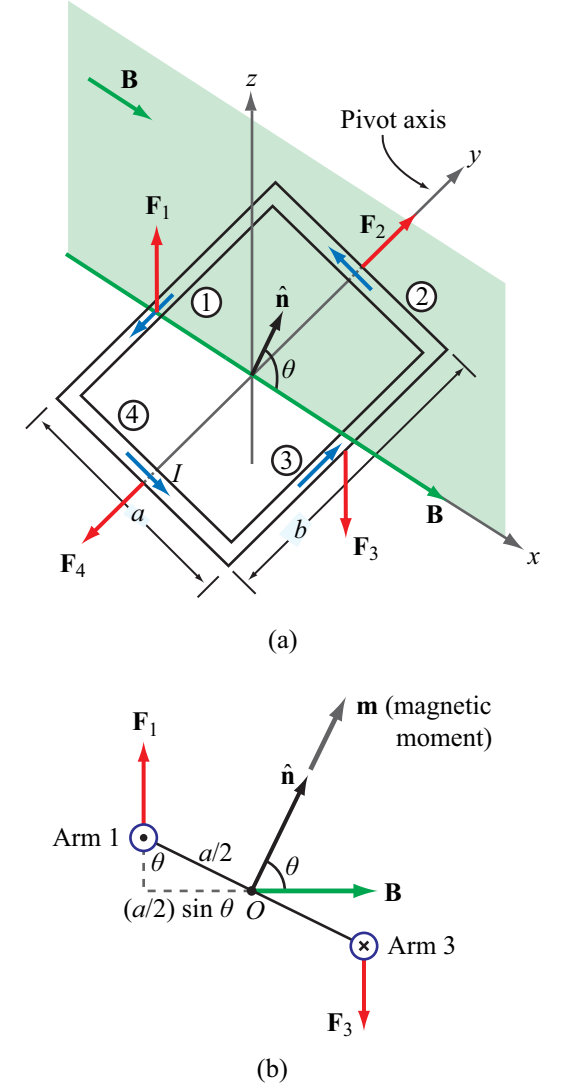
and $\mathbf{F}_4$ are equal in magnitude and opposite in direction and are along the rotation axis; hence, the net torque contributed by their combination is zero. The directions of the currents in arms 1 and 3 are always perpendicular to $\mathbf{B}$ regardless of the magnitude of θ . Hence, $\mathbf{F}_1$ and $\mathbf{F}_3$ have the same expressions given previously by Eqs. (5.15a,b), and for $0 \leq \theta \leq \pi/2$, their moment arms are of magnitude $(a/2)\sin\theta$, as illustrated in Fig. 5-7(b). Consequently, the magnitude of the net torque



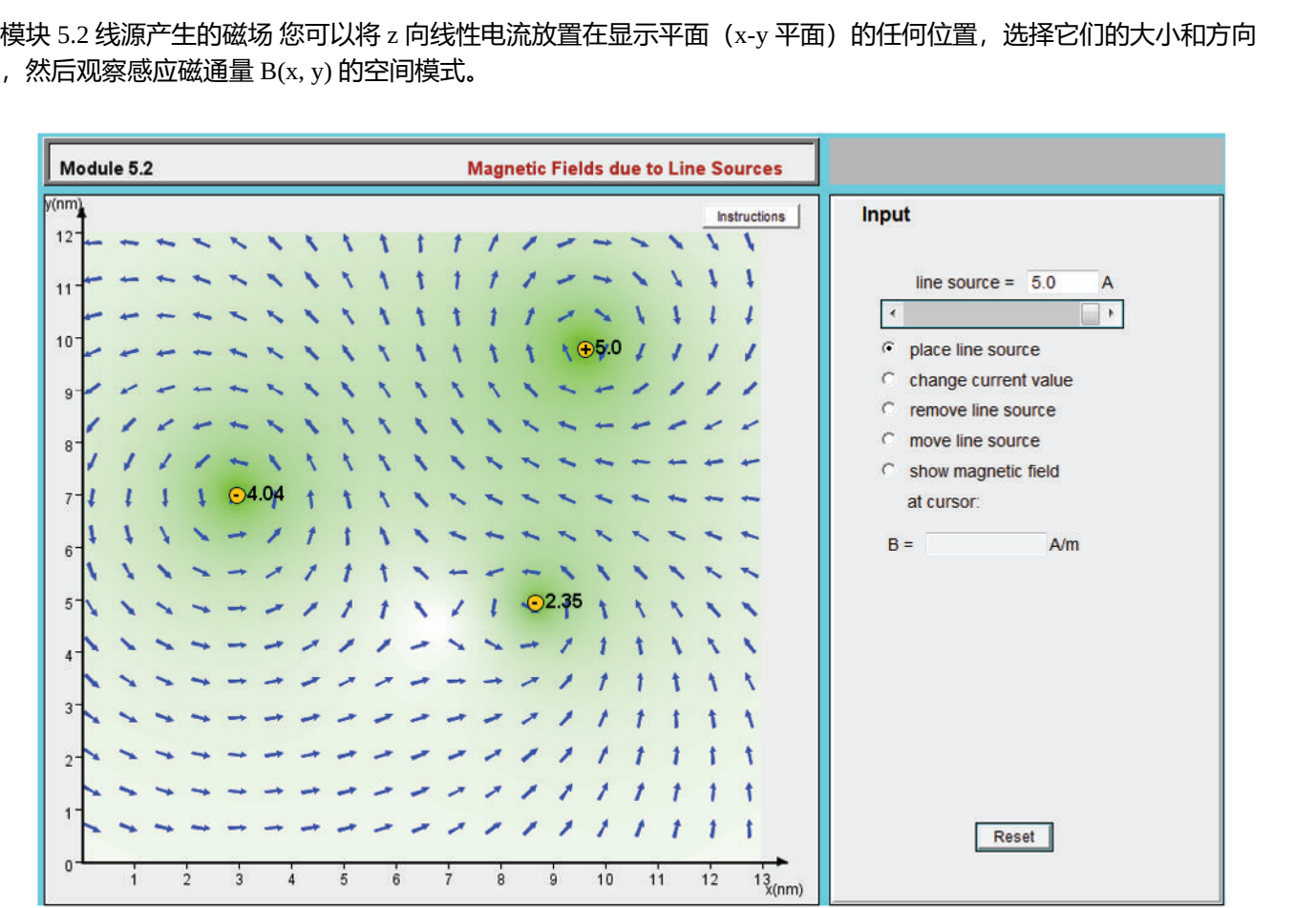
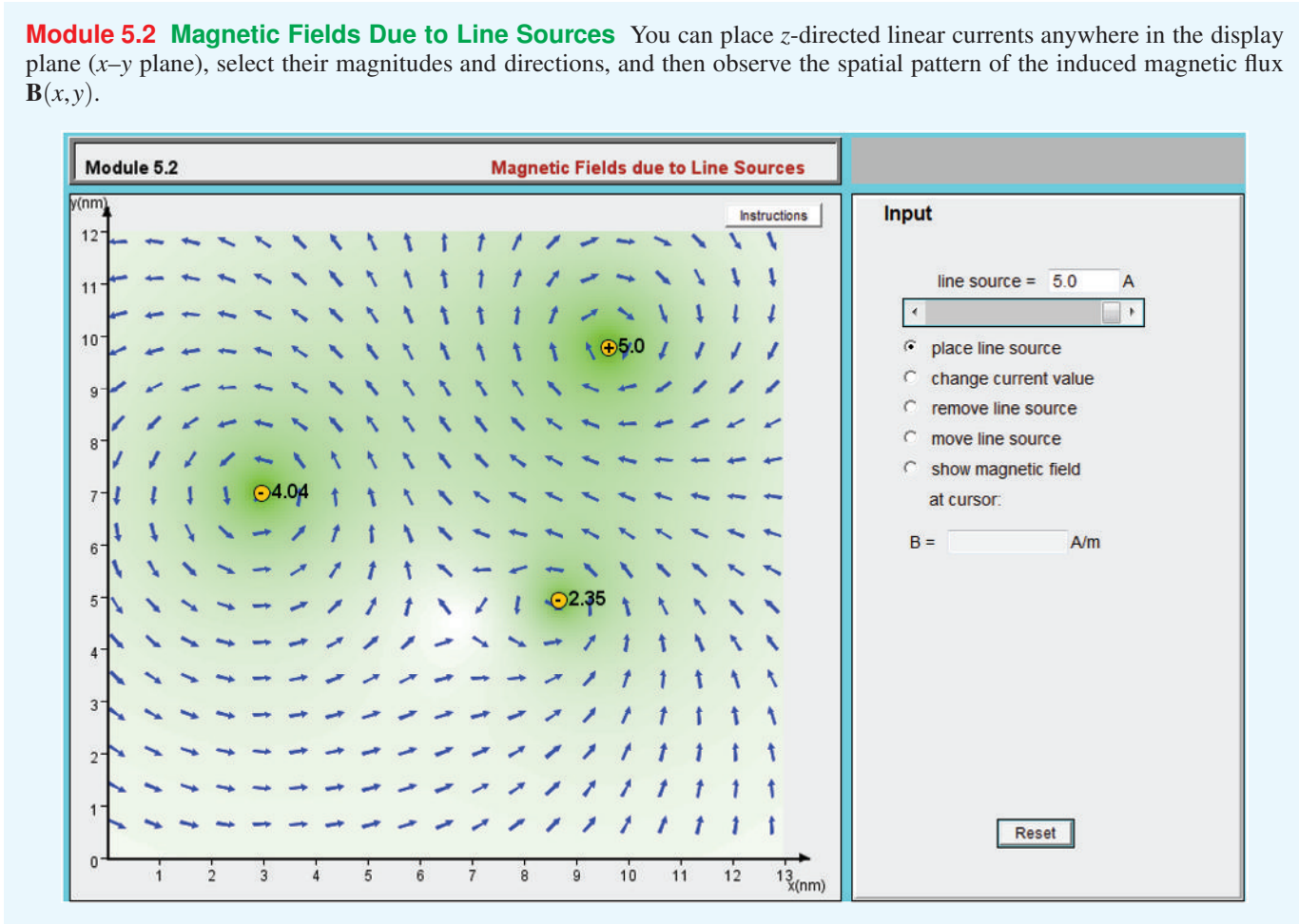
其中 $A = ab$ 是环的面积。右手定则告诉我们，旋转方向是顺时针方向。方程给出的结果。(5.16) 仅当磁场 $\mathbf{B}$ 平行于环路平面时才有效。一旦环路开始旋转，扭矩 T 就会减小，并且在四分之一完整旋转结束时，扭矩变为零，如下所述。

垂直于矩形环轴的磁场

对于图 5-7 所示的情况，其中 $\mathbf{B} = \hat{x}B_0$ ，场仍然垂直于环的旋转轴，但由于其方向可能相对于环的表面法线 $\hat{n}$ 成任何角度 θ ，因此我们现在可以在矩形环的所有四个臂上施加非零力。然而，力 $\mathbf{F}_2$



$\mathbf{F}_4$ 大小相等，方向相反，沿旋转轴；因此，它们的组合贡献的净扭矩为零。无论 θ 的大小如何，臂 1 和 3 中的电流方向始终垂直于 $\mathbf{B}$ 。因此， $\mathbf{F}_1$ 和 $\mathbf{F}_3$ 具有与之前方程 1 给出的相同表达式。(5.15a,b)，对于 $0 \leq \theta \leq \pi/2$ ，它们的力臂大小为 $(a/2)\sin\theta$ ，如图 5-7(b) 所示。因此，净扭矩的大小



exerted by the magnetic field about the axis of rotation is the same as that given by Eq. (5.16), but modified by $\sin \theta$:

$$T = IAB_0 \sin \theta. \quad (5.17)$$

According to Eq. (5.17), the torque is maximum when the magnetic field is parallel to the plane of the loop ($\theta = 90^\circ$) and zero when the field is perpendicular to the plane of the loop ($\theta = 0$). If the loop consists of N turns with each contributing a torque given by Eq. (5.17), then the total torque is

$$T = NIAB_0 \sin \theta. \quad (5.18)$$

The quantity NIA is called the **magnetic moment** m of the loop. Now, consider the vector

$$\mathbf{m} = \hat{\mathbf{n}} NIA = \hat{\mathbf{n}} m \quad (\text{A}\cdot\text{m}^2), \quad (5.19)$$

where $\hat{\mathbf{n}}$ is the surface normal of the loop and governed by the following **right-hand rule**: *When the four fingers of the right hand advance in the direction of the current I , the direction of the thumb specifies the direction of $\hat{\mathbf{n}}$.* In terms of $\mathbf{m}$, the torque vector $\mathbf{T}$ can be written as

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} \quad (\text{N}\cdot\text{m}). \quad (5.20)$$

Even though the derivation leading to Eq. (5.20) was obtained for $\mathbf{B}$ perpendicular to the axis of rotation of a rectangular loop, the expression is valid for any orientation of $\mathbf{B}$ and for a loop of any shape.

Concept Question 5-3: How is the direction of the magnetic moment of a loop defined?

围绕旋转轴施加的磁场与式 (1) 给出的相同。 (5.16) , 但由 $\sin \theta$ 修改:

$$T = IAB_0 \sin \theta. \quad (5.17)$$

根据方程。根据方程 (5.17) , 当磁场平行于环路平面 ($\theta = 90^\circ$) 时, 扭矩最大; 当磁场垂直于环路平面 ($\theta = 0$) 时, 扭矩为零。如果环路由 N 匝组成, 每匝都贡献由方程式给出的扭矩。 (5.17) 则总扭矩为

$$T = NIAB_0 \sin \theta. \quad (5.18)$$

量 NIA 称为环路的磁矩 m 。现在, 考虑向量

$$\mathbf{m} = \hat{\mathbf{n}} NIA = \hat{\mathbf{n}} m \quad (\text{A}\cdot\text{m}^2), \quad (5.19)$$

其中, $\hat{\mathbf{n}}$ 是环的表面法线, 并受以下右手定则约束: 当右手的四个手指沿电流 I 的方向前进时, 拇指的方向指定 $\hat{\mathbf{n}}$ 的方向。用 m 表示, 扭矩矢量 $\mathbf{T}$ 可以写为

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} \quad (\text{N}\cdot\text{m}). \quad (5.20)$$

即使推导出方程。 (5.20) 是针对垂直于矩形环的旋转轴的 $\mathbf{B}$ 获得的, 该表达式对于 $\mathbf{B}$ 的任何方向和任何形状的环都有效。

概念问题 5-3: 如何定义环路磁矩的方向?

Concept Question 5-4: If one of two wires of equal length is formed into a closed square loop and the other into a closed circular loop, and if both wires are carrying equal currents and both loops have their planes parallel to a uniform magnetic field, which loop would experience the greater torque?

Exercise 5-5: A square coil of 100 turns and 0.5-m long sides is in a region with a uniform magnetic flux density of 0.2 T. If the maximum magnetic torque exerted on the coil is 4×10^{-2} (N·m), what is the current flowing in the coil?

Answer: $I = 8$ mA. (See [EM](#).)

5-2 The Biot–Savart Law

In the preceding section, we elected to use the magnetic flux density **B** to denote the presence of a magnetic field in a given region of space. We now work with the magnetic field intensity **H** instead. We do this in part to remind the reader that for most materials the flux and field are linearly related by $\mathbf{B} = \mu \mathbf{H}$; therefore, knowledge of one implies knowledge of the other (assuming that μ is known).

Through his experiments on the deflection of compass needles by current-carrying wires, Hans Oersted established that currents induce magnetic fields that form closed loops around the wires (see Section 1-3.3). Building upon Oersted’s results, Jean Biot and Félix Savart arrived at an expression that relates the magnetic field **H** at any point in space to the current *I* that generates **H**. The **Biot–Savart law** states that the differential magnetic field *dH* generated by a steady current *I* flowing through a differential length vector *dI* is

$$d\mathbf{H} = \frac{I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} \quad (\text{A/m}), \quad (5.21)$$

where $\mathbf{R} = \hat{\mathbf{R}}R$ is the distance vector between *dI* and the observation point *P*, as shown in [Fig. 5-8](#). The SI unit for **H** is ampere·m/m² = (A/m). It is important to remember that Eq. (5.21) assumes that *dI* is along the direction of the current *I* and the unit vector $\hat{\mathbf{R}}$ points *from* the current element *to* the observation point.

According to Eq. (5.21), *dH* varies as R^{-2} , which is similar to the distance dependence of the electric field induced by an electric charge. However, unlike the electric field vector **E**, whose direction is *along* the distance vector **R** joining the charge to the observation point, the magnetic field **H** is *orthogonal* to the plane containing the direction of the current element *dI* and the distance vector **R**. At point *P* in [Fig. 5-8](#),

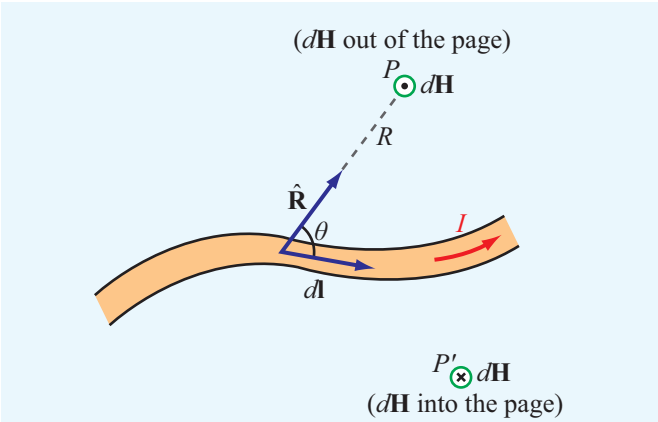


Figure 5-8 Magnetic field *dH* generated by a current element *I dI*. The direction of the field induced at point *P* is opposite to that induced at point *P'*.

the direction of *dH* is out of the page, whereas at point *P'* the direction of *dH* is into the page.

To determine the total magnetic field **H** due to a conductor of finite size, we need to sum up the contributions due to all the current elements making up the conductor. Hence, the Biot–Savart law becomes

$$\mathbf{H} = \frac{I}{4\pi} \int_l \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} \quad (\text{A/m}), \quad (5.22)$$

where *l* is the line path along which *I* exists.

5-2.1 Magnetic Field Due to Surface and Volume Current Distributions

The Biot–Savart law also may be expressed in terms of distributed current sources ([Fig. 5-9](#)) such as the **volume current density J**, measured in (A/m²), or the **surface current density J_s**, measured in (A/m). The surface current density **J_s** applies to currents that flow on the surface of a conductor in the form of a sheet of effectively zero thickness. When current sources are specified in terms of **J_s** over a surface *S* or in terms of **J** over a volume *V*, we can use the equivalence given by

$$I d\mathbf{l} \quad \longleftrightarrow \quad \mathbf{J}_s ds \quad \longleftrightarrow \quad \mathbf{J} d\mathbf{v} \quad (5.23)$$

概念题 5-4：将两根等长导线分别弯成闭合正方形回路和闭合圆形回路。若两回路中电流相等，且回路平面均平行于匀强磁场，哪个回路受到的力矩更大？

练习 5-5：一个 100 匝、边长为 0.5 m 的正方形线圈置于磁通密度为 0.2 T 的匀强磁场中。若线圈所受最大磁力矩为 4×10^{-2} N·m，求线圈中的电流。

答案：I = 8 mA。（参见配套 EM 资源。）

5-2 毕奥–萨伐尔定律

上一节用磁通密度 **B** 描述空间中的磁场。现在改用磁场强度 **H**，以提醒读者：对于大多数材料，二者具有线性关系 $\mathbf{B} = \mu \mathbf{H}$ 。因此，在磁导率 μ 已知的条件下，知道其中一个量便可确定另一个量。

奥斯特通过载流导线使磁针偏转的实验，发现电流产生的磁场围绕导线形成闭合磁力线（见第 1-3.3 节）。在此基础上，毕奥和萨伐尔得出了空间任意点的磁场 **H** 与产生该磁场的电流 *I* 之间的关系。毕奥–萨伐尔定律指出：稳恒电流 *I* 流经微小有向线元 *dI* 时，产生的磁场微元 *dH* 为

$$d\mathbf{H} = \frac{I}{4\pi} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} \quad (\text{A/m}), \quad (5.21)$$

其中，矢量 **R** 从线元 *dI* 指向观察点 *P*，如图 5-8 所示；其长度为 *R*。**H** 的 SI 单位为 A/m。式 (5.21) 中，*dI* 沿电流 *I* 的方向，**R** 上加帽号表示从电流元指向观察点的单位矢量。

由式 (5.21) 可知，*dH* 与 R^2 成反比，其距离关系与点电荷产生的电场相似。但电场 **E** 沿电荷指向观察点的位移矢量 **R**，而电流元产生的磁场 *dH* 垂直于 *dI* 与 **R** 所在的平面。在图 5-8 中的 *P* 点，

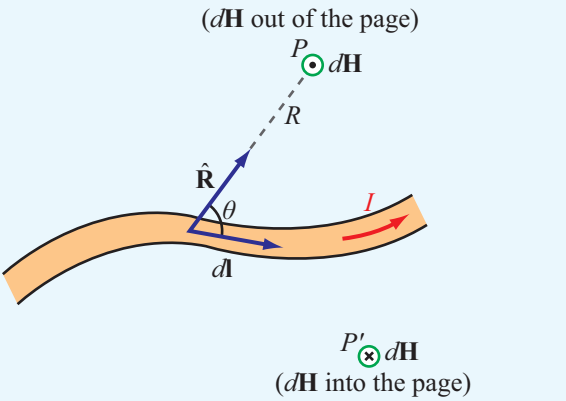


图 5-8 电流元 *I dI* 产生的磁场 *dH*。*P* 点与 *P'* 点的磁场方向相反。图中 out of the page 表示垂直纸面向外，into the page 表示垂直纸面向内。

dH 垂直纸面向外；在 *P'* 点，*dH* 垂直纸面向内。

为求有限尺寸导体产生的总磁场 **H**，需要将组成导体的所有电流元产生的磁场叠加。因此，毕奥–萨伐尔定律可写为

$$\mathbf{H} = \frac{I}{4\pi} \int_l \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} \quad (\text{A/m}), \quad (5.22)$$

其中，*l* 为电流 *I* 所沿的积分路径。

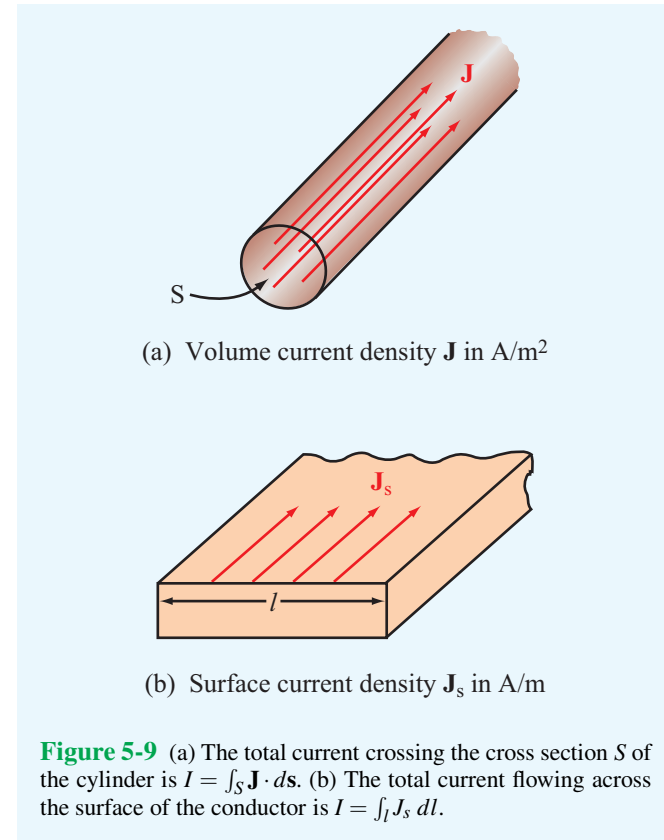
5-2.1 面电流和体电流分布产生的磁场

毕奥–萨伐尔定律也可用分布电流源表示（图 5-9），例如体电流密度 **J**（单位 A/m²）或面电流密度 **J_s**（单位 A/m）。**J** 用于描述沿导体表面流动、可视为厚度为零的薄层电流。当电流源由曲面 *S* 上的 **J_s** 或体积 *v* 内的 **J** 给定时，可采用以下等效关系：

$$I d\mathbf{l} \quad \longleftrightarrow \quad \mathbf{J}_s ds \quad \longleftrightarrow \quad \mathbf{J} d\mathbf{v} \quad (5.23)$$

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to express the Biot-Savart law as

$$\mathbf{H} = \frac{1}{4\pi} \int_S \frac{\mathbf{J}_s \times \hat{\mathbf{R}}}{R^2} ds, \quad (5.24a)$$

(surface current)

$$\mathbf{H} = \frac{1}{4\pi} \int_v \frac{\mathbf{J} \times \hat{\mathbf{R}}}{R^2} dv. \quad (5.24b)$$

(volume current)

Example 5-2: Magnetic Field of a Linear Conductor

A free-standing linear conductor of length l carries a current I along the z axis as shown in Fig. 5-10. Determine the magnetic flux density $\mathbf{B}$ at a point P located at a distance r in the x - y plane. The wire is, of course, part of a closed-loop circuit, but our current interest is only in the conducting wire.

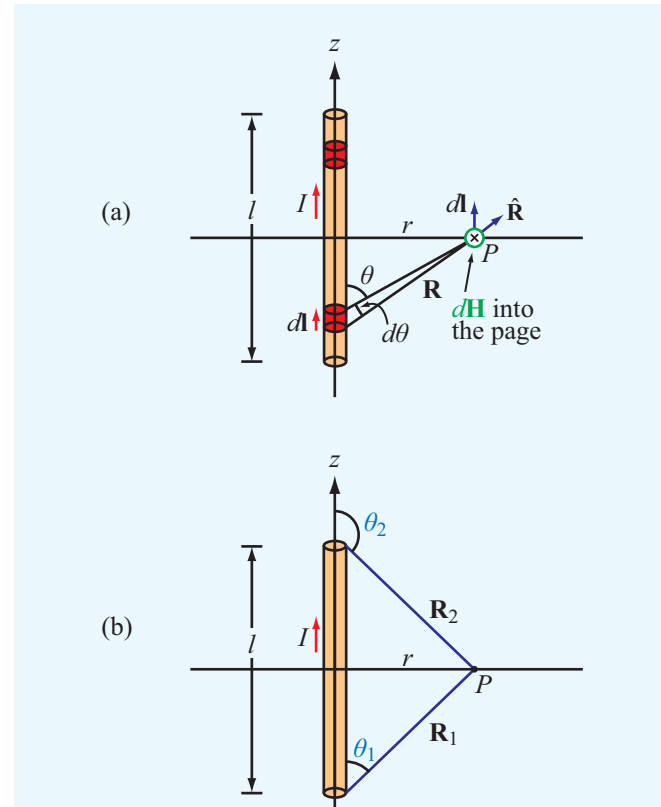


Figure 5-10 Linear conductor of length l carrying a current I . (a) The field $d\mathbf{H}$ at point P due to incremental current element $d\mathbf{l}$. (b) Limiting angles θ_1 and θ_2 , each measured between vector $I d\mathbf{l}$ and the vector connecting the end of the conductor associated with that angle to point P (Example 5-2).

Solution: From Fig. 5-10, the direction of I is along $+\hat{\mathbf{z}}$. Hence, the differential length vector is $d\mathbf{l} = \hat{\mathbf{z}} dz$, and $d\mathbf{l} \times \hat{\mathbf{R}} = dz (\hat{\mathbf{z}} \times \hat{\mathbf{R}}) = \hat{\phi} \sin\theta dz$, where $\hat{\phi}$ is the azimuth direction and θ is the angle between $d\mathbf{l}$ and $\hat{\mathbf{R}}$. Application of Eq. (5.22) gives

$$\mathbf{H} = \frac{I}{4\pi} \int_{z=-l/2}^{z=l/2} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} = \hat{\phi} \frac{I}{4\pi} \int_{-l/2}^{l/2} \frac{\sin\theta}{R^2} dz. \quad (5.25)$$

Both R and θ are dependent on the integration variable z , but the radial distance r is not. For convenience, we convert the integration variable from z to θ by using the transformations

$$R = r \csc \theta, \quad (5.26a)$$

$$z = -r \cot \theta, \quad (5.26b)$$

$$dz = r \csc^2 \theta d\theta. \quad (5.26c)$$

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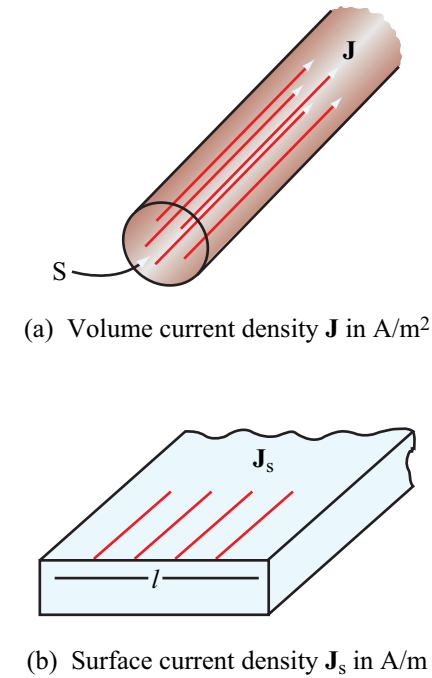


图5-9 (a)穿过圆柱体横截面S的总电流为 $I = \int_S \mathbf{J} \cdot d\mathbf{s}$ 。(b)流过导体表面的总电流为 $I = \int_l J_s dl$ 。

将毕奥-萨伐尔定律表示为

$$\mathbf{H} = \frac{1}{4\pi} \int_S \frac{\mathbf{J}_s \times \hat{\mathbf{R}}}{R^2} ds, \quad (5.24a)$$

(表面电流)

$$\mathbf{H} = \frac{1}{4\pi} \int_v \frac{\mathbf{J} \times \hat{\mathbf{R}}}{R^2} dv. \quad (5.24b)$$

(音量电流)

示例 5-2: 线性导体的磁场

长度为 l 的独立线性导体沿 z 轴承载电流 I , 如图 5-10 所示。确定 x - y 平面中距离 r 处的点 P 处的磁通密度 $\mathbf{B}$ 。当然, 导线是闭环电路的一部分, 但我们当前的兴趣仅在于导线。

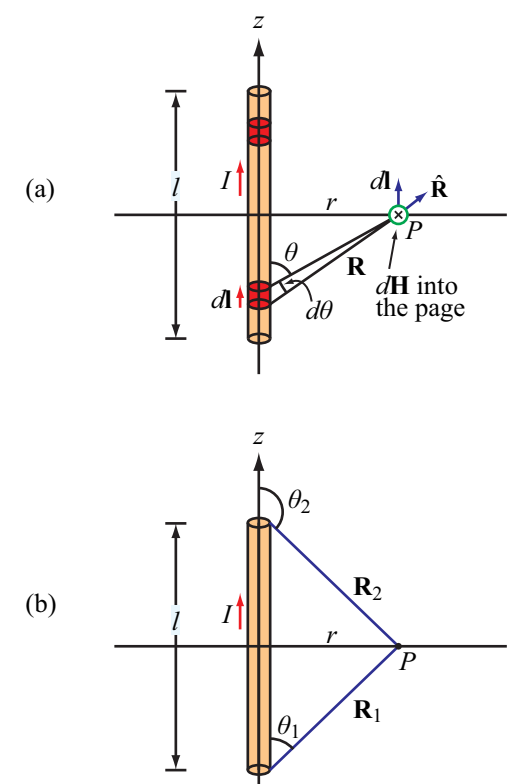


图 5-10 长度为 l 的线性导体, 承载电流 I 。(a) 由于增量电流元件 $d\mathbf{l}$, P 点处的磁场 $d\mathbf{H}$ 。(b) 限制角度 θ_1 和 θ_2 , 每个角度在矢量 $I d\mathbf{l}$ 和连接与该角度相关的导体末端到点 P 的矢量之间测量 (示例 5-2)。

解: 由图5-10可知, I 的方向是沿 $+\hat{\mathbf{z}}$ 。因此, 差分长度向量为 $d\mathbf{l} = \hat{\mathbf{z}} dz$, $d\mathbf{l} \times \hat{\mathbf{R}} = dz (\hat{\mathbf{z}} \times \hat{\mathbf{R}}) = \hat{\phi} \sin\theta dz$, 其中 $\hat{\phi}$ 是方位角方向, θ 是 $d\mathbf{l}$ 和 $\hat{\mathbf{R}}$ 之间的角度。方程的应用(5.22) 给出

$$\mathbf{H} = \frac{I}{4\pi} \int_{z=-l/2}^{z=l/2} \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} = \hat{\phi} \frac{I}{4\pi} \int_{-l/2}^{l/2} \frac{\sin\theta}{R^2} dz. \quad (5.25)$$

R 和 θ 都取决于积分变量 z , 但径向距离 r 则不然。为了方便起见, 我们使用转换将积分变量从 z 转换为 θ

$$R = r \csc \theta, \quad (5.26a)$$

$$z = -r \cot \theta, \quad (5.26b)$$

$$dz = r \csc^2 \theta d\theta. \quad (5.26c)$$

Upon inserting Eqs. (5.26a) and (5.26c) into Eq. (5.25), we have

$$\begin{aligned}\mathbf{H} &= \hat{\phi} \frac{I}{4\pi} \int_{\theta_1}^{\theta_2} \frac{\sin \theta \, r \csc^2 \theta \, d\theta}{r^2 \csc^2 \theta} \\ &= \hat{\phi} \frac{I}{4\pi r} \int_{\theta_1}^{\theta_2} \sin \theta \, d\theta = \hat{\phi} \frac{I}{4\pi r} (\cos \theta_1 - \cos \theta_2),\end{aligned}\quad (5.27)$$

where θ_1 and θ_2 are the limiting angles at $z = -l/2$ and $z = l/2$, respectively. From the right triangle in **Fig. 5-10(b)**, it follows that

$$\cos \theta_1 = \frac{l/2}{\sqrt{r^2 + (l/2)^2}},\quad (5.28a)$$

$$\cos \theta_2 = -\cos \theta_1 = \frac{-l/2}{\sqrt{r^2 + (l/2)^2}}.\quad (5.28b)$$

Hence,

$$\mathbf{B} = \mu_0 \mathbf{H} = \hat{\phi} \frac{\mu_0 I l}{2\pi r \sqrt{4r^2 + l^2}} \quad (\text{T}).\quad (5.29)$$

For an infinitely long wire with $l \gg r$, Eq. (5.29) reduces to

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}. \quad (\text{infinitely long wire})\quad (5.30)$$

► This is a very important and useful expression to keep in mind. It states that in the neighborhood of a linear conductor carrying a current I , the induced magnetic field forms concentric circles around the wire (**Fig. 5-11**), and its intensity is directly proportional to I and inversely proportional to the distance r . ◀

Example 5-3: Magnetic Field of a Circular Loop

A circular loop of radius a carries a steady current I . Determine the magnetic field $\mathbf{H}$ at a point on the axis of the loop.

Solution: Let us place the loop in the x - y plane (**Fig. 5-12**). Our task is to obtain an expression for $\mathbf{H}$ at point $P(0, 0, z)$.

We start by noting that any element $d\mathbf{l}$ on the circular loop is perpendicular to the distance vector $\mathbf{R}$ and that all elements around the loop are at the same distance R from P with $R = \sqrt{a^2 + z^2}$. From Eq. (5.21), the magnitude of $d\mathbf{H}$ due to current element $d\mathbf{l}$ is

$$dH = \frac{I}{4\pi R^2} |d\mathbf{l} \times \hat{\mathbf{R}}| = \frac{I dl}{4\pi(a^2 + z^2)},\quad (5.31)$$

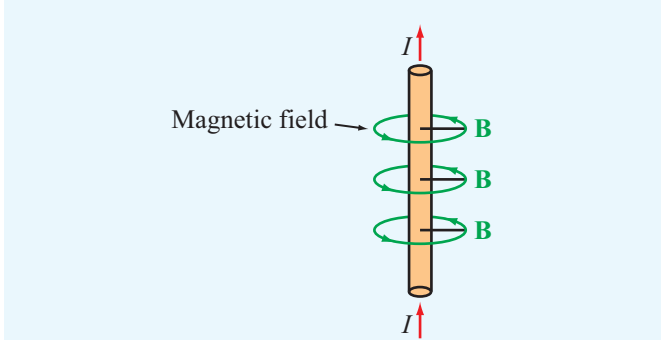


Figure 5-11 Magnetic field surrounding a long, linear current-carrying conductor.

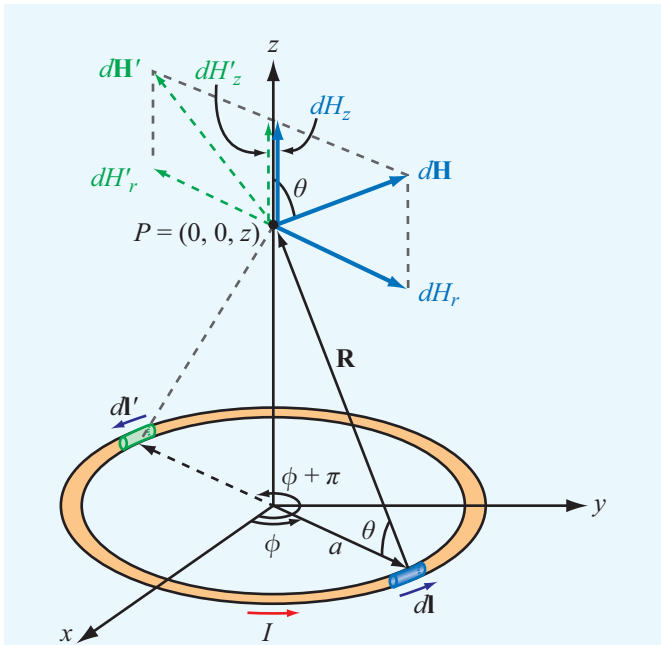


Figure 5-12 Circular loop carrying a current I (Example 5-3).

and the direction of $d\mathbf{H}$ is perpendicular to the plane containing $\mathbf{R}$ and $d\mathbf{l}$. $d\mathbf{H}$ is in the r - z plane (**Fig. 5-12**); therefore, it has components dH_r and dH_z . If we consider element $d\mathbf{l}'$ located diametrically opposite to $d\mathbf{l}$, we observe that the z components of the magnetic fields due to $d\mathbf{l}$ and $d\mathbf{l}'$ add because they are in the same direction, but their r components cancel because they are in opposite directions. Hence, the non-vanishing

插入方程式后。 (5.26a) 和 (5.26c) 代入方程。 (5.25) , 我们有

$$\begin{aligned}\mathbf{H} &= \hat{\phi} \frac{I}{4\pi} \int_{\theta_1}^{\theta_2} \frac{\sin \theta \, r \csc^2 \theta \, d\theta}{r^2 \csc^2 \theta} \\ &= \hat{\phi} \frac{I}{4\pi r} \int_{\theta_1}^{\theta_2} \sin \theta \, d\theta = \hat{\phi} \frac{I}{4\pi r} (\cos \theta_1 - \cos \theta_2),\end{aligned}\quad (5.27)$$

其中 θ_1 和 θ_2 分别是 $z = -l/2$ 和 $z = l/2$ 处的极限角。由图5-10(b)中的直角三角形可知

$$\cos \theta_1 = \frac{l/2}{\sqrt{r^2 + (l/2)^2}},\quad (5.28a)$$

$$\cos \theta_2 = -\cos \theta_1 = \frac{-l/2}{\sqrt{r^2 + (l/2)^2}}.\quad (5.28b)$$

因此,

$$\mathbf{B} = \mu_0 \mathbf{H} = \hat{\phi} \frac{\mu_0 I l}{2\pi r \sqrt{4r^2 + l^2}} \quad (\text{T}).\quad (5.29)$$

对于具有 $l \gg r$ 的无限长导线, 方程: (5.29) 减少为

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}. \quad (\text{infinitely long wire})\quad (5.30)$$

► 这是一个非常重要且有用的表达方式, 需要牢记在心。它指出, 在承载电流 I 的线性导体附近, 感应磁场在导线周围形成同心圆 (图 5-11), 其强度与 I 成正比, 与距离 r 成反比。◀

例 5-3: 圆环的磁场

半径为 a 的圆环流过稳定电流 I 。确定环轴上一点的磁场 $\mathbf{H}$ 。

解决方案: 让我们将环放置在 x - y 平面上 (图 5-12)。我们的任务是获得点 $P(0, 0, z)$ 处的 $\mathbf{H}$ 表达式。

我们首先注意到圆形环上的任何元素 $d\mathbf{l}$ 都垂直于距离向量 $\mathbf{R}$, 并且环周围的所有元素与 P 的距离 R 相同, 其中 $R = \sqrt{a^2 + z^2}$ 。从方程。 (5.21) 电流元素 $d\mathbf{l}$ 引起的 $d\mathbf{H}$ 的大小为

$$dH = \frac{I}{4\pi R^2} |d\mathbf{l} \times \hat{\mathbf{R}}| = \frac{I dl}{4\pi(a^2 + z^2)},\quad (5.31)$$

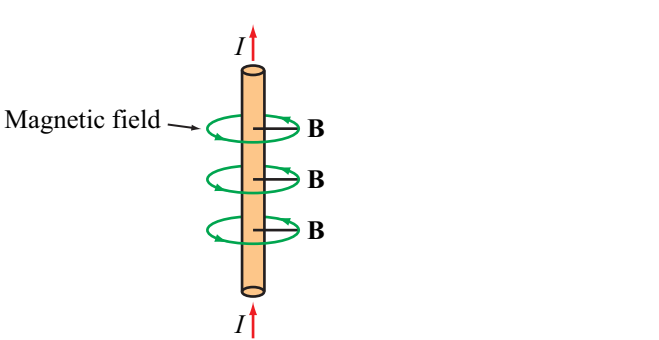


图 5-11 长线性载流导体周围的磁场。

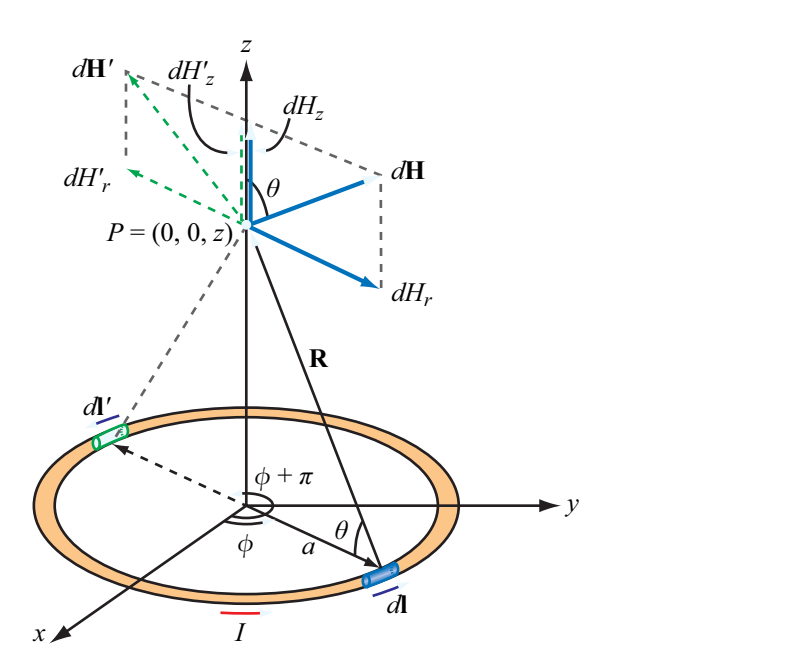
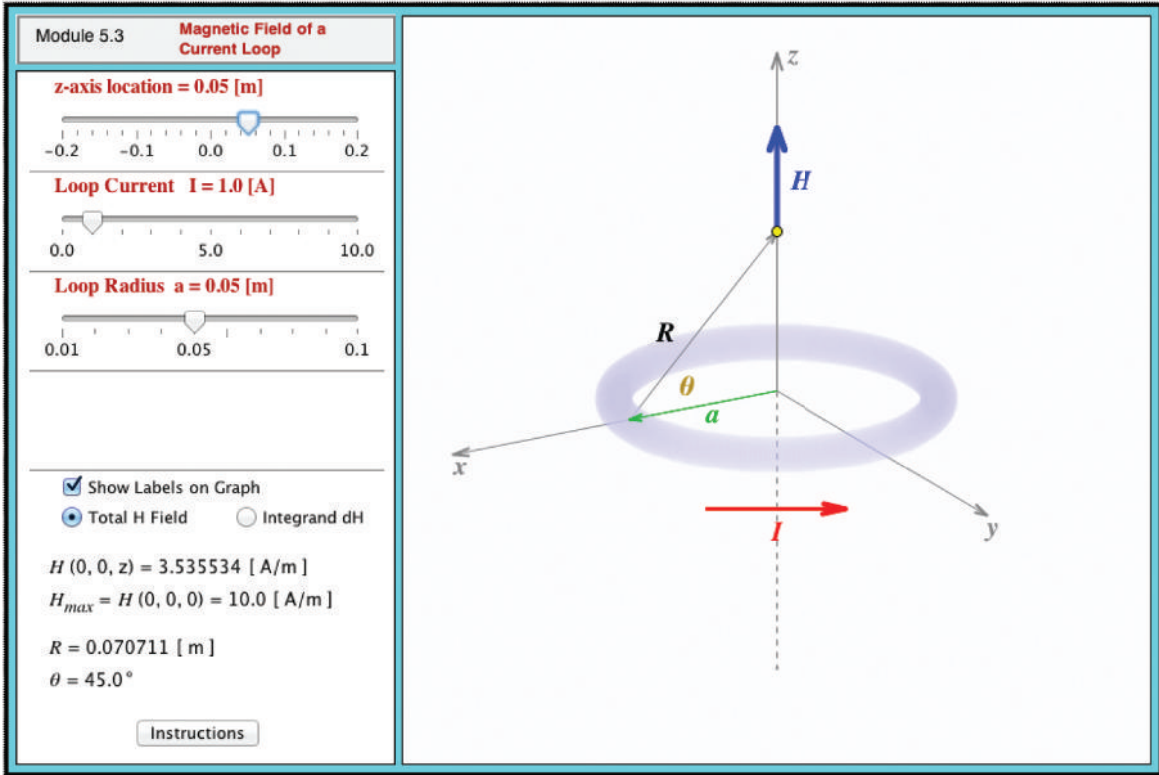
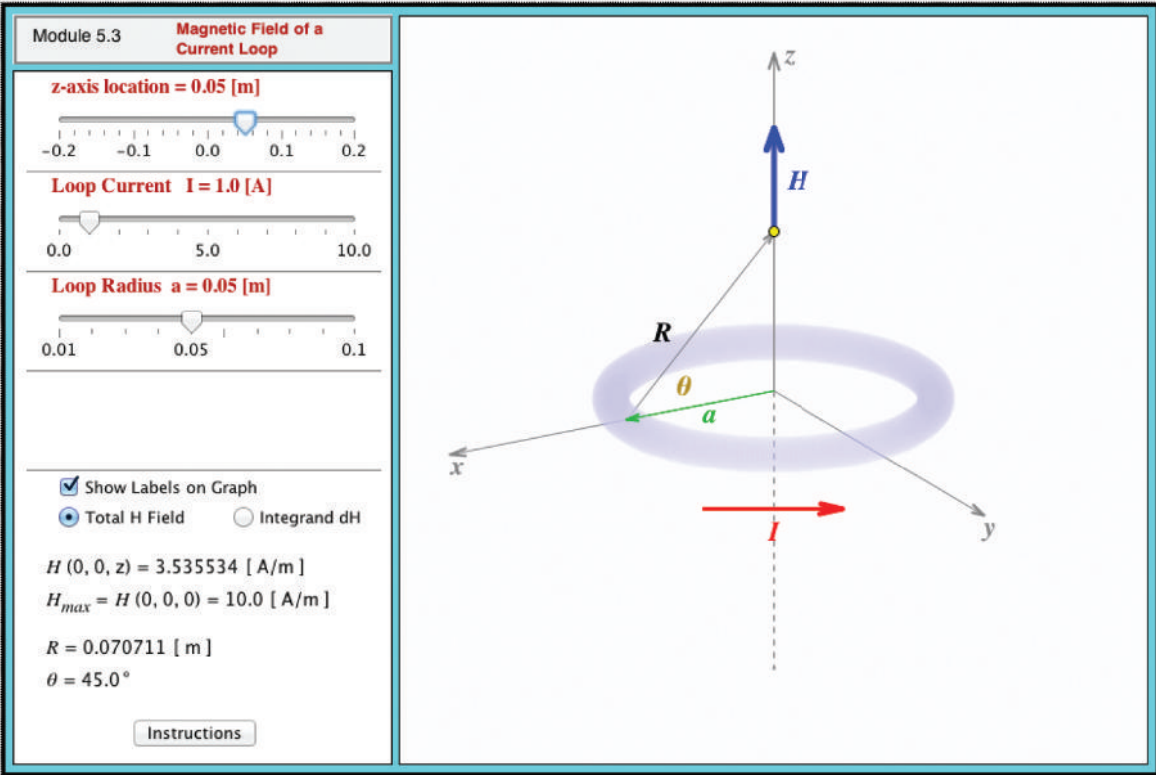


图 5-12 承载电流 I 的环形回路 (示例 5-3)。

$d\mathbf{H}$ 的方向垂直于包含 $\mathbf{R}$ 和 $d\mathbf{l}$ 的平面。 $d\mathbf{H}$ 位于 r - z 平面 (图 5-12); 因此, 它具有分量 dH_r 和 dH_z 。如果我们考虑元素 $d\mathbf{l}'$ 与 $d\mathbf{l}$ 完全相反, 我们会观察到 $d\mathbf{l}$ 和 $d\mathbf{l}'$ 产生的磁场 z 分量相加, 因为它们方向相同, 但它们的 r 分量抵消, 因为它们方向相反。因此, 不消失的

Module 5.3 Magnetic Field of a Current Loop Examine how the field along the loop axis changes with loop parameters.



component of the magnetic field is along z only. That is,

$$\begin{aligned} d\mathbf{H} &= \hat{\mathbf{z}} dH_z = \hat{\mathbf{z}} dH \cos \theta \\ &= \hat{\mathbf{z}} \frac{I \cos \theta}{4\pi(a^2 + z^2)} dl. \end{aligned} \quad (5.32)$$

For a fixed point $P(0, 0, z)$ on the axis of the loop, all quantities in Eq. (5.32) are constant, except for dl . Hence, integrating Eq. (5.32) over a circle of radius a gives

$$\begin{aligned} \mathbf{H} &= \hat{\mathbf{z}} \frac{I \cos \theta}{4\pi(a^2 + z^2)} \oint dl \\ &= \hat{\mathbf{z}} \frac{I \cos \theta}{4\pi(a^2 + z^2)} (2\pi a). \end{aligned} \quad (5.33)$$

Upon using the relation $\cos \theta = a/(a^2 + z^2)^{1/2}$, we obtain

$$\mathbf{H} = \hat{\mathbf{z}} \frac{Ia^2}{2(a^2 + z^2)^{3/2}} \quad (\text{A/m}). \quad (5.34)$$

At the center of the loop ($z = 0$), Eq. (5.34) reduces to

$$\mathbf{H} = \hat{\mathbf{z}} \frac{I}{2a} \quad (\text{at } z = 0), \quad (5.35)$$

and at points very far away from the loop such that $z^2 \gg a^2$, Eq. (5.34) simplifies to

$$\mathbf{H} = \hat{\mathbf{z}} \frac{Ia^2}{2|z|^3} \quad (\text{at } |z| \gg a). \quad (5.36)$$

5-2.2 Magnetic Field of a Magnetic Dipole

In view of the definition given by Eq. (5.19) for the magnetic moment $\mathbf{m}$ of a current loop, a single-turn loop situated in the x - y plane (Fig. 5-12) has magnetic moment $\mathbf{m} = \hat{\mathbf{z}}m$ with $m = I\pi a^2$. Consequently, Eq. (5.36) may be expressed as

$$\mathbf{H} = \hat{\mathbf{z}} \frac{m}{2\pi|z|^3} \quad (\text{at } |z| \gg a). \quad (5.37)$$

This expression applies to a point P far away from the loop and on its axis. Had we solved for $\mathbf{H}$ at any distant point

磁场的分量仅沿 z 方向。那是,

$$\begin{aligned} d\mathbf{H} &= \hat{\mathbf{z}} dH_z = \hat{\mathbf{z}} dH \cos \theta \\ &= \hat{\mathbf{z}} \frac{I \cos \theta}{4\pi(a^2 + z^2)} dl. \end{aligned} \quad (5.32)$$

对于循环轴上的固定点 $P(0, 0, z)$, 方程式中的所有量。(5.32) 是常数, 除了 dl 之外。因此, 整合方程。(5.32) 在半径为 a 的圆上给出

$$\begin{aligned} \mathbf{H} &= \hat{\mathbf{z}} \frac{I \cos \theta}{4\pi(a^2 + z^2)} \oint dl \\ &= \hat{\mathbf{z}} \frac{I \cos \theta}{4\pi(a^2 + z^2)} (2\pi a). \end{aligned} \quad (5.33)$$

使用关系 $\cos \theta = a/(a^2 + z^2)^{1/2}$, 我们得到

$$\mathbf{H} = \hat{\mathbf{z}} \frac{Ia^2}{2(a^2 + z^2)^{3/2}} \quad (\text{A/m}). \quad (5.34)$$

在循环的中心 ($z = 0$), 方程: (5.34) 减少为

$$\mathbf{H} = \hat{\mathbf{z}} \frac{I}{2a} \quad (\text{at } z = 0), \quad (5.35)$$

并且在距离循环很远的点处, 使得 $z^2 \gg a^2$, 方程: (5.34) 简化为

$$\mathbf{H} = \hat{\mathbf{z}} \frac{Ia^2}{2|z|^3} \quad (\text{at } |z| \gg a). \quad (5.36)$$

5-2.2 磁偶极子的磁场

鉴于方程给出的定义。(5.19) 对于电流环路的磁矩 $\mathbf{m}$, 位于 x - y 平面 (图 5-12) 的单匝环路的磁矩为 $\mathbf{m} = \hat{\mathbf{z}}m$, 其中 $m = I\pi a^2$ 。因此, 等式。(5.36) 可表示为

$$\mathbf{H} = \hat{\mathbf{z}} \frac{m}{2\pi|z|^3} \quad (\text{at } |z| \gg a). \quad (5.37)$$

该表达式适用于远离环且位于其轴上的点 P 。我们是否可以求解任意远点的 $\mathbf{H}$

$P = (R, \theta, \phi)$ in a spherical coordinate system with R the distance between the center of the loop and point P , we would have obtained the expression

$$\mathbf{H} = \frac{m}{4\pi R^3} (\hat{\mathbf{R}} 2 \cos \theta + \hat{\boldsymbol{\theta}} \sin \theta) \quad (\text{for } R \gg a). \quad (5.38)$$

► A current loop with dimensions much smaller than the distance between the loop and the observation point is called a **magnetic dipole**. This is because the pattern of its magnetic field lines is similar to that of a permanent magnet as well as to the pattern of the electric field lines of the electric dipole (Fig. 5-13). ◀

Concept Question 5-5: Two infinitely long parallel wires carry currents of equal magnitude. What is the resultant magnetic field due to the two wires at a point midway between the wires—compared with the magnetic field due to one of them alone—if the currents are (a) in the same direction and (b) in opposite directions?

Concept Question 5-6: Devise a right-hand rule for the direction of the magnetic field due to a linear current-carrying conductor.

Concept Question 5-7: What is a magnetic dipole? Describe its magnetic field distribution.

Exercise 5-6: A semi-infinite linear conductor extends between $z = 0$ and $z = \infty$ along the z axis. If the current I in the conductor flows along the positive z direction, find $\mathbf{H}$ at a point in the x - y plane at a radial distance r from the conductor.

Answer: $\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{I}{4\pi r}$ (A/m). (See EM.)

Exercise 5-7: A wire carrying a current of 4 A is formed into a circular loop. If the magnetic field at the center of the loop is 20 A/m, what is the radius of the loop if the loop has (a) only one turn and (b) 10 turns?

Answer: (a) $a = 10$ cm, (b) $a = 1$ m. (See EM.)

Exercise 5-8: A wire is formed into a square loop and placed in the x - y plane with its center at the origin and each of its sides parallel to either the x or y axes. Each side is 40 cm in length, and the wire carries a current of 5 A whose direction is clockwise when the loop is viewed from above. Calculate the magnetic field at the center of the loop.

Answer: $\mathbf{H} = -\hat{\mathbf{z}} \frac{4I}{\sqrt{2}\pi l} = -\hat{\mathbf{z}} 11.25$ A/m. (See EM.)

$P = (R, \theta, \phi)$ 在球坐标系中, R 为环中心与点 P 之间的距离, 我们可以得到表达式

$$\mathbf{H} = \frac{m}{4\pi R^3} (\hat{\mathbf{R}} 2 \cos \theta + \hat{\boldsymbol{\theta}} \sin \theta) \quad (\text{for } R \gg a). \quad (5.38)$$

► 尺寸远小于环路与观察点之间的距离的电流环路称为磁偶极子。这是因为它的磁场线图案与永磁体的磁场线图案以及电偶极子的电场线图案相似 (图5-13) 。 ◀

概念问题 5-5: 两条无限长的平行导线承载大小相等的电流。如果电流 (a) 方向相同且 (b) 方向相反, 那么与单独一根导线产生的磁场相比, 两根导线在导线中间点产生的磁场是多少?

概念问题 5-6: 为线性载流导体产生的磁场方向设计右手定则。

概念问题 5-7: 什么是磁偶极子? 描述其磁场分布。

练习 5-6: 半无限线性导体沿 z 轴在 $z = 0$ 和 $z = \infty$ 之间延伸。如果导体中的电流 I 沿 z 轴正方向流动, 则在 x - y 平面上距导体径向距离 r 处的点处找到 $\mathbf{H}$ 。

Answer: $\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{I}{4\pi r}$ (A/m). (See EM.)

练习 5-7: 将承载 4 A 电流的导线形成一个环形回路。如果环路中心的磁场为 20 A/m, 则环路 (a) 只有一匝和 (b) 10 匝时, 环路的半径是多少?

Answer: (a) $a = 10$ cm, (b) $a = 1$ m. (See EM.)

练习 5-8: 将一根导线形成一个方形环, 并将其放置在 x - y 平面中, 其中心位于原点, 其每条边平行于 x 轴或 y 轴。每边长 40 厘米, 导线承载 5 安培的电流, 从上方看, 电流方向为顺时针方向。计算环中心的磁场。

Answer: $\mathbf{H} = -\hat{\mathbf{z}} \frac{4I}{\sqrt{2}\pi l} = -\hat{\mathbf{z}} 11.25$ A/m. (See EM.)

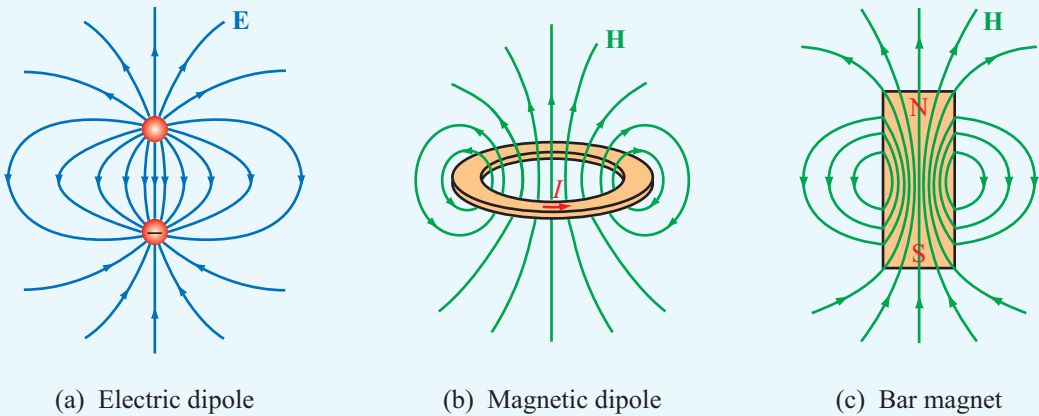


Figure 5-13 Patterns of (a) the electric field of an electric dipole, (b) the magnetic field of a magnetic dipole, and (c) the magnetic field of a bar magnet. Far away from the sources, the field patterns are similar in all three cases.

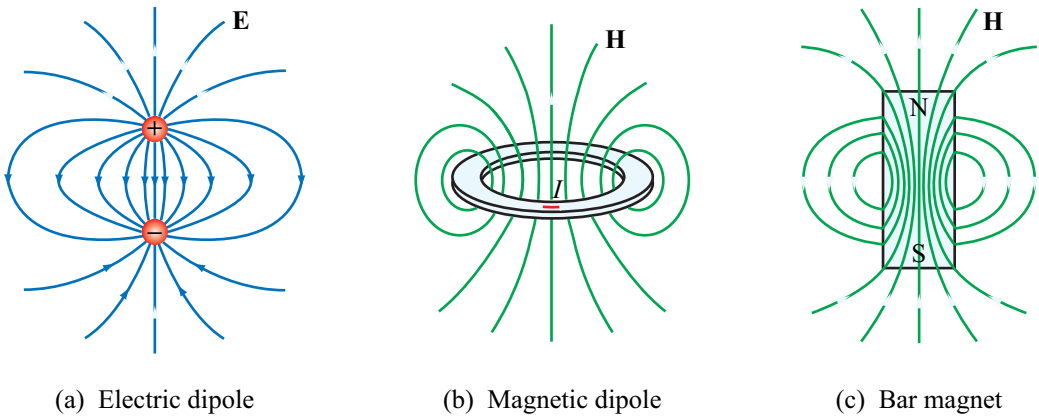
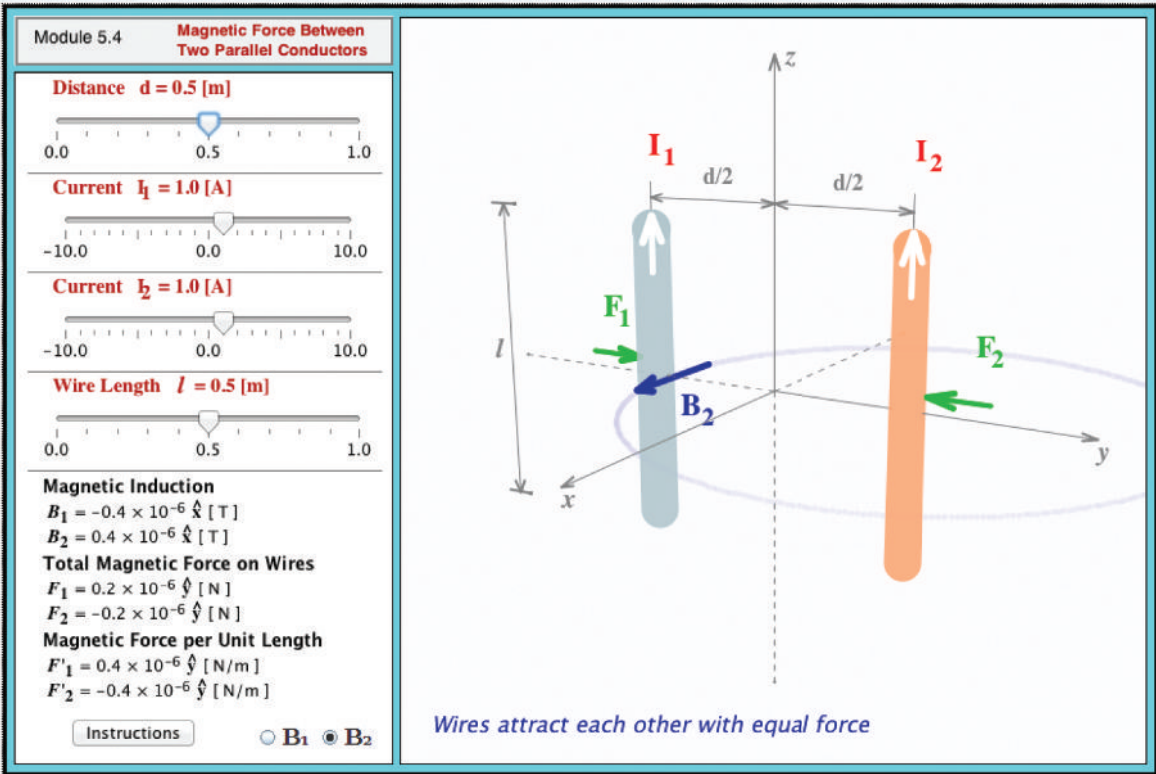
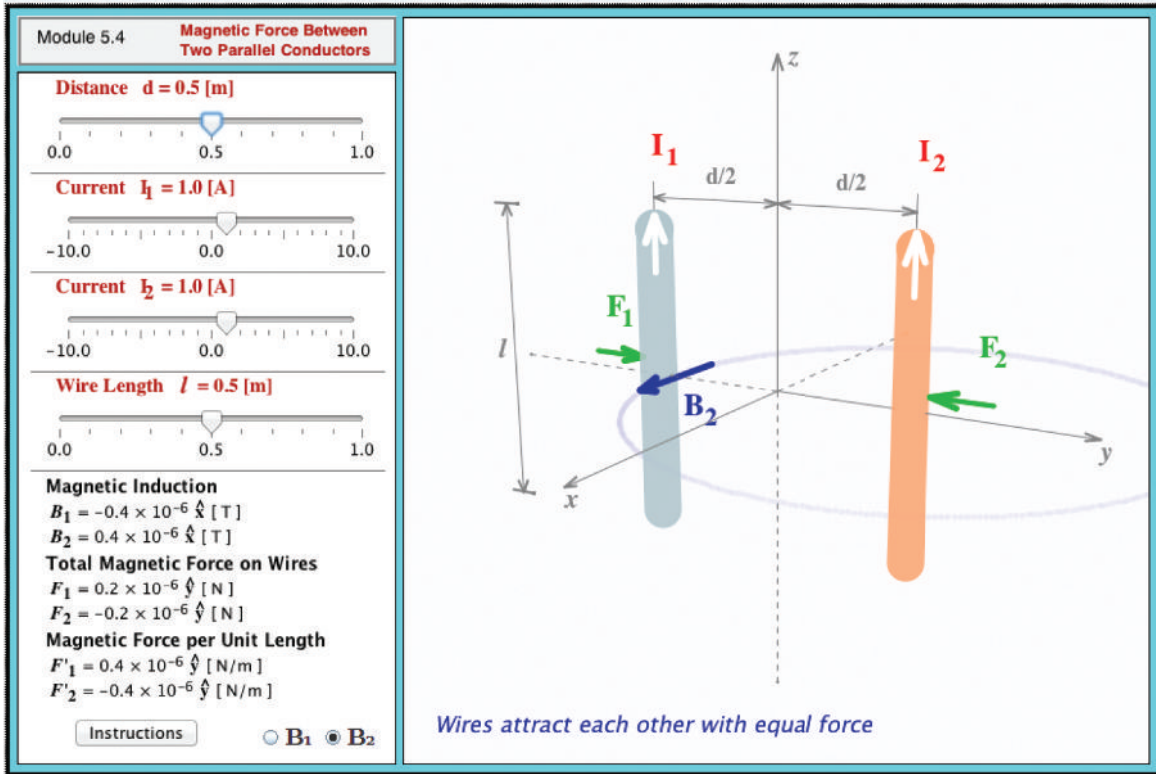


图 5-13 (a) 电偶极子的电场、(b) 磁偶极子的磁场和 (c) 条形磁铁的磁场的图案。远离源头, 所有三种情况下的场模式都是相似的。

Module 5.4 Magnetic Force between Two Parallel Conductors Observe the direction and magnitude of the force exerted on parallel current-carrying wires.



模块 5.4 两个平行导体之间的磁力 观察施加在平行载流线上的力的方向和大小。



5-2.3 Magnetic Force Between Two Parallel Conductors

In Section 5-1.1, we examined the magnetic force $\mathbf{F}_m$ that acts on a current-carrying conductor when placed in an external magnetic field. The current in the conductor, however, also generates its own magnetic field. Hence, if two current-carrying conductors are placed in each other's vicinity, each will exert a magnetic force on the other. Let us consider two very long (or effectively infinitely long), straight, free-standing, parallel wires separated by a distance d and carrying currents I_1 and I_2 in the z direction (Fig. 5-14) at $y = -d/2$ and $y = d/2$, respectively. We denote by $\mathbf{B}_1$ the magnetic field due to current I_1 , defined at the location of the wire carrying current I_2 and, conversely, by $\mathbf{B}_2$ the field due to I_2 at the location of the wire carrying current I_1 . From Eq. (5.30), with $l = I_1$, $r = d$, and $\hat{\phi} = -\hat{x}$ at the location of I_2 , the field $\mathbf{B}_1$ is

$$\mathbf{B}_1 = -\hat{x} \frac{\mu_0 I_1}{2\pi d}. \quad (5.39)$$

The force $\mathbf{F}_2$ exerted on a length l of wire I_2 due to its presence in field $\mathbf{B}_1$ may be obtained by applying Eq. (5.12):

$$\mathbf{F}_2 = I_2 l \hat{z} \times \mathbf{B}_1 = I_2 l \hat{z} \times (-\hat{x}) \frac{\mu_0 I_1}{2\pi d} = -\hat{y} \frac{\mu_0 I_1 I_2 l}{2\pi d}, \quad (5.40)$$

and the corresponding force per unit length is

$$\mathbf{F}'_2 = \frac{\mathbf{F}_2}{l} = -\hat{y} \frac{\mu_0 I_1 I_2}{2\pi d}. \quad (5.41)$$

A similar analysis performed for the force per unit length exerted on the wire carrying I_1 leads to

$$\mathbf{F}'_1 = \hat{y} \frac{\mu_0 I_1 I_2}{2\pi d}. \quad (5.42)$$

► Thus, two parallel wires carrying currents in the same direction attract each other with equal force. If the currents are in opposite directions, the wires would repel one another with equal force. ◀

5-2.3 两平行导体之间的磁力

在第 5-1.1 节中，我们研究了当放置在外磁场中时作用在载流导体上的磁力 $\mathbf{F}_m$ 。然而，导体中的电流也会产生自己的磁场。因此，如果两个载流导体放置在彼此附近，每个导体都会对另一个导体施加磁力。让我们考虑两条非常长（或实际上无限长）、直的、独立的、平行的导线，其间隔距离 d ，并在 z 方向（图 5-14）分别在 $y = -d/2$ 和 $y = d/2$ 处承载电流 I_1 和 I_2 。我们用 $\mathbf{B}_1$ 表示电流 I_1 产生的磁场，定义在承载电流 I_2 的导线位置，相反，用 $\mathbf{B}_2$ 表示在承载电流 I_1 的导线位置处由 I_2 产生的磁场。从方程 (5.30)，当 $l = I_1$ 、 $r = d$ 和 $\hat{\phi} = -\hat{x}$ 在 I_2 位置时，域 $\mathbf{B}_1$ 为

$$\mathbf{B}_1 = -\hat{x} \frac{\mu_0 I_1}{2\pi d}. \quad (5.39)$$

由于存在于磁场 $\mathbf{B}_1$ 中，施加在长度为 l 的导线 I_2 上的力 $\mathbf{F}_2$ 可以通过应用等式获得：(5.12)：

$$\mathbf{F}_2 = I_2 l \hat{z} \times \mathbf{B}_1 = I_2 l \hat{z} \times (-\hat{x}) \frac{\mu_0 I_1}{2\pi d} = -\hat{y} \frac{\mu_0 I_1 I_2 l}{2\pi d}, \quad (5.40)$$

单位长度对应的力为

$$\mathbf{F}'_2 = \frac{\mathbf{F}_2}{l} = -\hat{y} \frac{\mu_0 I_1 I_2}{2\pi d}. \quad (5.41)$$

对施加在承载 I_1 的导线上的每单位长度的力进行类似的分析，得出

$$\mathbf{F}'_1 = \hat{y} \frac{\mu_0 I_1 I_2}{2\pi d}. \quad (5.42)$$

► 因此，沿相同方向承载电流的两条平行导线以相等的力相互吸引。如果电流方向相反，电线将以相等的力相互排斥。 ◀

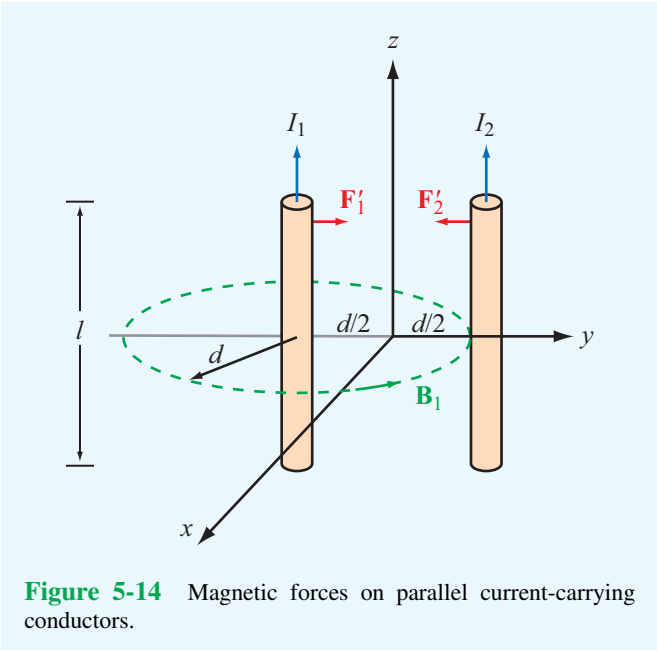


Figure 5-14 Magnetic forces on parallel current-carrying conductors.

Exercise 5-9: Suppose that the conductor carrying I_2 in **Fig. 5-14** is rotated so that it is parallel to the x axis. What would $\mathbf{F}_2$ be in that case?

Answer: $\mathbf{F}_2 = 0$. (See [EM.1](#).)

5-3 Maxwell’s Magnetostatic Equations

Thus far, we have introduced the Biot–Savart law for finding the magnetic flux density $\mathbf{B}$ and field $\mathbf{H}$ due to any distribution of electric currents in free space, and we examined how magnetic fields can exert magnetic forces on moving charged particles and current-carrying conductors. We now examine two additional important properties of magnetostatic fields.

5-3.1 Gauss’s Law for Magnetism

In Chapter 4, we learned that the net outward flux of the electric flux density $\mathbf{D}$ through a closed surface equals the enclosed net charge Q . We referred to this property as Gauss’s law (for electricity) and expressed it mathematically in differential and integral forms as

$$\nabla \cdot \mathbf{D} = \rho_v \quad \longleftrightarrow \quad \oint_S \mathbf{D} \cdot d\mathbf{s} = Q. \quad (5.43)$$

Conversion from differential to integral form was accomplished by applying the divergence theorem to a volume $\mathcal{V}$

that is enclosed by a surface S and contains a total charge $Q = \int_{\mathcal{V}} \rho_v d\mathcal{V}$ (Section 4-4).

The magnetostatic counterpart of Eq. (5.43), often called **Gauss’s law for magnetism**, is

$$\nabla \cdot \mathbf{B} = 0 \quad \longleftrightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{s} = 0. \quad (5.44)$$

The differential form is one of Maxwell’s four equations, and the integral form is obtained with the help of the divergence theorem. Note that the right-hand side of Gauss’s law for magnetism is zero, reflecting the fact that the magnetic equivalence of an electric point charge does not exist in nature.

▶ The hypothetical magnetic analogue to an electric point charge is called a **magnetic monopole**. Magnetic monopoles, however, always occur in pairs (that is, as dipoles). ◀

No matter how many times a permanent magnet is subdivided, each new piece will always have a north and a south pole, even if the process were to be continued down to the atomic level. Consequently, there is no magnetic equivalence to an electric charge q or charge density ρ_v .

Formally, the name “Gauss’s law” refers to the electric case, even when no specific reference to electricity is indicated. *The property described by Eq. (5.44) has been called “the law of nonexistence of isolated monopoles,” “the law of conservation of magnetic flux,” and “Gauss’s law for magnetism,” among others.* We prefer the last of the three cited names because it reminds us of the parallelism, as well as the differences, between the electric and magnetic laws of nature.

The difference between Gauss’s law for electricity and its magnetic counterpart can be elucidated in terms of field lines. Electric field lines originate from positive electric charges and terminate on negative ones. Hence, for the electric field lines of the electric dipole shown in **Fig. 5-15(a)**, the electric flux through a closed surface surrounding one of the charges is nonzero. In contrast, *magnetic field lines always form continuous closed loops*. As we saw in Section 5-2, the magnetic field lines due to currents do not begin or end at any point; this is true for the linear conductor of **Fig. 5-11** and the circular loop of **Fig. 5-12**, as well as for any current distribution. It is also true for a bar magnet (**Fig. 5-15(b)**). Because the magnetic field lines form closed loops, the net magnetic flux through any closed surface surrounding the south pole of the magnet (or through any other closed surface) is always zero, regardless of its shape.

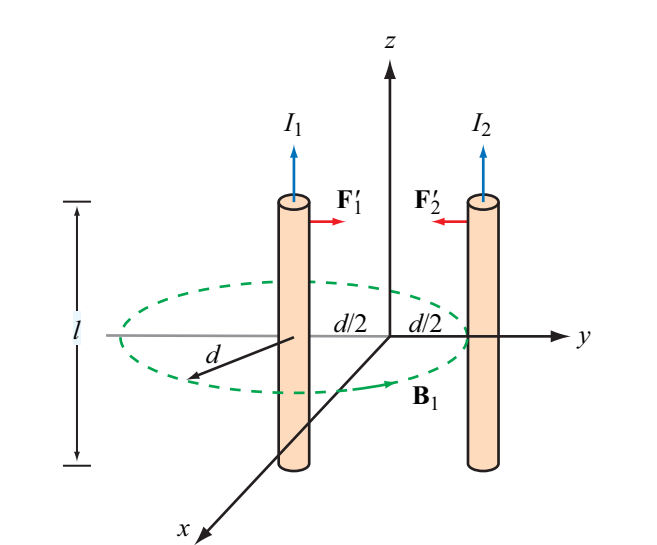


图 5-14 平行载流导体上的磁力。

练习5-9：假设图5-14中带有 I_2 的导体被旋转，使其平行于 x 轴。在这种情况下 $\mathbf{F}_2$ 会是什么？

答案： $\mathbf{F}_2 = 0$ 。（参见^{EM.1}。）

5-3 麦克斯韦静磁方程

到目前为止，我们已经介绍了毕奥-萨伐尔定律，用于计算自由空间中任何电流分布引起的磁通密度 $\mathbf{B}$ 和磁场 $\mathbf{H}$ ，并且我们研究了磁场如何对移动的带电粒子和载流导体施加磁力。我们现在研究静磁场的另外两个重要特性。

5-3.1 高斯磁定律

在第 4 章中，我们了解到通过封闭表面的电通量密度 $\mathbf{D}$ 的净向外通量等于封闭的净电荷 Q 。我们将这一性质称为高斯定律（对于电），并以微分和积分形式在数学上将其表示为

$$\nabla \cdot \mathbf{D} = \rho_v \quad \longleftrightarrow \quad \oint_S \mathbf{D} \cdot d\mathbf{s} = Q. \quad (5.43)$$

从微分形式到积分形式的转换是通过将散度定理应用于体积 $\mathcal{V}$ 来完成的

被表面 S 包围并包含总电荷 $Q = \int_{\mathcal{V}} \rho_v d\mathcal{V}$ (第 4-4 节)。

等式的静磁对应物。(5.43)，通常称为高斯磁定律，是

$$\nabla \cdot \mathbf{B} = 0 \quad \longleftrightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{s} = 0. \quad (5.44)$$

微分形式是麦克斯韦四个方程之一，积分形式是借助散度定理得到的。请注意，高斯磁性定律的右侧为零，反映了自然界中不存在点电荷的磁性等价这一事实。

▶ 点电荷的假设磁模拟称为磁单极子。然而，磁单极子总是成对出现（即偶极子）。◀

无论永磁体被细分多少次，每个新块都将始终有一个北极和一个南极，即使这个过程要继续到原子水平。因此，不存在与电荷 q 或电荷密度 ρ_v 的磁等价。

从形式上讲，“高斯定律”这个名称指的是电力情况，即使没有具体指出电力。方程描述的属性。(5.44) 被称为“孤立单极子不存在定律”、“磁通量守恒定律”和“高斯磁定律”等。我们更喜欢引用的三个名称中的最后一个，因为它提醒我们自然电定律和磁定律之间的相似性和差异。

高斯电定律和磁定律之间的差异可以通过场线来阐明。电场线源自正电荷并终止于负电荷。因此，对于图 5-15(a) 所示的电偶极子的电场线，通过围绕一个电荷的闭合表面的电通量不为零。相反，磁场线总是形成连续的闭环。正如我们在第 5-2 节中看到的，电流产生的磁力线不会在任何点开始或结束；对于图 5-11 的线性导体和图 5-12 的圆形回路以及任何电流分布都是如此。对于条形磁铁也是如此（图 5-15(b)）。由于磁力线形成闭环，因此通过围绕磁体南极的任何闭合表面（或通过任何其他闭合表面）的净磁通量始终为零，无论其形状如何。

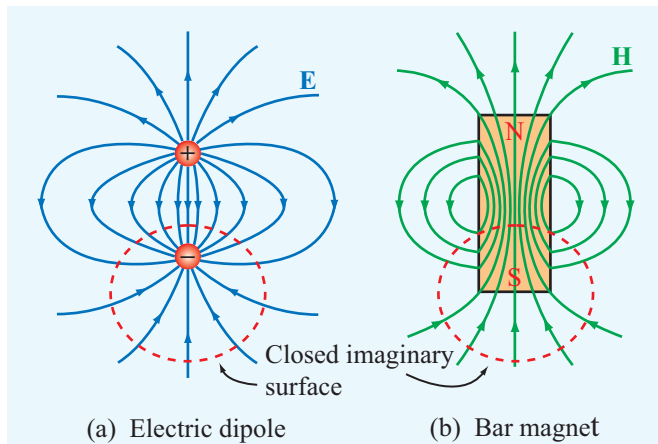


Figure 5-15 Whereas (a) the net electric flux through a closed surface surrounding a charge is not zero, and (b) the net magnetic flux through a closed surface surrounding one of the poles of a magnet is zero.

5-3.2 Ampère's Law

In Chapter 4 we learned that the electrostatic field is conservative, meaning that its line integral along a closed contour always vanishes. This property of the electrostatic field was expressed in differential and integral forms as

$$\nabla \times \mathbf{E} = 0 \quad \longleftrightarrow \quad \oint_C \mathbf{E} \cdot d\mathbf{l} = 0. \quad (5.45)$$

Conversion of the differential to integral form was accomplished by applying Stokes's theorem to a surface S with contour C .

The magnetostatic counterpart of Eq. (5.45), known as **Ampère's law**, is

$$\nabla \times \mathbf{H} = \mathbf{J} \quad \longleftrightarrow \quad \oint_C \mathbf{H} \cdot d\mathbf{l} = I, \quad (5.46)$$

where I is the total current passing through S . The differential form again is one of Maxwell's equations, and the integral form is obtained by integrating both sides of Eq. (5.46) over an open surface S ,

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} = \int_S \mathbf{J} \cdot d\mathbf{s}, \quad (5.47)$$

and then invoking Stokes's theorem with $I = \int \mathbf{J} \cdot d\mathbf{s}$.

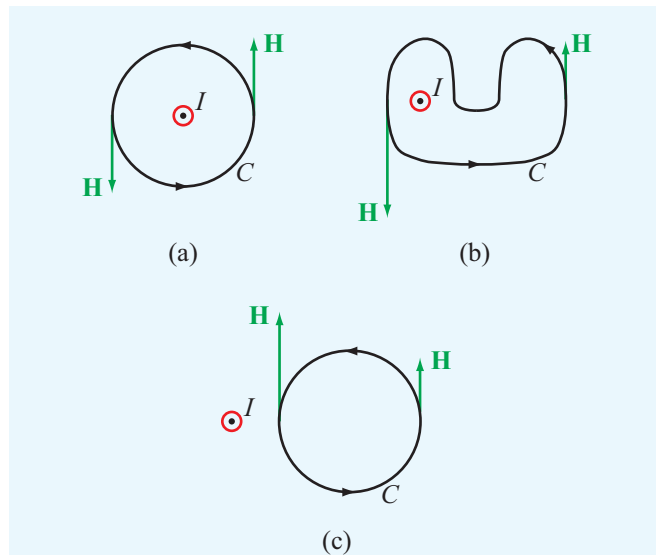


Figure 5-16 Ampère's law states that the line integral of $\mathbf{H}$ around a closed contour C is equal to the current traversing the surface bounded by the contour. This is true for contours (a) and (b), but the line integral of $\mathbf{H}$ is zero for the contour in (c) because the current I (denoted by the symbol $\odot$) is not enclosed by the contour C .

► The sign convention for the direction of the contour path C in Ampère's law is taken so that I and $\mathbf{H}$ satisfy the right-hand rule defined earlier in connection with the Biot-Savart law. That is, if the direction of I is aligned with the direction of the thumb of the right hand, then the direction of the contour C should be chosen along that of the other four fingers. ◀

In words, *Ampère's circuital law states that the line integral of $\mathbf{H}$ around a closed path is equal to the current traversing the surface bounded by that path.* To apply Ampère's law, the current must flow through a closed path. By way of illustration, for both configurations shown in **Figs. 5-16(a)** and **(b)**, the line integral of $\mathbf{H}$ is equal to the current I , even though the paths have very different shapes and the magnitude of $\mathbf{H}$ is not uniform along the path of the configuration in part (b). By the same token, because path (c) in **Fig. 5-16** does not enclose the current I , the line integral of $\mathbf{H}$ along it vanishes, even though $\mathbf{H}$ is not zero along the path.

When we examined Gauss's law in Section 4-4, we discovered that in practice its usefulness for calculating the electric flux density $\mathbf{D}$ is limited to charge distributions that possess a certain degree of symmetry and that the calculation procedure

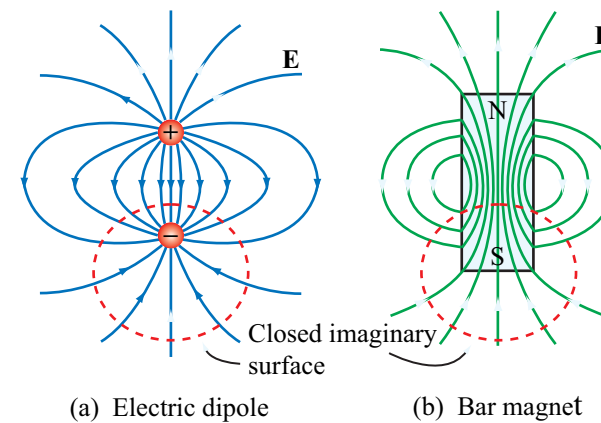


图 5-15 而 (a) 通过围绕电荷的封闭表面的净电通量不为零, 并且 (b) 通过围绕磁体之一的封闭表面的净磁通量为零。

5-3.2 安培环路定律

在第四章中, 我们了解到静电场是保守的, 这意味着它沿着闭合轮廓的线积分总是消失。静电场的这种性质可以用微分和积分形式表示为

$$\nabla \times \mathbf{E} = 0 \quad \longleftrightarrow \quad \oint_C \mathbf{E} \cdot d\mathbf{l} = 0. \quad (5.45)$$

通过将斯托克斯定理应用于具有轮廓 C 的曲面 S 来完成微分形式到积分形式的转换。

等式的静磁对应物。 (5.45), 称为安培环路定律, 是

$$\nabla \times \mathbf{H} = \mathbf{J} \quad \longleftrightarrow \quad \oint_C \mathbf{H} \cdot d\mathbf{l} = I, \quad (5.46)$$

其中 I 是通过 S 的总电流。微分形式又是麦克斯韦方程组之一, 积分形式是通过对方程两边积分得到。 (5.46) 在开放表面 S 上,

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} = \int_S \mathbf{J} \cdot d\mathbf{s}, \quad (5.47)$$

然后用 $I = \int \mathbf{J} \cdot d\mathbf{s}$ 调用斯托克斯定理。

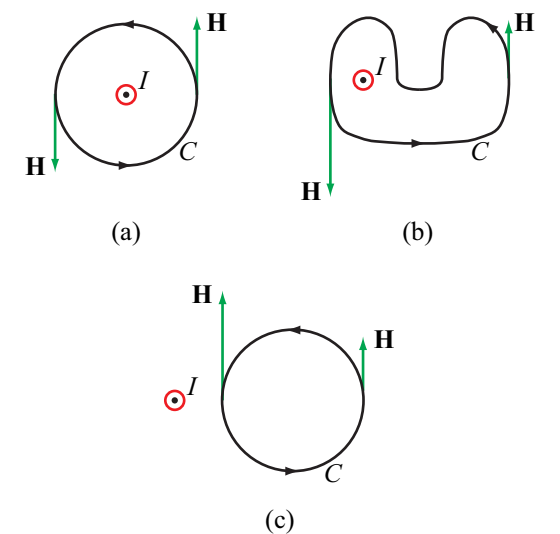


图 5-16 安培环路定律 指出, $\mathbf{H}$ 围绕闭合轮廓 C 的线积分等于穿过由轮廓界定的表面的电流。对于轮廓 (a) 和 (b) 来说是这样, 但对于 (c) 中的轮廓, $\mathbf{H}$ 的线积分为零, 因为电流 I (用符号 $\odot$ 表示) 没有被轮廓 C 包围。

► 轮廓路径 C 方向的符号约定 (以 Amp 为单位) 采用埃伦定律, 使得 I 和 $\mathbf{H}$ 满足先前定义的与毕奥-萨伐尔定律相关的右手定则。也就是说, 如果 I 的方向与右手大拇指的方向一致, 则轮廓 C 的方向应沿着其他四个手指的方向选择。◀

用言语来说, 放大器埃雷环路定律指出, $\mathbf{H}$ 围绕闭合路径的线积分等于流经该路径所界定的表面的电流。应用放大器根据定律, 电流必须流经闭合路径。作为说明, 对于图 1 和 2 中所示的两种配置, 如图 5-16(a) 和 (b) 所示, $\mathbf{H}$ 的线积分等于电流 I , 尽管路径的形状非常不同, 并且 $\mathbf{H}$ 的大小沿着 (b) 部分的配置路径并不均匀。出于同样的原因, 由于图 5-16 中的路径 (c) 不包含电流 I , 因此即使 $\mathbf{H}$ 沿路径不为零, $\mathbf{H}$ 沿其的线积分也会消失。

当我们研究第 4-4 节中的高斯定律时, 我们发现实际上它计算电通量密度 $\mathbf{D}$ 的有用性仅限于具有一定程度对称性的电荷分布, 并且计算过程

is subject to the proper choice of a Gaussian surface enclosing the charges. A similar restriction applies to Ampère's law: its usefulness is limited to symmetric current distributions that allow the choice of convenient **Ampèrian contours** around them, as illustrated by Examples 5-4 to 5-6.

Example 5-4: Magnetic Field of a Long Wire

A long (practically infinite) straight wire of radius a carries a steady current I that is uniformly distributed over its cross section. Determine the magnetic field $\mathbf{H}$ a distance r from the wire axis for (a) $r \leq a$ (inside the wire) and (b) $r \geq a$ (outside the wire).

Solution: (a) We choose I to be along the $+z$ direction (Fig. 5-17(a)). To determine $\mathbf{H}_1 = \mathbf{H}$ at a distance $r = r_1 \leq a$, we choose the Ampèrian contour C_1 to be a circular path of radius $r = r_1$ (Fig. 5-17(b)). In this case, Ampère's law takes the form

$$\oint_{C_1} \mathbf{H}_1 \cdot d\mathbf{l}_1 = I_1, \quad (5.48)$$

where I_1 is the fraction of the total current I flowing through C_1 . From symmetry, $\mathbf{H}_1$ must be constant in magnitude and parallel to the contour at any point along the path. Furthermore, to satisfy the right-hand rule and given that I is along the z direction, $\mathbf{H}_1$ must be in the $+\phi$ direction. Hence, $\mathbf{H}_1 = \hat{\phi}H_1$, $d\mathbf{l}_1 = \hat{\phi}r_1 d\phi$, and the left-hand side of Eq. (5.48) becomes

$$\oint_{C_1} \mathbf{H}_1 \cdot d\mathbf{l}_1 = \int_0^{2\pi} H_1 (\hat{\phi} \cdot \hat{\phi}) r_1 d\phi = 2\pi r_1 H_1.$$

The current I_1 flowing through the area enclosed by C_1 is equal to the total current I multiplied by the ratio of the area enclosed by C_1 to the total cross-sectional area of the wire:

$$I_1 = \left(\frac{\pi r_1^2}{\pi a^2} \right) I = \left(\frac{r_1}{a} \right)^2 I.$$

Equating both sides of Eq. (5.48) and then solving for $\mathbf{H}_1$ yields

$$\mathbf{H}_1 = \hat{\phi}H_1 = \hat{\phi} \frac{r_1}{2\pi a^2} I \quad (\text{for } r_1 \leq a). \quad (5.49a)$$

(b) For $r = r_2 \geq a$, we choose path C_2 , which encloses all the current I . Hence, $\mathbf{H}_2 = \hat{\phi}H_2$, $d\mathbf{l}_2 = \hat{\phi}r_2 d\phi$, and

$$\oint_{C_2} \mathbf{H}_2 \cdot d\mathbf{l}_2 = 2\pi r_2 H_2 = I,$$

which yields

$$\mathbf{H}_2 = \hat{\phi}H_2 = \hat{\phi} \frac{I}{2\pi r_2} \quad (\text{for } r_2 \geq a). \quad (5.49b)$$

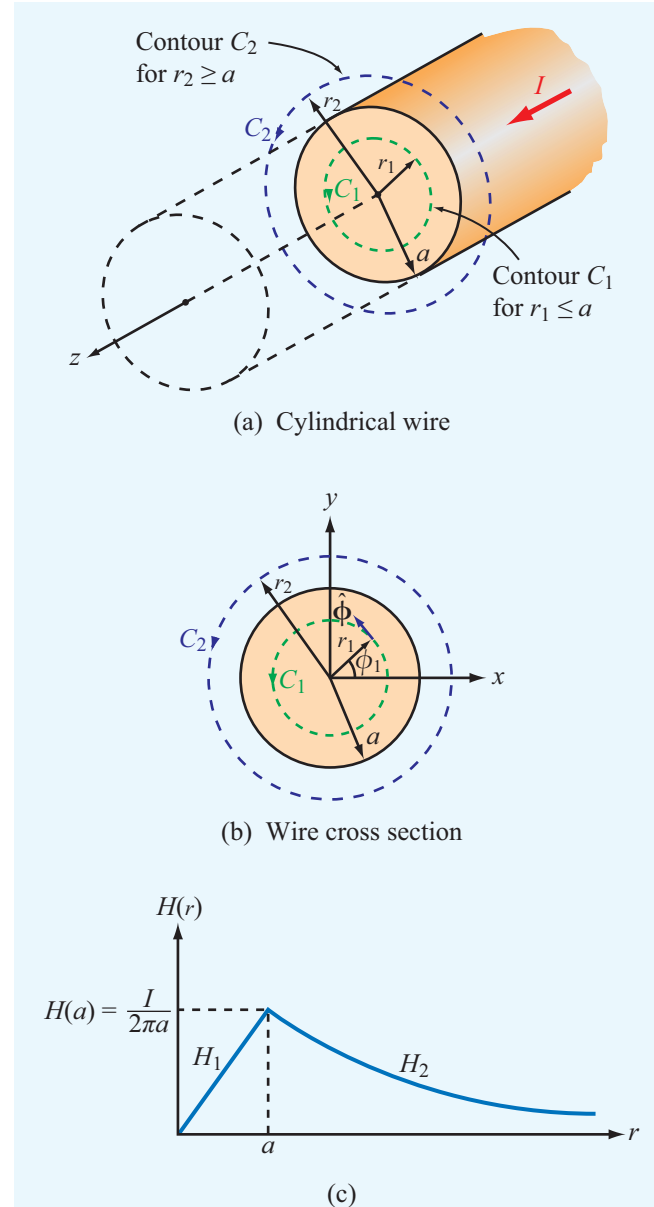


Figure 5-17 Infinitely long wire of radius a carrying a uniform current I along the $+z$ direction: (a) general configuration showing contours C_1 and C_2 ; (b) cross-sectional view; and (c) a plot of H versus r (Example 5-4).

Ignoring the subscript 2, we observe that Eq. (5.49b) provides the same expression for $\mathbf{B} = \mu_0 \mathbf{H}$ as Eq. (5.30), which was derived on the basis of the Biot-Savart law.

The variation of the magnitude of H as a function of r is

(续上页) 这取决于能否选取适当的、包围电荷的高斯面。安培环路定律也有类似限制: 只有当电流分布具有足够对称性、能够选取方便的安培环路时, 它才便于直接求场, 如例 5-4至5-6所示。

例 5-4: 长导线的磁场

半径为 a 的长 (几乎无限) 直导线承载均匀分布在其横截面上的稳定电流 I 。确定 (a) $r \leq a$ (导线内部) 和 (b) $r \geq a$ (导线外部) 的磁场 H 距导线轴的距离 r 。

解: (a) 令电流 I 沿 $+z$ 方向 (图5-17(a))。为求距轴线 $r = r_1 \leq a$ 处的磁场 H_1 , 选取半径为 r_1 的圆形安培环路 C_1 (图5-17(b))。安培环路定律给出

$$\oint_{C_1} \mathbf{H}_1 \cdot d\mathbf{l}_1 = I_1, \quad (5.48)$$

其中 I_1 是流过 C_1 的总电流 I 的比例。从对称性来看, H_1 的大小必须恒定, 并且沿路径的任何点都与轮廓平行。此外, 为了满足右手定则并且假设 I 沿 z 方向, H_1 必须沿 $+\phi$ 方向。因此, $\mathbf{H}_1 = \hat{\phi}H_1$, $d\mathbf{l}_1 = \hat{\phi}r_1 d\phi$ 以及等式的左侧: (5.48) 变为

$$\oint_{C_1} \mathbf{H}_1 \cdot d\mathbf{l}_1 = \int_0^{2\pi} H_1 (\hat{\phi} \cdot \hat{\phi}) r_1 d\phi = 2\pi r_1 H_1.$$

流过 C_1 围成的面积的电流 I_1 等于总电流 I 乘以 C_1 围成的面积与导线总横截面积的比值:

$$I_1 = \left(\frac{\pi r_1^2}{\pi a^2} \right) I = \left(\frac{r_1}{a} \right)^2 I.$$

等式两边相等。 (5.48) 然后求解 H_1 得出

$$\mathbf{H}_1 = \hat{\phi}H_1 = \hat{\phi} \frac{r_1}{2\pi a^2} I \quad (\text{for } r_1 \leq a). \quad (5.49a)$$

(b) 当 $r = r_2 \geq a$ 时, 选取包围全部电流 I 的环路 C_2 。 H_2 与有向线元均沿方位角方向, 故

$$\oint_{C_2} \mathbf{H}_2 \cdot d\mathbf{l}_2 = 2\pi r_2 H_2 = I,$$

这产生

$$\mathbf{H}_2 = \hat{\phi}H_2 = \hat{\phi} \frac{I}{2\pi r_2} \quad (\text{for } r_2 \geq a). \quad (5.49b)$$

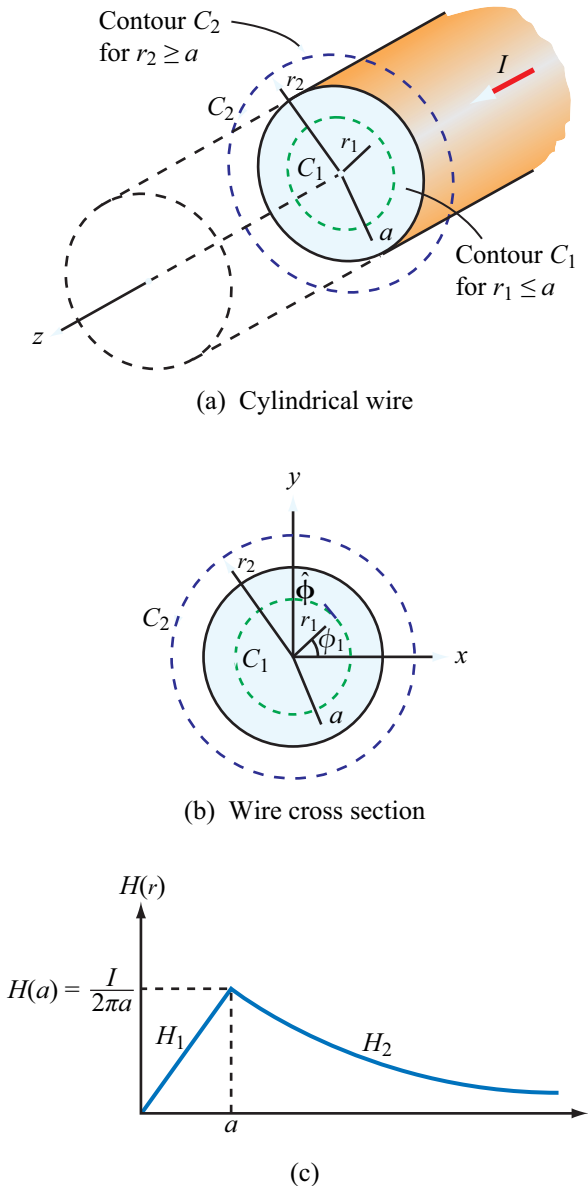


图 5-17 半径为 a 的无限长导线, 沿 $+z$ 方向承载均匀电流 I : (a) 显示轮廓 C_1 和 C_2 的一般配置; (b) 剖面图; (c) H 与 r 的关系图 (示例 5-4)。

忽略下标 2, 我们观察到等式: (5.49b) 为 $\mathbf{B} = \mu_0 \mathbf{H}$ 提供了与等式 1 相同的表达式。 (5.30), 这是根据毕奥-萨伐尔定律推导出来的。

H 的大小作为 r 的函数的变化为

plotted in Fig. 5-17(c); H increases linearly between $r = 0$ and $r = a$ (inside the conductor), and then decreases as $1/r$ for $r > a$ (outside the conductor).

Exercise 5-10: A current I flows in the inner conductor of a long coaxial cable and returns through the outer conductor. What is the magnetic field in the region outside the coaxial cable and why?

Answer: $\mathbf{H} = 0$ outside the coaxial cable because the net current enclosed by an Ampèrian contour enclosing the cable is zero.

Exercise 5-11: The metal niobium becomes a superconductor with zero electrical resistance when it is cooled to below 9 K, but its superconductive behavior ceases when the magnetic flux density at its surface exceeds 0.12 T. Determine the maximum current that a 0.1-mm diameter niobium wire can carry and remain superconductive.

Answer: $I = 30$ A. (See )

Example 5-5: Magnetic Field inside a Toroidal Coil

A **toroidal coil** (also called a torus or toroid) is a doughnut-shaped structure (called the core) wrapped in closely spaced turns of wire (Fig. 5-18). For clarity, we show the turns in the figure as spaced far apart, but in practice, they are wound in a closely spaced arrangement to form approximately circular loops. The toroid is used to magnetically couple multiple circuits and to measure the magnetic properties of materials, as illustrated later in Fig. 5-31. For a toroid with N turns carrying a current I , determine the magnetic field $\mathbf{H}$ in each of the following three regions: $r < a$, $a < r < b$, and $r > b$ with all in the azimuthal symmetry plane of the toroid.

Solution: From symmetry, it is clear that $\mathbf{H}$ is uniform in the azimuthal direction. If we construct a circular Ampèrian contour with center at the origin and radius $r < a$, there will be no current flowing through the surface of the contour. Therefore,

$$\mathbf{H} = 0 \quad \text{for } r < a.$$

Similarly, for an Ampèrian contour with radius $r > b$, the *net* current flowing through its surface is zero because an equal number of current coils cross the surface in both directions; hence,

$$\mathbf{H} = 0 \quad \text{for } r > b \text{ (region exterior to the toroidal coil).}$$

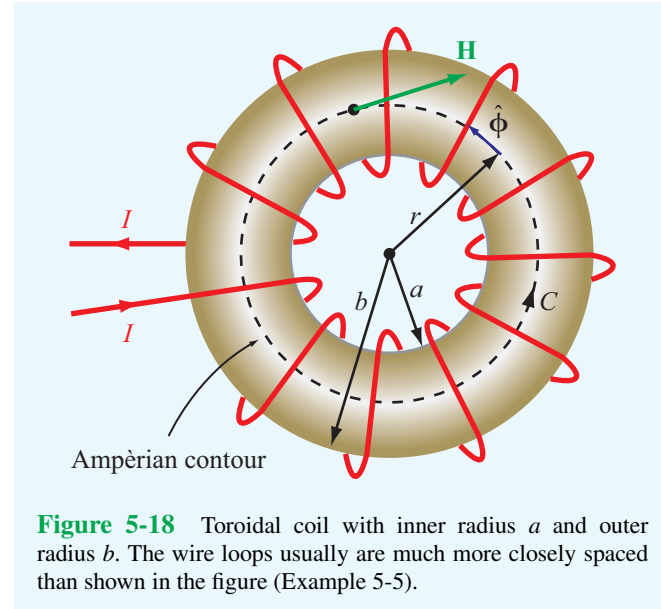


Figure 5-18 Toroidal coil with inner radius a and outer radius b . The wire loops usually are much more closely spaced than shown in the figure (Example 5-5).

For the region inside the core, we construct a path of radius r (Fig. 5-18). For each loop, we know from Example 5-3 that the field $\mathbf{H}$ at the center of the loop points along the axis of the loop, which in this case is the ϕ direction, and in view of the direction of the current I shown in Fig. 5-18, the right-hand rule tells us that $\mathbf{H}$ must be in the $-\phi$ direction. Hence, $\mathbf{H} = -\hat{\phi}H$. The total current crossing the surface of the contour with radius r is NI and its direction is into the page. According to the right-hand rule associated with Ampère's law, the current is positive if it crosses the surface of the contour in the direction of the four fingers of the right hand when the thumb is pointing along the direction of the contour C . Hence, the current through the surface spanned by the contour is $-NI$. Application of Ampère's law then gives

$$\begin{aligned} \oint_C \mathbf{H} \cdot d\mathbf{l} &= \int_0^{2\pi} (-\hat{\phi}H) \cdot \hat{\phi}r d\phi \\ &= -2\pi rH \\ &= -NI. \end{aligned}$$

Hence, $H = NI/(2\pi r)$ and

$$\mathbf{H} = -\hat{\phi}H = -\hat{\phi} \frac{NI}{2\pi r} \quad (\text{for } a < r < b). \quad (5.50)$$

绘制在图5-17(c)中; H 在 $r = 0$ 和 $r = a$ (导体内部) 之间线性增加, 然后对于 $r > a$ (导体外部) 以 $1/r$ 形式减小。

练习 5-10: 电流 I 在长同轴电缆的内导体中流动, 并通过外导体返回。同轴电缆外部区域的磁场是什么? 为什么?

答案: 电缆外部 $H = 0$, 因为包围同轴电缆的安培环路所包围的净电流为零。

练习 5-11: 当金属铌冷却到 9 K 以下时, 它会成为电阻为零的超导体, 但当其表面磁通密度超过 0.12 T 时, 其超导行为就会停止。确定 0.1 毫米直径的铌丝可以承载并保持超导性的最大电流。

答案: $I = 30$ A. (见 )

示例 5-5: 环形线圈内的磁场

环形线圈 (也称为环面或环形线圈) 是一种环形结构 (称为磁芯), 包裹在紧密间隔的线圈中 (图 5-18)。为了清楚起见, 我们将图中的匝间隔得很远, 但实际上, 它们以紧密间隔的排列方式缠绕, 形成近似圆形的环。环形线圈用于磁耦合多个电路并测量材料的磁性, 如后面的图 5-31 所示。对于承载电流 I 的 N 匝环形线圈, 确定以下三个区域中每个区域的磁场 H : $r < a$ 、 $a < r < b$ 和 $r > b$, 所有区域均位于环形线圈的方位角对称平面内。

解: 由对称性, H 的大小不随方位角变化。选取以原点为圆心、半径 $r < a$ 的圆形安培环路, 则没有电流穿过该环路所围的曲面。因此,

$$\mathbf{H} = 0 \quad \text{for } r < a.$$

同理, 对于半径 $r > b$ 的安培环路, 正、反方向穿过所围曲面的电流相互抵消, 净电流为零。因此,

$$H = 0 \quad \text{对于 } r > b \text{ (环形线圈外部的区域) }。$$

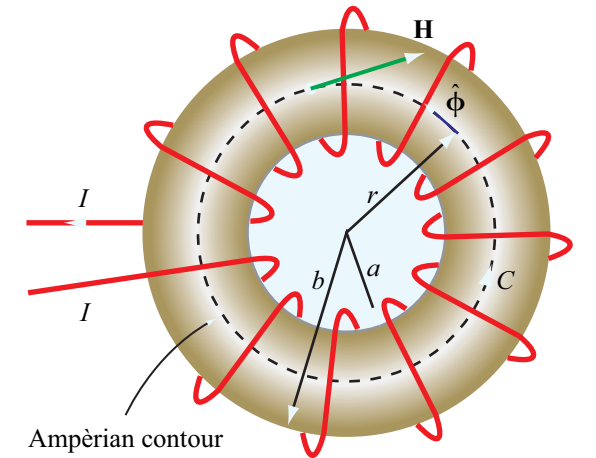


图 5-18 内半径为 a 、外半径为 b 的环形线圈。线圈的间距通常比图中所示的要紧密得多 (示例 5-5)。

在磁芯内部, 选取半径为 r 的圆形环路 (图5-18)。由例 5-3, 每匝线圈在其中心产生的磁场沿该匝的轴线, 即方位角方向。根据图示电流方向及右手定则, H 沿负方位角方向。穿过所围曲面的电流大小为 NI , 方向垂直纸面向内。确定电流正号时, 右手四指沿环路 C 的正向弯曲, 拇指所指为曲面正法向; 据此, 本例的有向电流为 $-NI$ 。代入安培环路定律得 (原书的拇指、四指说明互换, 此处已校正):

$$\begin{aligned} \oint_C \mathbf{H} \cdot d\mathbf{l} &= \int_0^{2\pi} (-\hat{\phi}H) \cdot \hat{\phi}r d\phi \\ &= -2\pi rH \\ &= -NI. \end{aligned}$$

因此, $H = NI/(2\pi r)$ 和

$$\mathbf{H} = -\hat{\phi}H = -\hat{\phi} \frac{NI}{2\pi r} \quad (\text{for } a < r < b). \quad (5.50)$$

► The magnetic field induced by a current-carrying coil wound around a toroid is confined entirely to the toroid volume:

$$\mathbf{H} = \begin{cases} 0 & \text{for } r < a, \\ -\hat{\phi} \frac{NI}{2\pi r} & \text{for } a < r < b, \\ 0 & \text{for } r > b. \end{cases}$$

Example 5-6: Magnetic Field of an Infinite Current Sheet

The x - y plane contains an infinite current sheet with surface current density $\mathbf{J}_s = \hat{\mathbf{x}}J_s$ (Fig. 5-19). Find the magnetic field $\mathbf{H}$ everywhere in space.

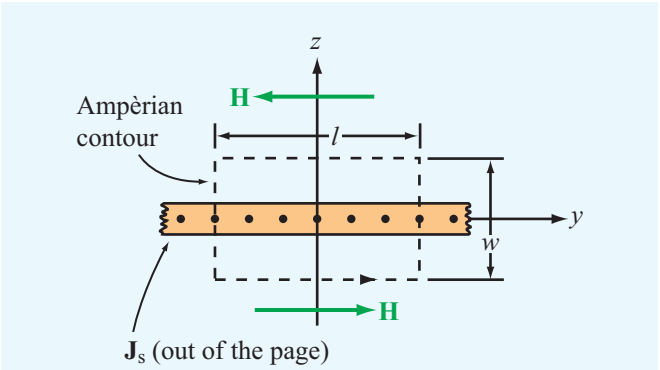


Figure 5-19 A thin current sheet in the x - y plane carrying a surface current density $\mathbf{J}_s = \hat{\mathbf{x}}J_s$ (Example 5-6).

Solution: From symmetry considerations and the right-hand rule, for $z > 0$ and $z < 0$ $\mathbf{H}$ must be in the directions shown in the figure. That is,

$$\mathbf{H} = \begin{cases} -\hat{\mathbf{y}}H & \text{for } z > 0, \\ \hat{\mathbf{y}}H & \text{for } z < 0. \end{cases}$$

To evaluate the line integral in Ampère’s law, we choose a rectangular Ampèrian path around the sheet with dimensions l and w (Fig. 5-19). Recalling that J_s represents current per unit length along the y direction, the total current crossing the surface of the rectangular loop is $I = J_s l$. Hence, applying Ampère’s law over the loop in a counterclockwise direction, while noting that $\mathbf{H}$ is perpendicular to the paths of length w , we have

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = 2Hl = J_s l,$$

from which we obtain the result

$$\mathbf{H} = \begin{cases} -\hat{\mathbf{y}} \frac{J_s}{2} & \text{for } z > 0, \\ \hat{\mathbf{y}} \frac{J_s}{2} & \text{for } z < 0. \end{cases} \quad (5.51)$$

Concept Question 5-8: What are the fundamental differences between electric and magnetic fields?

Concept Question 5-9: If the line integral of $\mathbf{H}$ over a closed contour is zero, does it follow that $\mathbf{H} = 0$ at every point on the contour? If not, what then does it imply?

Concept Question 5-10: Compare the utility of applying the Biot–Savart law versus applying Ampère’s law for computing the magnetic field due to current-carrying conductors.

Concept Question 5-11: What is a toroid? What is the magnetic field outside the toroid?

5-4 Vector Magnetic Potential

In our treatment of electrostatic fields in Chapter 4, we defined the electrostatic potential V as the line integral of the electric field $\mathbf{E}$ and found that V and $\mathbf{E}$ are related by $\mathbf{E} = -\nabla V$. This relationship proved useful not only in relating electric field distributions in circuit elements (such as resistors and capacitors) to the voltages across them but also to determine $\mathbf{E}$ for a given charge distribution by first computing V using Eq. (4.48). We now explore a similar approach in connection with the magnetic flux density $\mathbf{B}$.

According to Eq. (5.44), $\nabla \cdot \mathbf{B} = 0$. We wish to define $\mathbf{B}$ in terms of a magnetic potential with the constraint that such a definition guarantees that the divergence of $\mathbf{B}$ is always zero. This can be realized by taking advantage of the vector identity given by Eq. (3.106b), which states that, for any vector $\mathbf{A}$,

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0. \quad (5.52)$$

Hence, by introducing the **vector magnetic potential** $\mathbf{A}$ such that

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (\text{Wb/m}^2), \quad (5.53)$$

we are guaranteed that $\nabla \cdot \mathbf{B} = 0$. The SI unit for $\mathbf{B}$ is the tesla (T). An equivalent unit is webers per square meter (Wb/m^2). Consequently, the SI unit for $\mathbf{A}$ is (Wb/m).

► 缠绕在环形线圈上的载流线圈感应的磁场完全限制在环形线圈体积内:

$$\mathbf{H} = \begin{cases} 0 & \text{for } r < a, \\ -\hat{\phi} \frac{NI}{2\pi r} & \text{for } a < r < b, \\ 0 & \text{for } r > b. \end{cases}$$

例 5-6: 无限电流片的磁场

x - y 平面包含表面电流密度为 $\mathbf{J}_s = \hat{\mathbf{x}}J_s$ (的无限电流片。 5-19)。求空间中各处的磁场 $\mathbf{H}$ 。

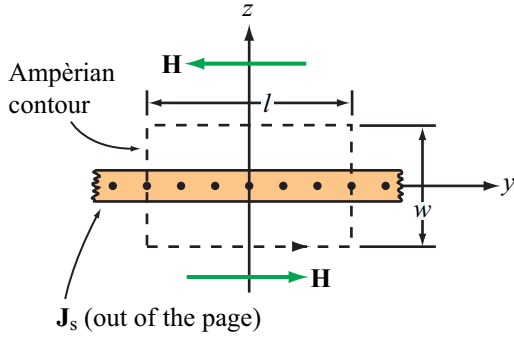


图 5-19 x - y 平面上的薄电流片，其表面电流密度为 $\mathbf{J}_s = \hat{\mathbf{x}}J_s$ (示例 5-6)。

解: 从对称性考虑和右手定则, 对于 $z > 0$ 和 $z < 0$ $\mathbf{H}$ 必须处于图中所示的方向。那是,

$$\mathbf{H} = \begin{cases} -\hat{\mathbf{y}}H & \text{for } z > 0, \\ \hat{\mathbf{y}}H & \text{for } z < 0. \end{cases}$$

评估 Amp 中的线路积分根据这条定律, 我们选择一个矩形放大器尺寸为 l 和 w 的板材周围的埃里安路径 (图 5-19)。回想起 J_s 表示沿 y 方向单位长度的电流, 穿过矩形环路表面的总电流为 $I=J_s l$ 因此, 应用放大器逆时针方向的环路遵循埃雷定律, 同时注意到 $\mathbf{H}$ 垂直于长度为 w 的路径, 我们有

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = 2Hl = J_s l,$$

从中我们得到结果

$$\mathbf{H} = \begin{cases} -\hat{\mathbf{y}} \frac{J_s}{2} & \text{for } z > 0, \\ \hat{\mathbf{y}} \frac{J_s}{2} & \text{for } z < 0. \end{cases} \quad (5.51)$$

概念问题 5-8: 电场和磁场之间的根本区别是什么?

概念问题 5-9: 如果 $\mathbf{H}$ 在闭合轮廓上的线积分为零, 是否可以得出轮廓上每个点的 $\mathbf{H} = 0$? 如果不是, 那么这意味着什么?

概念问题 5-10: 比较应用毕奥-萨伐尔定律与应用安培环路定律计算载流导体产生的磁场的效用。

概念问题 5-11: 什么是环形线圈? 环形线圈外的磁场是多少?

5-4 矢量磁势

在第 4 章静电场的处理中, 我们将静电势 V 定义为电场 $\mathbf{E}$ 的线积分, 并发现 V 和 $\mathbf{E}$ 通过 $\mathbf{E} = -\nabla V$ 相关。事实证明, 这种关系不仅有助于将电路元件 (例如电阻器和电容器) 中的电场分布与它们两端的电压联系起来, 而且还可以通过首先使用方程式计算 V 来确定给定电荷分布的 $\mathbf{E}$ 。(4.48)。我们现在探索与磁通密度 $\mathbf{B}$ 相关的类似方法。

根据方程。(5.44), $\nabla \cdot \mathbf{B} = 0$ 。我们希望用磁势来定义 $\mathbf{B}$, 并限制这样的定义保证 $\mathbf{B}$ 的散度始终为零。这可以通过利用等式给出的向量恒等式来实现。(3.106b), 它指出, 对于任何向量 $\mathbf{A}$,

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0. \quad (5.52)$$

因此, 通过引入矢量磁势 $\mathbf{A}$ 使得

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (\text{Wb/m}^2), \quad (5.53)$$

我们保证 $\nabla \cdot \mathbf{B} = 0$ 。 $\mathbf{B}$ 的 SI 单位是特斯拉 (T)。等效单位是韦伯每平方米 (Wb/m^2)。因此, $\mathbf{A}$ 的 SI 单位为 (Wb/m)。

Technology Brief 10: Electromagnets

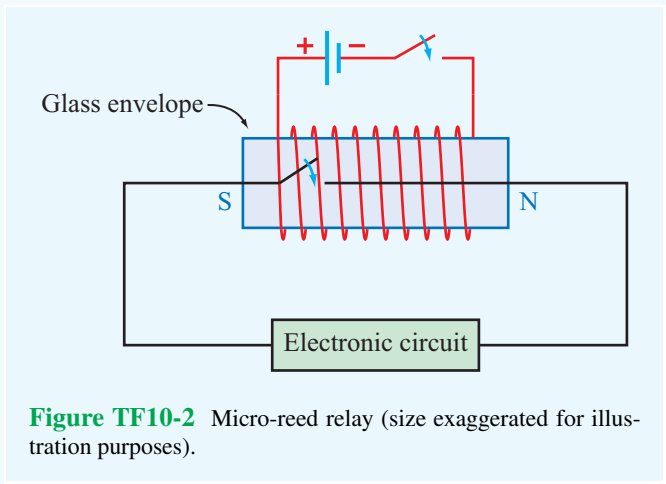
William Sturgeon developed the first practical **electromagnet** in the 1820s. Today, the principle of the electromagnet is used in motors, relay switches in read/write heads for hard disks and tape drives, loud speakers, magnetic levitation, and many other applications.

Basic Principle

Electromagnets can be constructed in various shapes, including the linear **solenoid** and **horseshoe** geometries depicted in **Fig. TF10-1**. In both cases, when an electric current flows through the insulated wire coiled around the central core, it induces a magnetic field with lines resembling those generated by a bar magnet. The strength of the magnetic field is proportional to the current, the number of turns, and the magnetic permeability of the core material. By using a **ferromagnetic core**, the field strength can be increased by several orders of magnitude, depending on the purity of the iron material. When subjected to a magnetic field, ferromagnetic materials, such as iron or nickel, get magnetized and act like magnets themselves.

Magnetic Relays

A magnetic relay is a **switch** or circuit breaker that can be activated into the “ON” and “OFF” positions magnetically. One example is the low-power **reed relay** used in telephone equipment, which consists of two flat nickel-iron blades separated by a small gap (**Fig. TF10-2**). The blades are shaped in such a way that, in the absence of an external force, they remain apart and unconnected (OFF position). Electrical contact between the blades



技术简介 10：电磁铁

William Sturgeon 在 1820 年代开发出第一个实用的电磁体。如今，电磁体原理已用于电机、硬盘和磁带驱动器读/写头中的继电器开关、扬声器、磁悬浮以及许多其他应用。

基本原理

电磁体可以构造成各种形状，包括图 TF10-1 中所示的线性螺线管和马蹄形几何形状。在这两种情况下，当电流流过缠绕在中心磁芯周围的绝缘线时，它会感应出一个磁场，其磁场线类似于条形磁铁产生的磁场线。磁场强度与电流、匝数和磁芯材料的磁导率成正比。通过使用铁磁芯，磁场强度可以增加几个数量级，具体取决于铁材料的纯度。当受到磁场作用时，铁磁材料（例如铁或镍）会被磁化，并且本身就像磁铁一样。

磁性继电器

磁继电器是一种开关或断路器，可以通过磁力激活到 “ON” 和 “OFF” 位置。一个例子是电话设备中使用的低功率簧片继电器，它由两个由小间隙隔开的扁平镍铁刀片组成（图 TF10-2）。叶片的形状使得在没有外力的情况下，它们保持分开且不连接（关闭位置）。刀片之间的电接触

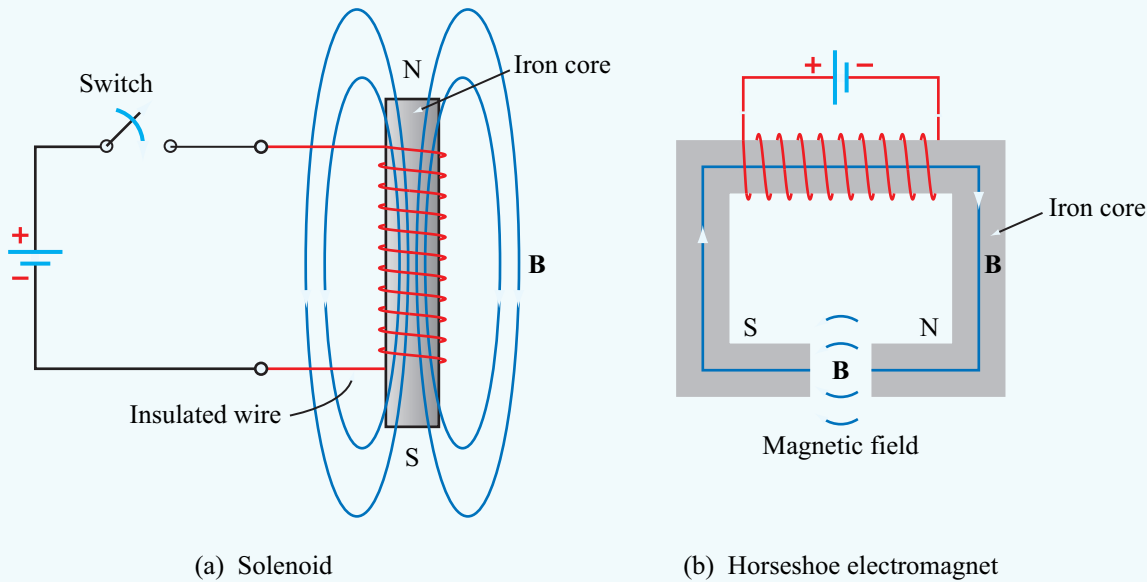
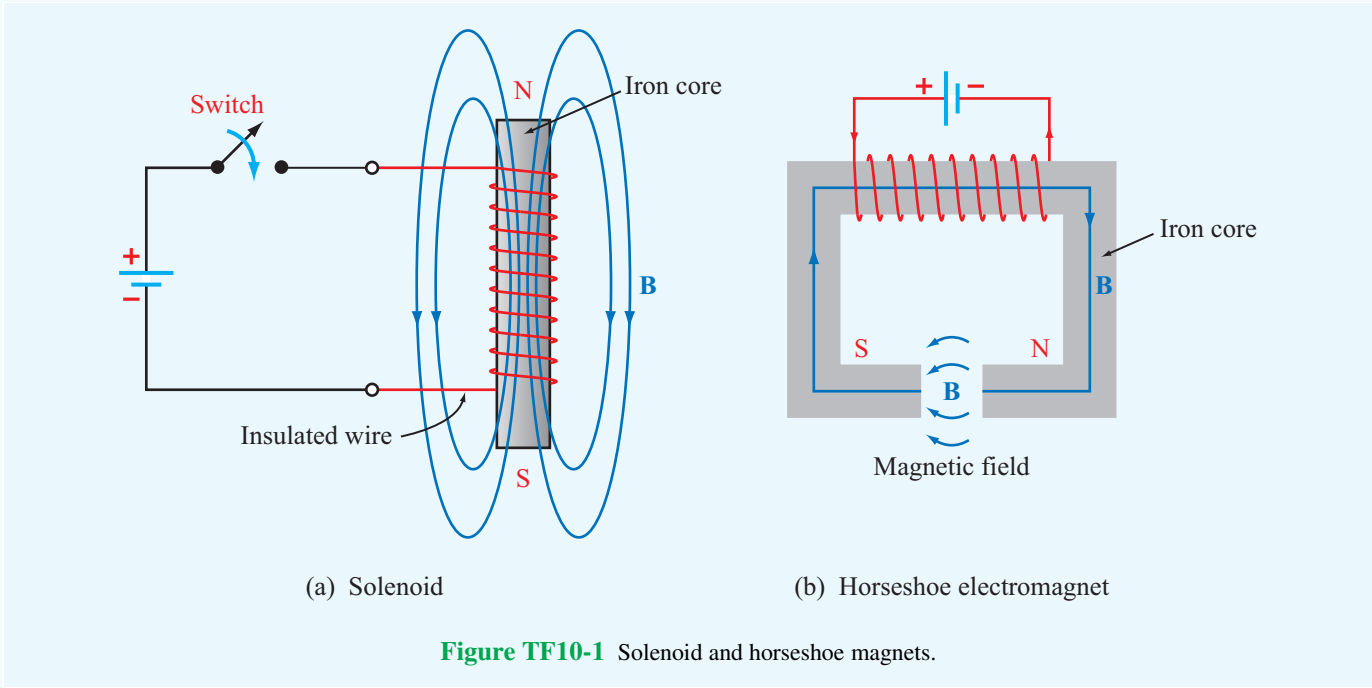
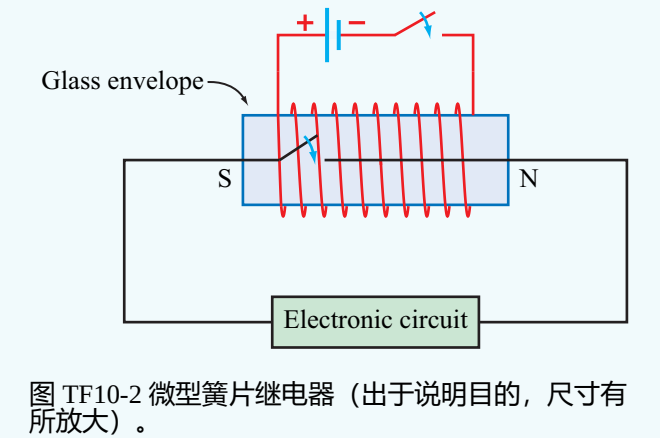


图 TF10-1 螺线管和马蹄形磁铁。

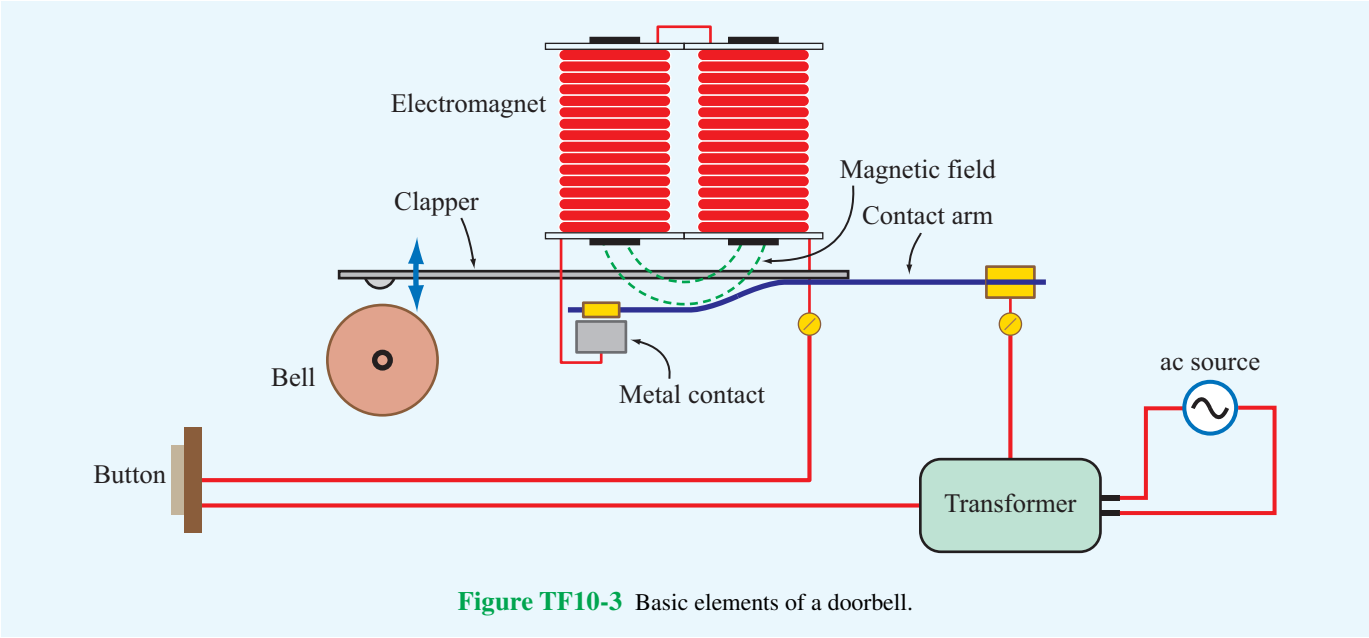
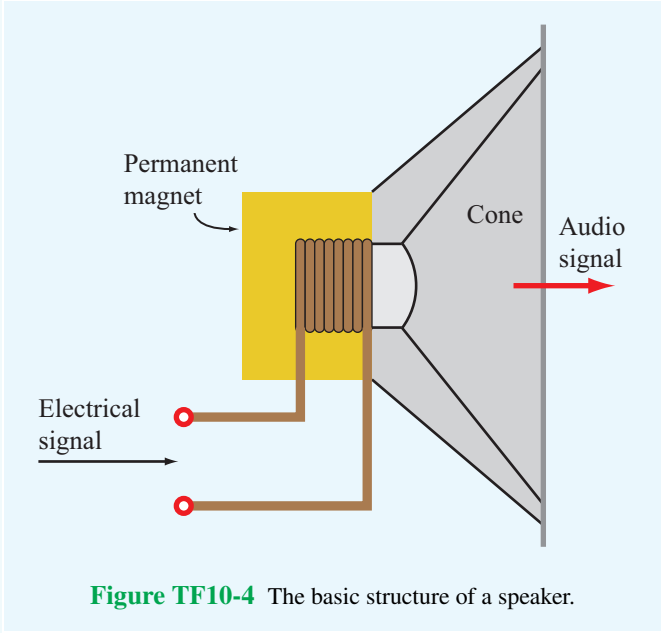
(ON position) is realized by applying a magnetic field along their length. The field, induced by a current flowing in the wire coiled around the glass envelope, causes the two blades to assume opposite magnetic polarities, thereby forcing them to attract together and close out the gap.

The Doorbell

In a doorbell circuit (Fig. TF10-3), the doorbell button is a switch; pushing and holding it down serves to connect the circuit to the household ac source through an appropriate **step-down transformer**. The current from the source flows through the electromagnet, via a contact arm with only one end anchored in place (and the other moveable), and onward to the switch. The magnetic field generated by the current flowing in the windings of the electromagnet pulls the unanchored end of the contact arm (which has an iron bar on it) closer in—in the direction of the electromagnet—thereby losing connection with the metal contact and severing current flow in the circuit. With no magnetic field to pull on the contact arm, it snaps back into its earlier position, re-establishing the current in the circuit. This back-and-forth cycle is repeated many times per second, so long as the doorbell button continues to be pushed down, and with every cycle, the clapper arm attached to the contact arm hits the metal bell and generates a ringing sound.

The Loudspeaker

By using a combination of a stationary, permanent magnet and a moveable electromagnet, the electromagnet/speaker-cone of the loudspeaker (Fig. TF10-4) can be made to move back and forth in response to the electrical



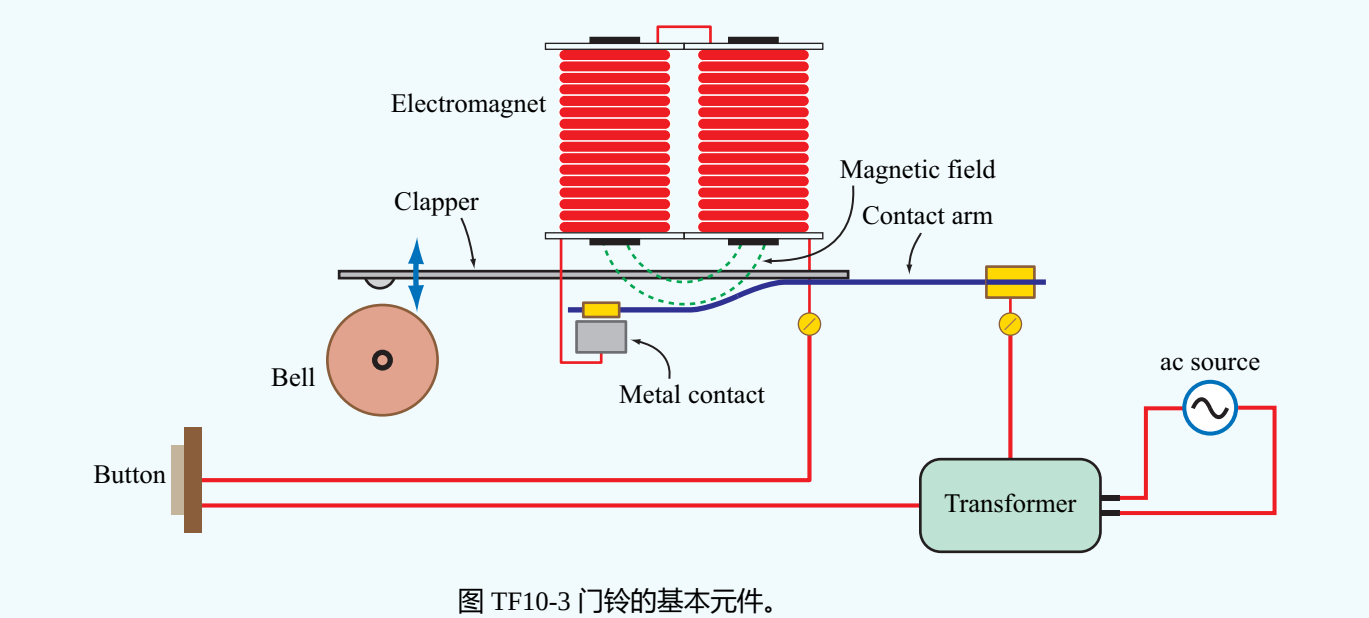
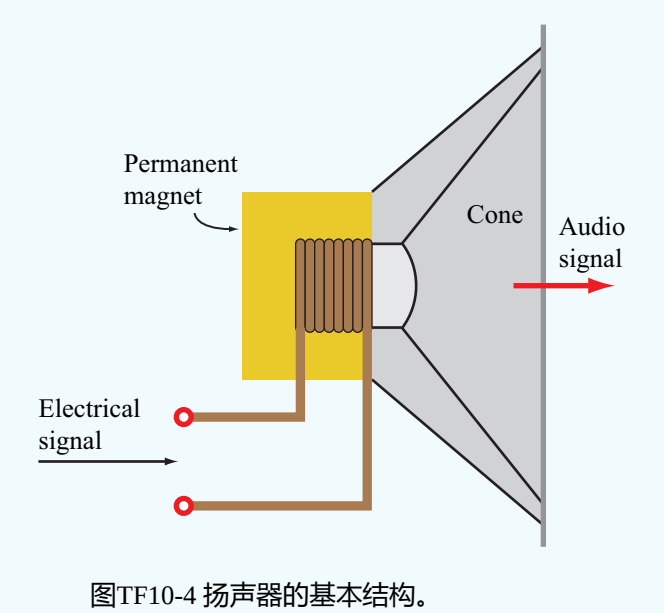
(ON 位置)是通过沿其长度施加磁场来实现的。由缠绕在玻璃外壳上的导线中流动的电流所产生的磁场,使两个叶片呈现相反的磁极性,从而迫使它们吸引在一起并关闭间隙。

门铃

在门铃电路中(图TF10-3),门铃按钮是一个开关;按下并按住它可以通过适当的降压变压器将电路连接到家用交流电源。来自源头的电流流过电磁体,通过只有一端固定到位(另一端可移动)的接触臂,然后流向开关。电磁体绕组中流动的电流产生的磁场将接触臂的非锚定端(上面有一个铁棒)拉近(沿电磁体的方向),从而失去与金属触点的连接并切断电路中的电流。由于没有磁场拉动接触臂,它会弹回原来的位置,从而在电路中重新建立电流。这种来回循环每秒重复多次,只要继续按下门铃按钮,每循环一次,连接到接触臂上的拍板臂就会撞击金属铃并产生铃声。

扩音器

通过使用固定的永磁体和可移动电磁体的组合,扬声器的电磁体/扬声器锥体(图TF10-4)可以根据电力来前后移动。



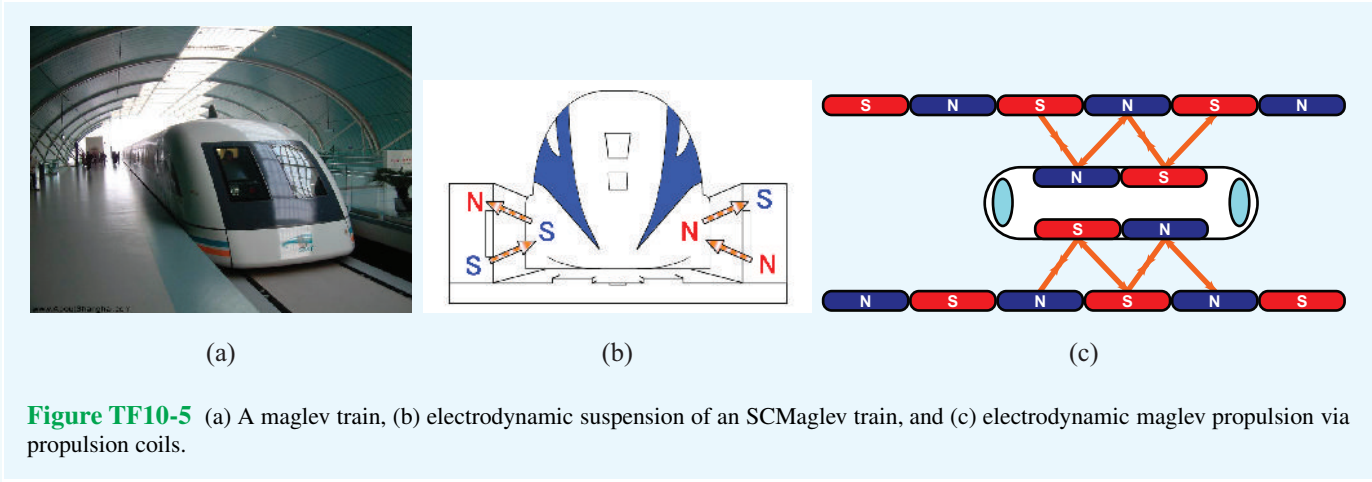


Figure TF10-5 (a) A maglev train, (b) electrodynamic suspension of an SCMaglev train, and (c) electrodynamic maglev propulsion via propulsion coils.

signal exciting the electromagnet. The vibrating movement of the cone generates sound waves with the same distribution of frequencies as contained in the spectrum of the electrical signal.

Magnetic Levitation

► Magnetically levitated trains (Fig. TF10-5(a)), called **maglevs** for short, can achieve speeds as high as 500 km/hr, primarily because there is no friction between the train and the track. ◀

The train basically floats at a height of 1 or more centimeters above the track, which is made possible by magnetic levitation (Fig. TF10-5(b)). The train carries superconducting electromagnets that induce currents in coils built into the guide rails alongside the train. The magnetic interaction between the train's superconducting electromagnets and the guide-rail coils serves not only to levitate the train but also to propel it along the track.

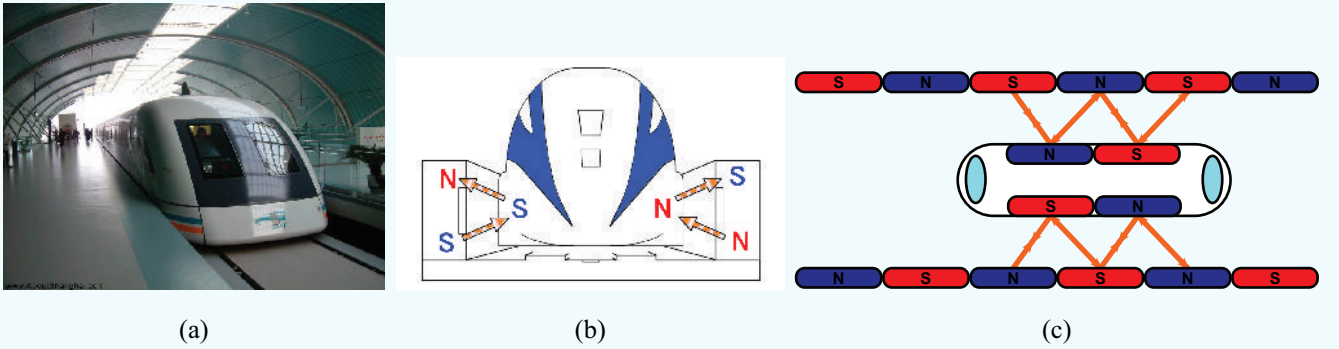


图 TF10-5 (a) 磁悬浮列车; (b) 超导磁悬浮列车的电动悬浮; (c) 利用推进线圈实现电动磁悬浮推进。

信号激励电磁铁。锥体的振动运动产生与电信号频谱中包含的频率分布相同的声波。

火车基本上漂浮在轨道上方 1 厘米或以上的高度，这是通过磁悬浮实现的（图 TF10-5 (b)）。火车装有超导体磁体，可在火车旁边内置于导轨中的线圈中感应电流。火车超导体磁体和导轨线圈之间的磁相互作用不仅可以使火车悬浮，还可以沿着轨道推动火车。

磁悬浮

► 磁悬浮列车（图TF10-5 (a)），简称磁悬浮，可以达到高达500公里/小时的速度，主要是因为列车与轨道之间没有摩擦。 ◀

5-4.1 **Vector Poisson’s Equation**

With $\mathbf{B} = \mu \mathbf{H}$, the differential form of Ampère’s law given by Eq. (5.46) can be written as

$$\nabla \times \mathbf{B} = \mu \mathbf{J}. \tag{5.54}$$

If we substitute Eq. (5.53) into Eq. (5.54), we obtain

$$\nabla \times (\nabla \times \mathbf{A}) = \mu \mathbf{J}. \tag{5.55}$$

For any vector $\mathbf{A}$, the Laplacian of $\mathbf{A}$ obeys the vector identity given by Eq. (3.113), that is,

$$\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times (\nabla \times \mathbf{A}), \tag{5.56}$$

where, by definition, $\nabla^2 \mathbf{A}$ in Cartesian coordinates is

$$\begin{aligned} \nabla^2 \mathbf{A} &= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \mathbf{A} \\ &= \hat{\mathbf{x}} \nabla^2 A_x + \hat{\mathbf{y}} \nabla^2 A_y + \hat{\mathbf{z}} \nabla^2 A_z. \end{aligned} \tag{5.57}$$

Combining Eq. (5.55) with Eq. (5.56) gives

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu \mathbf{J}. \tag{5.58}$$

This equation contains a term involving $\nabla \cdot \mathbf{A}$. It turns out that we have a fair amount of latitude in specifying a value or a mathematical form for $\nabla \cdot \mathbf{A}$ without conflicting with the requirement represented by Eq. (5.53). The simplest among these allowed restrictions on $\mathbf{A}$ is

$$\nabla \cdot \mathbf{A} = 0. \tag{5.59}$$

Using this choice in Eq. (5.58) leads to the ***vector Poisson’s equation***

$$\nabla^2 \mathbf{A} = -\mu \mathbf{J}. \tag{5.60}$$

Using the definition for $\nabla^2 \mathbf{A}$ given by Eq. (5.57), the vector Poisson’s equation can be decomposed into three scalar Poisson’s equations:

$$\nabla^2 A_x = -\mu J_x, \tag{5.61a}$$

$$\nabla^2 A_y = -\mu J_y, \tag{5.61b}$$

$$\nabla^2 A_z = -\mu J_z. \tag{5.61c}$$

In electrostatics, Poisson’s equation for the scalar potential V is given by Eq. (4.70) as

$$\nabla^2 V = -\frac{\rho_v}{\epsilon}, \tag{5.62}$$

and its solution for a volume charge distribution ρ_v occupying a volume $\mathcal{V}'$ was given by Eq. (4.71) as

$$V = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v}{R'} d\mathcal{V}'. \tag{5.63}$$

Poisson’s equations for A_x , A_y , and A_z are mathematically identical in form to Eq. (5.62). Hence, for a current density $\mathbf{J}$ with x component J_x distributed over a volume $\mathcal{V}'$, the solution for Eq. (5.61a) is

$$A_x = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{J_x}{R'} d\mathcal{V}' \quad (\text{Wb/m}). \tag{5.64}$$

Similar solutions can be written for A_y in terms of J_y and for A_z in terms of J_z . The three solutions can be combined into a vector equation:

$$\mathbf{A} = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\mathbf{J}}{R'} d\mathcal{V}' \quad (\text{Wb/m}). \tag{5.65}$$

In view of Eq. (5.23), if the current distribution is specified over a surface S' , then $\mathbf{J} d\mathcal{V}'$ should be replaced with $\mathbf{J}_s ds'$ and $\mathcal{V}'$ should be replaced with S' ; similarly, for a line distribution, $\mathbf{J} d\mathcal{V}'$ should be replaced with $I d\mathbf{l}'$, and the integration should be performed over the associated path l' .

► The vector magnetic potential provides a third approach for computing the magnetic field due to current-carrying conductors in addition to the methods suggested by the Biot–Savart and Ampère laws. ◀

For a specified current distribution, Eq. (5.65) can be used to find $\mathbf{A}$, and then Eq. (5.53) can be used to find $\mathbf{B}$. Except for simple current distributions with symmetrical geometries that lend themselves to the application of Ampère’s law, in practice we often use the approaches provided by the Biot–Savart law and the vector magnetic potential, and among these two, the latter often is more convenient to apply because it is easier to perform the integration in Eq. (5.65) than that in Eq. (5.22).

5-4.2 **Magnetic Flux**

The ***magnetic flux*** Φ linking a surface S is defined as the total magnetic flux density passing through it, or

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{Wb}). \tag{5.66}$$

If we insert Eq. (5.53) into Eq. (5.66) and then invoke Stokes’s theorem, we obtain

$$\Phi = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\mathbf{l} \quad (\text{Wb}), \tag{5.67}$$

5-4.1 矢量泊松方程

对于 $\mathbf{B} = \mu \mathbf{H}$, Ampère 定律的微分形式由方程式给出。(5.46) 可以写成

$$\nabla \times \mathbf{B} = \mu \mathbf{J}. \tag{5.54}$$

如果我们替换方程。(5.53) 代入等式。(5.54), 我们得到

$$\nabla \times (\nabla \times \mathbf{A}) = \mu \mathbf{J}. \tag{5.55}$$

对于任何向量 $\mathbf{A}$, $\mathbf{A}$ 的拉普拉斯算子服从式 (1) 给出的向量恒等式: (3.113), 即

$$\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times (\nabla \times \mathbf{A}), \tag{5.56}$$

其中, 根据定义, 笛卡尔坐标中的 $\nabla^2 \mathbf{A}$ 为

$$\begin{aligned} \nabla^2 \mathbf{A} &= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \mathbf{A} \\ &= \hat{\mathbf{x}} \nabla^2 A_x + \hat{\mathbf{y}} \nabla^2 A_y + \hat{\mathbf{z}} \nabla^2 A_z. \end{aligned} \tag{5.57}$$

结合方程。(5.55) 与等式。(5.56) 给出

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu \mathbf{J}. \tag{5.58}$$

这个方程包含一个涉及 $\nabla \cdot \mathbf{A}$ 的项。事实证明, 我们在为 $\nabla \cdot \mathbf{A}$ 指定一个值或数学形式方面有相当大的自由度, 而不与方程 1 所表示的要求相冲突。(5.53)。对 $\mathbf{A}$ 的这些允许的限制中最简单的是

$$\nabla \cdot \mathbf{A} = 0. \tag{5.59}$$

在方程式中使用这个选择。(5.58) 导出矢量泊松方程

$$\nabla^2 \mathbf{A} = -\mu \mathbf{J}. \tag{5.60}$$

使用方程式给出的 $\nabla^2 \mathbf{A}$ 的定义。(5.57) 式中, 矢量泊松方程可以分解为三个标量泊松方程:

$$\nabla^2 A_x = -\mu J_x, \tag{5.61a}$$

$$\nabla^2 A_y = -\mu J_y, \tag{5.61b}$$

$$\nabla^2 A_z = -\mu J_z. \tag{5.61c}$$

在静电学中, 标量势 V 的泊松方程由方程 1 给出。(4.70) 作为

$$\nabla^2 V = -\frac{\rho_v}{\epsilon}, \tag{5.62}$$

其占据体积 $\mathcal{V}'$ 的体积电荷分布 ρ_v 的解由式 (1) 给出。(4.71) 作为

$$V = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v}{R'} d\mathcal{V}'. \tag{5.63}$$

A_x 、 A_y 和 A_z 的泊松方程在数学上与方程 1 相同。(5.62)。因此, 对于 x 分量 J_x 分布在体积 $\mathcal{V}'$ 上的电流密度 $\mathbf{J}$, 方程 1 的解为: (5.61a) 是

$$A_x = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{J_x}{R'} d\mathcal{V}' \quad (\text{Wb/m}). \tag{5.64}$$

对于 A_y 可以根据 J_y 编写类似的解决方案, 对于 A_z 可以根据 J_z 编写类似的解决方案。这三个解可以组合成一个向量方程:

$$\mathbf{A} = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\mathbf{J}}{R'} d\mathcal{V}' \quad (\text{Wb/m}). \tag{5.65}$$

鉴于方程。(5.23), 如果在表面 S' 上指定电流分布, 则 $\mathbf{J} d\mathcal{V}'$ 应替换为 $\mathbf{J}_s ds'$, $\mathcal{V}'$ 应替换为 S' ; 类似地, 对于线分布, $\mathbf{J} d\mathcal{V}'$ 应替换为 $I d\mathbf{l}'$, 并且积分应在相关路径 l' 上执行。

► 除了 Biot-Savart 和 Amp 建议的方法之外, 矢量磁势提供了第三种计算载流导体产生的磁场的方法这是法律。◀

对于特定的电流分布, 方程。(5.65) 可用于求 $\mathbf{A}$, 然后可得方程 (5.65)。式 (5.53) 可以用来求 $\mathbf{B}$ 。除了具有对称几何形状的简单电流分布适合应用安培环路定律之外, 在实践中我们经常使用毕奥-萨伐尔定律和矢量磁势提供的方法, 而在这两者中, 后者通常更方便应用, 因为更容易执行式 (5.53) 中的积分。(5.65) 比等式中的值 (5.22) 。

5-4.2 磁通量

连接表面 S 的磁通量 Φ 定义为穿过该表面的总磁通量密度, 或者

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{Wb}). \tag{5.66}$$

如果我们插入方程。(5.53) 代入等式。(5.66) 然后调用斯托克斯定理, 我们得到

$$\Phi = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\mathbf{l} \quad (\text{WB}), \tag{5.67}$$

where C is the contour bounding the surface S . Thus, Φ can be determined by either Eq. (5.66) or Eq. (5.67), whichever is easier to integrate for the specific problem under consideration.

5-5 Magnetic Properties of Materials

Because of the similarity between the pattern of the magnetic field lines generated by a current loop and those exhibited by a permanent magnet, the loop can be regarded as a magnetic dipole with north and south poles (Section 5-2.2 and Fig. 5-13). The magnetic moment $\mathbf{m}$ of a loop of area A has magnitude $m = IA$ and a direction normal to the plane of the loop (in accordance with the right-hand rule). Magnetization in a material is due to atomic scale current loops associated with (1) orbital motions of the electrons and protons around and inside the nucleus and (2) electron spin. The magnetic moment due to proton motion typically is three orders of magnitude smaller than that of the electrons; therefore, the total orbital and spin magnetic moment of an atom is dominated by the sum of the magnetic moments of its electrons.

► The magnetic behavior of a material is governed by the interaction of the magnetic dipole moments of its atoms with an external magnetic field. The nature of the behavior depends on the crystalline structure of the material and is used as a basis for classifying materials as **diamagnetic**, **paramagnetic**, or **ferromagnetic**. ◀

The atoms of a diamagnetic material have no permanent magnetic moments. In contrast, both paramagnetic and ferromagnetic materials have atoms with permanent magnetic dipole moments, albeit with very different organizational structures.

5-5.1 Electron Orbital and Spin Magnetic Moments

This section presents a semiclassical, intuitive model of the atom, which provides quantitative insight into the origin of electron magnetic moments. An electron with charge of $-e$ moving at constant speed u in a circular orbit of radius r (Fig. 5-20(a)) completes one revolution in time $T = 2\pi r/u$. This circular motion of the electron constitutes a tiny loop with current I given by

$$I = -\frac{e}{T} = -\frac{eu}{2\pi r}. \quad (5.68)$$

The magnitude of the associated **orbital magnetic moment** $\mathbf{m}_o$ is

$$m_o = IA = \left(-\frac{eu}{2\pi r}\right)(\pi r^2) = -\frac{eur}{2} = -\left(\frac{e}{2m_e}\right)L_e, \quad (5.69)$$

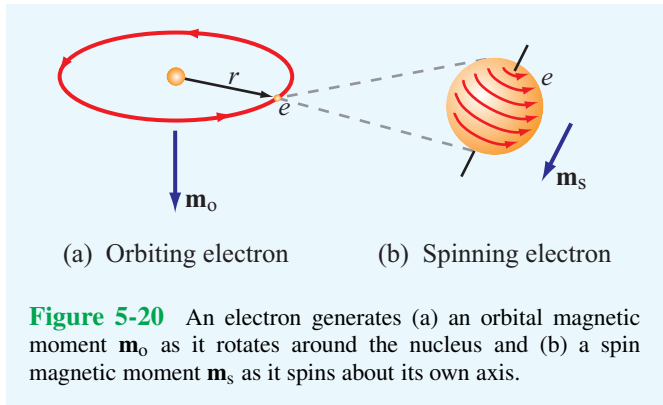


Figure 5-20 An electron generates (a) an orbital magnetic moment $\mathbf{m}_o$ as it rotates around the nucleus and (b) a spin magnetic moment $\mathbf{m}_s$ as it spins about its own axis.

where $L_e = m_e u r$ is the angular momentum of the electron and m_e is its mass. According to quantum physics, the orbital angular momentum is quantized; specifically, L_e is always some integer multiple of $\hbar = h/2\pi$, where h is Planck's constant. That is, $L_e = 0, \hbar, 2\hbar, \dots$. Consequently, the smallest nonzero magnitude of the orbital magnetic moment of an electron is

$$m_o = -\frac{e\hbar}{2m_e}. \quad (5.70)$$

Despite the fact that all materials contain electrons that exhibit magnetic dipole moments, most are effectively nonmagnetic. This is because, in the absence of an external magnetic field, the atoms of most materials are oriented *randomly*, as a result of which they exhibit a zero or very small net magnetic moment.

In addition to the magnetic moment due to its orbital motion, an electron has an intrinsic **spin magnetic moment** $\mathbf{m}_s$ due to its spinning motion about its own axis (Fig. 5-20(b)). The magnitude of $\mathbf{m}_s$ predicted by quantum theory is

$$m_s = -\frac{e\hbar}{2m_e}, \quad (5.71)$$

which is equal to the minimum orbital magnetic moment m_o . The electrons of an atom with an even number of electrons usually exist in pairs with the members of a pair having opposite spin directions, thereby canceling each others' spin magnetic moments. If the number of electrons is odd, the atom has a net nonzero spin magnetic moment due to its unpaired electron.

5-5.2 Magnetic Permeability

In Chapter 4, we learned that the relationship $\mathbf{D} = \epsilon_0 \mathbf{E}$, between the electric flux and field in free space, is modified to $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$ in a dielectric material. Likewise, the relationship

其中 C 是包围表面 S 的轮廓。因此, Φ 可以由等式 1 确定。(5.66) 或等式。(5.67), 对于所考虑的特定问题, 以更容易积分的方式为准。

5-5 材料的磁性

由于电流环产生的磁力线模式与永磁体产生的磁场线模式相似, 因此电流环可以被视为具有北极和南极的磁偶极子(第5-2.2节和图5-13)。面积为 A 的环路的磁矩 $\mathbf{m}$ 的大小为 $m = IA$, 方向垂直于环路平面(根据右手定则)。材料中的磁化是由于与(1)原子核周围和内部的电子和质子的轨道运动以及(2)电子自旋相关的原子级电流环引起的。质子运动产生的磁矩通常比电子的磁矩小三个数量级; 因此, 原子的总轨道磁矩和自旋磁矩由其电子磁矩之和决定。

► 材料的磁性行为由其原子的磁偶极矩与外部磁场的相互作用控制。行为的性质取决于材料的晶体结构, 并用作将材料分类为抗磁性、顺磁性或铁磁性的基础。◀

反磁性材料的原子没有永磁矩。相比之下, 顺磁和铁磁材料都具有具有永磁偶极矩的原子, 尽管组织结构非常不同。

5-5.1 电子轨道和自旋磁矩

本节介绍了一个半经典、直观的原子模型, 它提供了对电子磁矩起源的定量见解。电荷为 $-e$ 的电子在半径为 r 的圆形轨道上以恒定速度 u 运动(图 5-20 (a)), 在时间 $T = 2\pi r/u$ 内完成一圈。电子的圆周运动构成了一个微小的环路, 电流 I 由下式给出

$$I = -\frac{e}{T} = -\frac{eu}{2\pi r}. \quad (5.68)$$

相关轨道磁矩 $\mathbf{m}_o$ 的大小为

$$m_o = IA = \left(-\frac{eu}{2\pi r}\right)(\pi r^2) = -\frac{eur}{2} = -\left(\frac{e}{2m_e}\right)L_e, \quad (5.69)$$

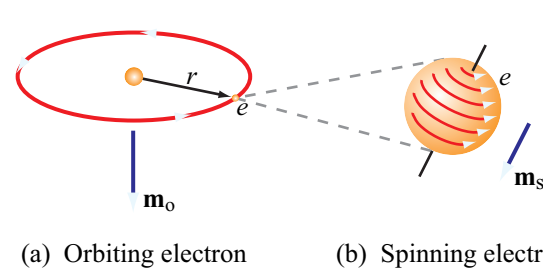


图 5-20 电子在绕原子核旋转时产生 (a) 轨道磁矩 $\mathbf{m}_o$ 和 (b) 在绕自身轴旋转时产生自旋磁矩 $\mathbf{m}_s$ 。

其中 $L_e = m_e u r$ 是电子的角动量, m_e 是电子的质量。根据量子物理学, 轨道角动量是量子化的; 具体来说, L_e 始终是 $\hbar = h/2\pi$ 的整数倍, 其中 h 是普朗克常数。即, $L_e = 0, \hbar, 2\hbar, \dots$ 。因此, 电子轨道磁矩的最小非零幅度为

$$m_o = -\frac{e\hbar}{2m_e}. \quad (5.70)$$

尽管所有材料都含有表现出磁偶极矩的电子, 但大多数材料实际上都是非磁性的。这是因为, 在没有外部磁场的情况下, 大多数材料的原子是随机取向的, 因此它们表现出零或非常小的净磁矩。

除了轨道运动产生的磁矩外, 电子还因其绕自身轴的自旋运动而具有固有的自旋磁矩 $\mathbf{m}_s$ (图 5-20 (b))。量子理论预测的 $\mathbf{m}_s$ 的大小为

$$m_s = -\frac{e\hbar}{2m_e}, \quad (5.71)$$

它等于最小轨道磁矩 m_o 。具有偶数个电子的原子中的电子通常与自旋方向相反的电子对中的成员成对存在, 从而抵消彼此的自旋磁矩。如果电子数量为奇数, 则原子由于其不成对电子而具有净非零自旋磁矩。

5-5.2 磁导率

在第 4 章中, 我们了解到自由空间中电通量与场之间的关系 $\mathbf{D} = \epsilon_0 \mathbf{E}$ 在介电材料中被修改为 $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$ 。同样, 关系

$\mathbf{B} = \mu_0 \mathbf{H}$ in free space is modified to

$$\mathbf{B} = \mu_0 \mathbf{H} + \mu_0 \mathbf{M} = \mu_0 (\mathbf{H} + \mathbf{M}), \quad (5.72)$$

where the *magnetization vector* $\mathbf{M}$ is defined as the vector sum of the magnetic dipole moments of the atoms contained in a unit volume of the material. Scale factors aside, the roles and interpretations of $\mathbf{B}$, $\mathbf{H}$, and $\mathbf{M}$ in Eq. (5.72) mirror those of $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{P}$ in Eq. (4.93). Moreover, just as in most dielectrics $\mathbf{P}$ and $\mathbf{E}$ are linearly related, in most magnetic materials

$$\mathbf{M} = \chi_m \mathbf{H}, \quad (5.73)$$

where χ_m is a dimensionless quantity called the *magnetic susceptibility* of the material. For diamagnetic and paramagnetic materials, χ_m is a (temperature-dependent) constant, resulting in a linear relationship between $\mathbf{M}$ and $\mathbf{H}$ at a given temperature. This is not the case for ferromagnetic substances; the relationship between $\mathbf{M}$ and $\mathbf{H}$ not only is nonlinear, but also depends on the “history” of the material, as explained in the next section.

Keeping this fact in mind, we can combine Eqs. (5.72) and (5.73) to get

$$\mathbf{B} = \mu_0 (\mathbf{H} + \chi_m \mathbf{H}) = \mu_0 (1 + \chi_m) \mathbf{H}, \quad (5.74)$$

or

$$\mathbf{B} = \mu \mathbf{H}, \quad (5.75)$$

where μ , the *magnetic permeability* of the material, relates to χ_m as

$$\mu = \mu_0 (1 + \chi_m) \quad (\text{H/m}). \quad (5.76)$$

Often it is convenient to define the magnetic properties of a material in terms of the *relative permeability* μ_r :

$$\mu_r = \frac{\mu}{\mu_0} = 1 + \chi_m. \quad (5.77)$$

A material usually is classified as diamagnetic, paramagnetic, or ferromagnetic on the basis of the value of its χ_m (Table 5-2). Diamagnetic materials have negative susceptibilities, whereas paramagnetic materials have positive ones. However, the absolute magnitude of χ_m is on the order of 10^{-5} for both classes of materials, which for most applications allows us to ignore χ_m relative to 1 in Eq. (5.77).

► Thus, $\mu_r \approx 1$ or $\mu \approx \mu_0$ for diamagnetic and paramagnetic substances, which include dielectric materials and most metals. In contrast, $|\mu_r| \gg 1$ for ferromagnetic materials; $|\mu_r|$ of purified iron, for example, is on the order of 2×10^5 . ◀

Ferromagnetic materials are discussed next.

Exercise 5-12: The magnetic vector $\mathbf{M}$ is the vector sum of the magnetic moments of all the atoms contained in a unit volume (1m^3). If a certain type of iron with 8.5×10^{28} atoms/ m^3 contributes one electron per atom to align its spin magnetic moment along the direction of the applied field, find (a) the spin magnetic moment of a single electron given that $m_e = 9.1 \times 10^{-31}$ (kg) and $\hbar = 1.06 \times 10^{-34}$ (J·s) and (b) the magnitude of $\mathbf{M}$.

Answer: (a) $m_s = 9.3 \times 10^{-24}$ (A·m²), (b) $M = 7.9 \times 10^5$ (A/m). (See ◀.)

5-5.3 Magnetic Hysteresis of Ferromagnetic Materials

Ferromagnetic materials, which include iron, nickel, and cobalt, exhibit unique magnetic properties due to the fact that their magnetic moments tend to readily align along the direction of an external magnetic field. Moreover, such materials remain partially magnetized even after the external field is removed. Because of these peculiar properties, ferromagnetic materials are used in the fabrication of permanent magnets.

A key to understanding the properties of ferromagnetic materials is the notion of *magnetized domains*, that are microscopic regions (on the order of 10^{-10}m^3) within which the magnetic moments of all atoms (typically on the order of 10^{19} atoms) are permanently aligned with each other. This alignment, which occurs in all ferromagnetic materials, is due to strong coupling forces between the magnetic dipole moments constituting an individual domain. In the absence of an external magnetic field, the domains take on random orientations relative to each other (Fig. 5-21(a)), resulting in zero net magnetization. The *domain walls* forming the boundaries between adjacent domains consist of thin transition regions. When an unmagnetized sample of a ferromagnetic material is placed in an external magnetic field, the domains partially align with the external field, as illustrated in Fig. 5-21(b). A quantitative understanding of how the domains form and how they behave under the influence of an external magnetic field requires a heavy dose of quantum mechanics and is outside the scope of the present treatment. Hence, we confine our discussion to a qualitative description of the magnetization process and its implications.

The magnetization behavior of a ferromagnetic material is described in terms of its *B–H magnetization curve*, where B and H refer to the amplitudes of the $\mathbf{B}$ flux and $\mathbf{H}$ field in the material. Suppose that we start with an unmagnetized sample of iron, denoted by point O in Fig. 5-22. When we increase $\mathbf{H}$ continuously, for example, by increasing the current passing

自由空间中的 $\mathbf{B} = \mu_0 \mathbf{H}$ 被修改为

$$\mathbf{B} = \mu_0 \mathbf{H} + \mu_0 \mathbf{M} = \mu_0 (\mathbf{H} + \mathbf{M}), \quad (5.72)$$

其中磁化矢量 $\mathbf{M}$ 定义为材料单位体积中所含原子磁偶极矩的矢量和。撇开比例因素不谈，方程中 $\mathbf{B}$ 、 $\mathbf{H}$ 和 $\mathbf{M}$ 的作用和解释。(5.72)反映了方程式中的 $\mathbf{D}$ 、 $\mathbf{E}$ 和 $\mathbf{P}$ 。(4.93)。此外，正如在大多数电介质中 $\mathbf{P}$ 和 $\mathbf{E}$ 呈线性相关一样，在大多数磁性材料中

$$\mathbf{M} = \chi_m \mathbf{H}, \quad (5.73)$$

其中 χ_m 是无量纲量，称为材料的磁化率。对于抗磁性和顺磁性材料， χ_m 是一个（与温度相关的）常数，导致给定温度下 $\mathbf{M}$ 和 $\mathbf{H}$ 之间呈线性关系。铁磁物质则不然。 $\mathbf{M}$ 和 $\mathbf{H}$ 之间的关系不仅是非线性的，而且还取决于材料的“历史”，如下一节所述。

记住这一事实，我们可以结合等式。(5.72)和(5.73)得到

$$\mathbf{B} = \mu_0 (\mathbf{H} + \chi_m \mathbf{H}) = \mu_0 (1 + \chi_m) \mathbf{H}, \quad (5.74)$$

or

$$\mathbf{B} = \mu \mathbf{H}, \quad (5.75)$$

其中 μ （材料的磁导率）与 χ_m 的关系为

$$\mu = \mu_0 (1 + \chi_m) \quad (\text{H/m}). \quad (5.76)$$

通常，用相对磁导率 μ_r 来定义材料的磁性是很方便的：

$$\mu_r = \frac{\mu}{\mu_0} = 1 + \chi_m. \quad (5.77)$$

材料通常根据其 χ_m (表 5-2) 的值分为抗磁性、顺磁性或铁磁性。抗磁材料具有负磁化率，而顺磁材料具有正磁化率。然而，对于两类材料， χ_m 的绝对量值约为 10^{-5} ，这对于大多数应用来说允许我们忽略等式 1 中相对于 1 的 χ_m 。(5.77)。

► 因此， $\mu_r \approx 1$ 或 $\mu \approx \mu_0$ 用于抗磁性和顺磁性物质，其中包括介电材料和大多数金属。相反，对于铁磁材料， $|\mu_r| \gg 1$ ；例如，精铁的 $|\mu_r|$ 约为 2×10^5 。◀

接下来讨论铁磁材料。

练习5-12：磁矢量 $\mathbf{M}$ 是单位体积（ 1m^3 ）中包含的所有原子的磁矩矢量和。如果某种类型的铁具有 8.5×10^{28} 原子/ m^3 ，每个原子贡献一个电子以沿施加场的方向排列其自旋磁矩，则在 $m_e = 9.1 \times 10^{-31}$ (kg) 和 $\hbar = 1.06$ 的情况下，求 (a) 单个电子的自旋磁矩 $\times 10^{-34}$ (J·s) 和 (b) $\mathbf{M}$ 的大小。

答案：(a) $m_s = 9.3 \times 10^{-24}$ (A·m²), (b) $M = 7.9 \times 10^5$ (A/m)。(参见^{EM}。)

5-5.3 铁磁材料的磁滞现象

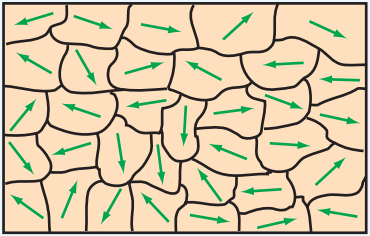
铁磁材料（包括铁、镍和钴）表现出独特的磁性，因为它们的磁矩很容易沿着外部磁场的方向排列。此外，即使在外部磁场移除后，此类材料仍保持部分磁化。由于这些特殊的性质，铁磁材料被用于制造永磁体。

理解铁磁材料特性的关键是磁畴的概念，磁畴是微观区域（大约为 10^{-10}m^3 ），其中所有原子（通常大约为 10^{19} 原子）的磁矩彼此永久对齐。这种排列出现在所有铁磁材料中，是由于构成单个磁域的磁偶极矩之间的强耦合力造成的。在没有外部磁场的情况下，磁畴相对于彼此呈现随机方向（图5-21（a）），导致净磁化强度为零。形成相邻域之间的边界的域壁由薄过渡区域组成。当铁磁材料的未磁化样品放置在外部磁场中时，磁畴部分与外部磁场对齐，如图 5-21(b) 所示。对磁畴如何形成以及它们在外部磁场影响下如何表现的定量理解需要大量的量子力学，并且超出了当前处理的范围。因此，我们的讨论仅限于磁化过程及其含义的定性描述。

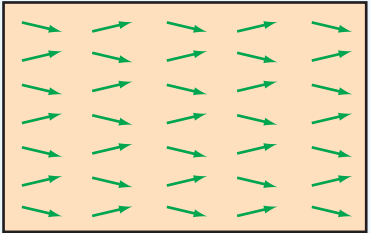
铁磁材料的磁化行为用其 $\mathbf{B}$ - $\mathbf{H}$ 磁化曲线来描述，其中 $\mathbf{B}$ 和 $\mathbf{H}$ 是指材料中 $\mathbf{B}$ 磁通和 $\mathbf{H}$ 场的振幅。假设我们从未磁化的铁样品开始，如图 5-22 中的 O 点所示。当我们不断增加 $\mathbf{H}$ 时，例如，通过增加通过的电流

Table 5-2 Properties of magnetic materials.

	Diamagnetism	Paramagnetism	Ferromagnetism
Permanent magnetic dipole moment	No	Yes, but weak	Yes, and strong
Primary magnetization mechanism	Electron orbital magnetic moment	Electron spin magnetic moment	Magnetized domains
Direction of induced magnetic field (relative to external field)	Opposite	Same	Hysteresis [see Fig. 5-22]
Common substances	Bismuth, copper, diamond, gold, lead, mercury, silver, silicon	Aluminum, calcium, chromium, magnesium, niobium, platinum, tungsten	Iron, nickel, cobalt
Typical value of χ_m Typical value of μ_r	$\approx -10^{-5}$ ≈ 1	$\approx 10^{-5}$ ≈ 1	$ \chi_m \gg 1$ and hysteretic $ \mu_r \gg 1$ and hysteretic



(a) Unmagnetized domains



(b) Magnetized domains

Figure 5-21 Comparison of (a) unmagnetized and (b) magnetized domains in a ferromagnetic material.

through a wire wound around the sample, **B** increases also along the *B–H* curve from point *O* to point *A*₁, at which nearly all the domains have become aligned with **H**. Point *A*₁

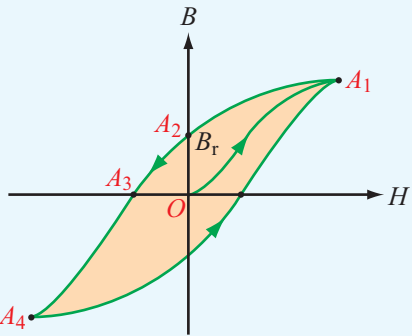
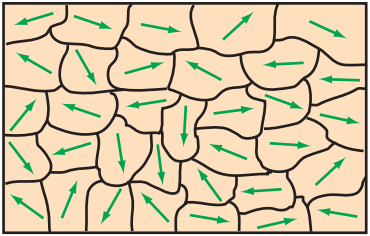


Figure 5-22 Typical hysteresis curve for a ferromagnetic material.

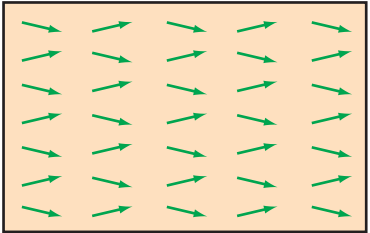
represents a saturation condition. If we then decrease **H** from its value at point *A*₁ back to zero (by reducing the current through the wire), the magnetization curve follows the path from *A*₁ to *A*₂. At point *A*₂, the external field **H** is zero (owing to the fact that the current through the wire is zero), but the flux density **B** in the material is not. The magnitude of **B** at *A*₂ is called the *residual flux density* *B_r*. The iron material is now magnetized and ready to be used as a permanent magnet owing to the fact that a large fraction of its magnetized domains have remained aligned. Reversing the direction of **H** and increasing its intensity causes **B** to decrease from *B_r* at point *A*₂ to zero at point *A*₃, and if the intensity of **H** is increased further

表5-2 磁性材料的性质

磁性特征	抗磁性	顺磁性	铁磁性
永久磁偶极矩	无	有，较弱	有，较强
主要磁化机制	电子轨道磁矩	电子自旋磁矩	磁畴的磁化
感应磁场相对外场的方向	相反	相同	有磁滞，见图5-22
常见物质	铋、铜、金刚石、金、铅、汞、银、硅	铝、钙、铬、镁、铌、铂、钨	铁、镍、钴
χ_m 的典型值	约 -10^{-5}	约 10^{-5}	$ \chi_m $ 远大于1，有磁滞
μ_r 的典型值	约1	约1	$ \mu_r $ 远大于1，有磁滞



(a) 未磁化状态下的磁畴



(b) 磁化状态下的磁畴

图5-21 铁磁材料在 (a) 未磁化状态与 (b) 磁化状态下的磁畴。

通过缠绕在样品上的导线，**B** 也沿着 *B–H* 曲线从 *O* 点到 *A*₁ 点增加，在该点几乎所有域都与 **H** 对齐。 *A*₁ 点

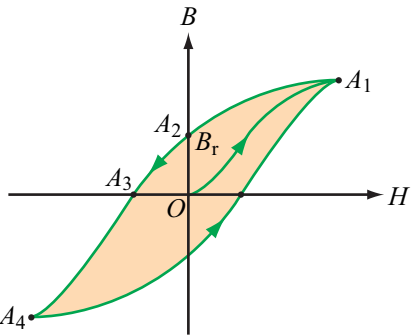


图 5-22 铁磁材料的典型磁滞曲线。

表示饱和条件。如果我们随后将 **H** 从 *A*₁ 点的值减小到零（通过减少通过导线的电流），磁化曲线将遵循从 *A*₁ 到 *A*₂ 的路径。在点 *A*₂ 处，外场 **H** 为零（因为通过导线的电流为零），但材料中的磁通密度 **B** 并非为零。 *A*₂ 处 **B** 的大小称为剩余磁通密度 *B_r*。由于大部分磁畴 保持对齐，铁材料现在已被磁化，可以用作永磁体。反转 **H** 的方向并增加其强度会导致 **B** 从 *A*₂ 点处的 *B_r* 减少到 *A*₃ 点处的零，并且如果 **H** 的强度进一步增加

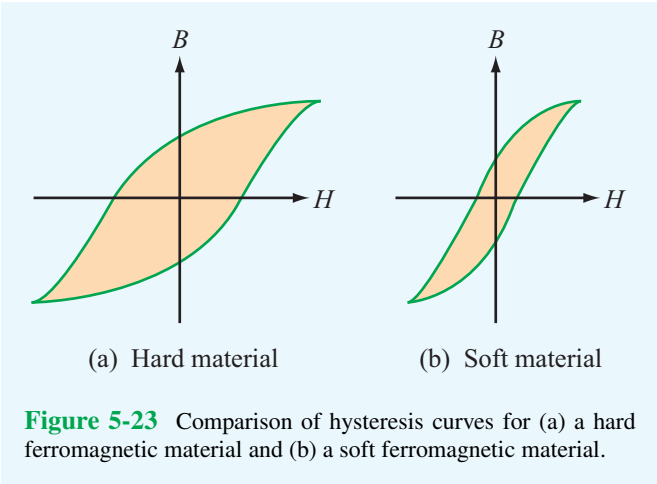


Figure 5-23 Comparison of hysteresis curves for (a) a hard ferromagnetic material and (b) a soft ferromagnetic material.

while maintaining its direction, the magnetization moves to the saturation condition at point A_4 . Finally, as $\mathbf{H}$ is made to return to zero and is then increased again in the positive direction, the curve follows the path from A_4 to A_1 . This process is called **magnetic hysteresis**. Hysteresis means “lag behind.” The existence of a **hysteresis loop** implies that the magnetization process in ferromagnetic materials depends not only on the magnetic field $\mathbf{H}$ but also on the magnetic history of the material. The shape and extent of the hysteresis loop depend on the properties of the ferromagnetic material and the peak-to-peak range over which $\mathbf{H}$ is made to vary. **Hard ferromagnetic materials** are characterized by wide hysteresis loops (**Fig. 5-23(a)**). They cannot be easily demagnetized by an external magnetic field because they have a large residual magnetization B_r . Hard ferromagnetic materials are used in the fabrication of permanent magnets for motors and generators. **Soft ferromagnetic materials** have narrow hysteresis loops (**Fig. 5-23(b)**); hence, they can be more easily magnetized and demagnetized. To demagnetize any ferromagnetic material, the material is subjected to several hysteresis cycles while gradually decreasing the peak-to-peak range of the applied field.

Concept Question 5-12: What are the three types of magnetic materials, and what are typical values of their relative permeabilities?

Concept Question 5-13: What causes magnetic hysteresis in ferromagnetic materials?

Concept Question 5-14: What does a magnetization curve describe? What is the difference between the magnetization curves of hard and soft ferromagnetic materials?

5-6 Magnetic Boundary Conditions

In Chapter 4, we derived a set of boundary conditions that describes how, at the boundary between two dissimilar contiguous media, the electric flux and field $\mathbf{D}$ and $\mathbf{E}$ in the first medium relate to those in the second medium. We now derive a similar set of boundary conditions for the magnetic flux and field $\mathbf{B}$ and $\mathbf{H}$. By applying Gauss’s law to a pill box that straddles the boundary, we determined that the difference between the normal components of the electric flux densities in two media equals the surface charge density ρ_s . That is,

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q \quad \rightarrow \quad D_{1n} - D_{2n} = \rho_s. \quad (5.78)$$

By analogy, application of Gauss’s law for magnetism, as expressed by Eq. (5.44), leads to the conclusion that

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \quad \rightarrow \quad B_{1n} = B_{2n}. \quad (5.79)$$

▶ Thus, the normal component of $\mathbf{B}$ is continuous across the boundary between two adjacent media. ◀

Because $\mathbf{B}_1 = \mu_1 \mathbf{H}_1$ and $\mathbf{B}_2 = \mu_2 \mathbf{H}_2$ for linear, isotropic media, the boundary condition for $\mathbf{H}$ corresponding to Eq. (5.79) is

$$\mu_1 H_{1n} = \mu_2 H_{2n}. \quad (5.80)$$

Comparison of Eqs. (5.78) and (5.79) reveals a striking difference between the behavior of the magnetic and electric fluxes across a boundary: *Whereas the normal component of $\mathbf{B}$ is continuous across the boundary, the normal component of $\mathbf{D}$ is not (unless $\rho_s = 0$).* The reverse applies to the tangential components of the electric and magnetic fields $\mathbf{E}$ and $\mathbf{H}$: *Whereas the tangential component of $\mathbf{E}$ is continuous across the boundary, the tangential component of $\mathbf{H}$ is not (unless the surface current density $\mathbf{J}_s = 0$).* To obtain the boundary condition for the tangential component of $\mathbf{H}$, we follow the same basic procedure used in Chapter 4 to establish the boundary condition for the tangential component of $\mathbf{E}$. With reference to **Fig. 5-24**, we apply Ampère’s law (Eq. (5.47)) to

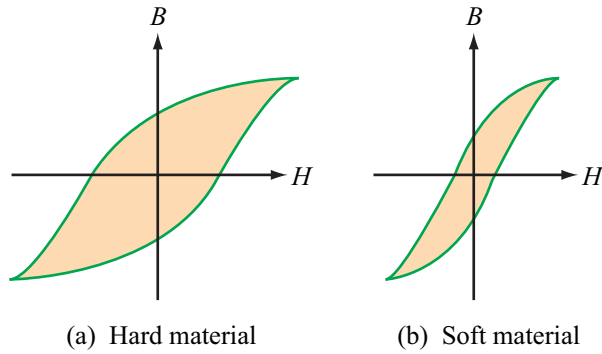


图 5-23 (a) 硬铁磁材料和 (b) 软铁磁材料的磁滞曲线比较。

在保持其方向的同时，磁化强度在 A_4 点处达到饱和状态。最后，随着 H 回到零，然后再次向正方向增加，曲线遵循从 A_4 到 A_1 的路径。这个过程称为磁滞。滞后意味着“落后”。磁滞回线的存在意味着铁磁材料的磁化过程不仅取决于磁场 H ，还取决于材料的磁历史。磁滞回线的形状和范围取决于铁磁材料的特性以及 H 变化的峰值范围。硬铁磁材料的特点是具有宽磁滞回线（图5-23 (a)）。它们不易被外部磁场消磁，因为它们具有较大的剩余磁化强度 B_r 。硬铁磁材料用于制造电动机和发电机的永磁体。软铁磁材料具有窄磁滞回线（图5-23 (b)）；因此，它们更容易被磁化和消磁。为了对任何铁磁材料进行消磁，该材料会经历几个磁滞循环，同时逐渐减小所施加磁场的峰峰值范围。

概念问题 5-12：磁性材料有哪三种类型？它们的相对磁导率的典型值是多少？

概念问题 5-13：铁磁材料中产生磁滞现象的原因是什么？

概念问题 5-14：磁化曲线描述什么？硬铁磁材料和软铁磁材料的磁化曲线有什么区别？

5-6 磁边界条件

在第四章中，我们导出了一组边界条件，描述了在两种不同的相邻介质之间的边界处，第一种介质中的电通量以及场 $\mathbf{D}$ 和 $\mathbf{E}$ 与第二种介质中的电通量和场 $\mathbf{D}$ 和 $\mathbf{E}$ 之间的关系。现在，我们推导出磁通量以及磁场 $\mathbf{B}$ 和 $\mathbf{H}$ 的一组类似的边界条件。通过将高斯定律应用于跨越边界的药丸盒，我们确定两种介质中电通量密度的法向分量之间的差等于表面电荷密度 ρ_s 。那是，

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q \quad \rightarrow \quad D_{1n} - D_{2n} = \rho_s. \quad (5.78)$$

以此类推，应用高斯磁定律，如方程所示。 (5.44) ，得出的结论是

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \quad \rightarrow \quad B_{1n} = B_{2n}. \quad (5.79)$$

▶ 因此， $\mathbf{B}$ 的法向分量在两个相邻介质之间的边界上是连续的。 ◀

由于 $\mathbf{B}_1 = \mu_1 \mathbf{H}_1$ 和 $\mathbf{B}_2 = \mu_2 \mathbf{H}_2$ 为线性各向同性介质，因此 $\mathbf{H}$ 的边界条件对应于等式： (5.79) 是

$$\mu_1 H_{1n} = \mu_2 H_{2n}. \quad (5.80)$$

等式的比较(5.78) 和 (5.79) 揭示了跨越边界的磁通量和电通量行为之间的显著差异：虽然 $\mathbf{B}$ 的法向分量在边界上是连续的，但 $\mathbf{D}$ 的法向分量却不是（除非 $\rho_s = 0$ ）。相反的情况适用于电场和磁场 $\mathbf{E}$ 和 $\mathbf{H}$ 的切向分量：虽然 $\mathbf{E}$ 的切向分量在边界上是连续的，但 $\mathbf{H}$ 的切向分量却不是（除非表面电流密度 $\mathbf{J}_s = 0$ ）。为了获得 $\mathbf{H}$ 切向分量的边界条件，我们遵循第 4 章中建立 $\mathbf{E}$ 切向分量边界条件的相同基本程序。参考图 5-24，我们应用 Ampère 定律（方程 (5.47) ）

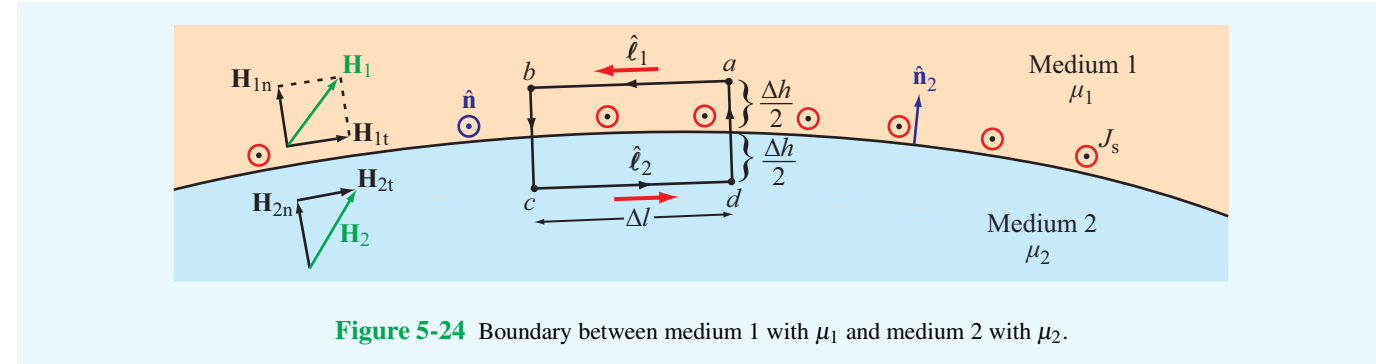


Figure 5-24 Boundary between medium 1 with μ_1 and medium 2 with μ_2 .

a closed rectangular path with sides of lengths Δl and Δh , and then let $\Delta h \rightarrow 0$ to obtain

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_a^b \mathbf{H}_1 \cdot \hat{\ell}_1 d\ell + \int_c^d \mathbf{H}_2 \cdot \hat{\ell}_2 d\ell = I, \quad (5.81)$$

where I is the net current crossing the surface of the loop in the direction specified by the right-hand rule (I is in the direction of the thumb when the fingers of the right hand extend in the direction of the loop C). As we let Δh of the loop approach zero, the surface of the loop approaches a thin line of length Δl . The total current flowing through this thin line is $I = J_s \Delta l$, where J_s is the magnitude of the component of the surface current density $\mathbf{J}_s$ normal to the loop. That is, $J_s = \mathbf{J}_s \cdot \hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the normal to the loop. In view of these considerations, Eq. (5.81) becomes

$$(\mathbf{H}_1 - \mathbf{H}_2) \cdot \hat{\ell}_1 \Delta l = \mathbf{J}_s \cdot \hat{\mathbf{n}} \Delta l. \quad (5.82)$$

The vector $\hat{\ell}_1$ can be expressed as $\hat{\ell}_1 = \hat{\mathbf{n}} \times \hat{\mathbf{n}}_2$, where $\hat{\mathbf{n}}$ and $\hat{\mathbf{n}}_2$ are the normals to the loop and to the surface of medium 2 (Fig. 5-24), respectively. Using this relation in Eq. (5.82) and then applying the vector identity $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A})$ leads to

$$\hat{\mathbf{n}} \cdot [\hat{\mathbf{n}}_2 \times (\mathbf{H}_1 - \mathbf{H}_2)] = \mathbf{J}_s \cdot \hat{\mathbf{n}}. \quad (5.83)$$

Since Eq. (5.83) is valid for any $\hat{\mathbf{n}}$, it follows that

$$\hat{\mathbf{n}}_2 \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s. \quad (5.84)$$

This equation implies that the tangential components of $\mathbf{H}$ parallel to $\mathbf{J}_s$ are continuous across the interface, whereas those orthogonal to $\mathbf{J}_s$ are discontinuous in the amount of $\mathbf{J}_s$.

Surface currents can exist only on the surfaces of perfect conductors and superconductors. Hence, *at the interface between media with finite conductivities*, $\mathbf{J}_s = 0$ and

$$H_{1t} = H_{2t}. \quad (5.85)$$

Example 5-7: Tilted-Plane Boundary

In Fig. 5-25 the plane defined by $y + 2x = 2$ separates medium 1 of permeability μ_1 from medium 2 of permeability μ_2 . If no surface current exists on the boundary and

$$\mathbf{B}_1 = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 \quad (\text{T}),$$

find $\mathbf{B}_2$ and then evaluate the result for $\mu_2 = 2\mu_1$.

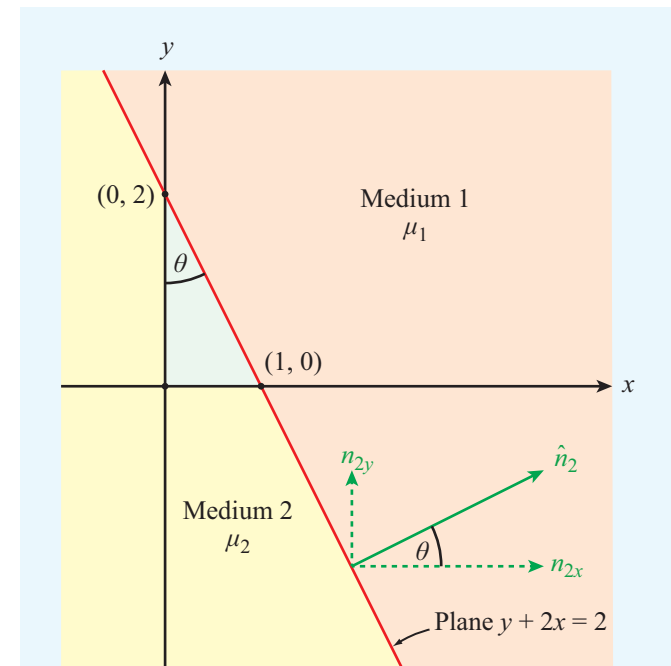


Figure 5-25 Magnetic media separated by the plane $y + 2x = 2$ (Example 5-7).

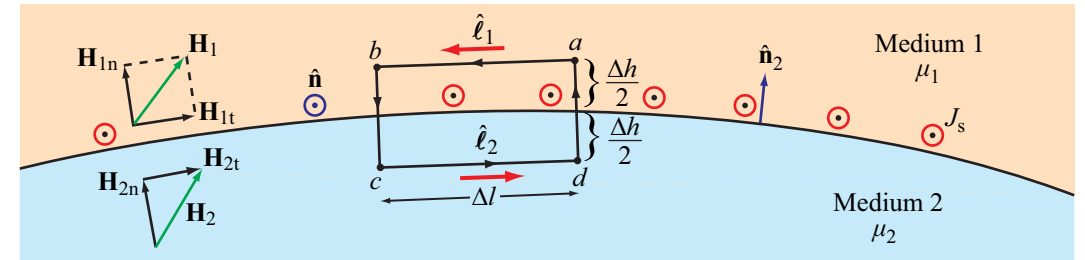


图 5-24 介质 1 与介质 1 之间的边界 μ_1 和中 2W 它 μ_2 。

边长为 Δl 和 Δh 的闭合矩形路径，然后令 $\Delta h \rightarrow 0$ 得到

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_a^b \mathbf{H}_1 \cdot \hat{\ell}_1 d\ell + \int_c^d \mathbf{H}_2 \cdot \hat{\ell}_2 d\ell = I, \quad (5.81)$$

其中 I 是按右手定则指定的方向穿过环路表面的净电流（当右手手指沿环路 C 的方向延伸时， I 为拇指方向）。当我们让环路的 Δh 接近零时，环路表面接近一条长度为 Δl 的细线。流过这条细线的总电流为 $I = J_s \Delta l$ ，其中 J_s 是垂直于环路的表面电流密度 J_s 分量的大小。即 $J_s = \mathbf{J}_s \cdot \hat{\mathbf{n}}$ ，其中 $\hat{\mathbf{n}}$ 是循环的法线。考虑到这些考虑，等式 (5.81) 变为

$$(\mathbf{H}_1 - \mathbf{H}_2) \cdot \hat{\ell}_1 \Delta l = \mathbf{J}_s \cdot \hat{\mathbf{n}} \Delta l. \quad (5.82)$$

矢量 $\hat{\ell}_1$ 可以表示为 $\hat{\ell}_1 = \hat{\mathbf{n}} \times \hat{\mathbf{n}}_2$ ，其中 $\hat{\mathbf{n}}$ 和 $\hat{\mathbf{n}}_2$ 分别是环路和介质2表面的法线（图5-24）。在等式中使用这个关系。(5.82) 然后应用向量恒等式 $(\mathbf{B} \times \mathbf{C}) \cdot \mathbf{A} = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A})$ 导致

$$\hat{\mathbf{n}} \cdot [\hat{\mathbf{n}}_2 \times (\mathbf{H}_1 - \mathbf{H}_2)] = \mathbf{J}_s \cdot \hat{\mathbf{n}}. \quad (5.83)$$

由于方程 (5.83) 对于任何 $\hat{\mathbf{n}}$ 都有效，因此

$$\hat{\mathbf{n}}_2 \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s. \quad (5.84)$$

该方程意味着平行于 J_s 的 H 切向分量在界面上是连续的，而正交于 J_s 的切向分量在 J_s 的量上是不连续的。

表面电流只能存在于理想导体和超导体的表面。因此，在具有有限电导率的介质之间的界面处， $J_s = 0$ 和

$$H_{1t} = H_{2t}. \quad (5.85)$$

示例 5-7：倾斜平面边界

在图 5-25 中，由 $y + 2x = 2$ 定义的平面将渗透率 μ_1 的介质 1 与渗透率 μ_2 的介质 2 分开。如果边界上不存在表面电流且

$$\mathbf{B}_1 = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 \quad (\text{T}),$$

找到 $\mathbf{B}_2$ ，然后计算 $\mu_2 = 2\mu_1$ 的结果。

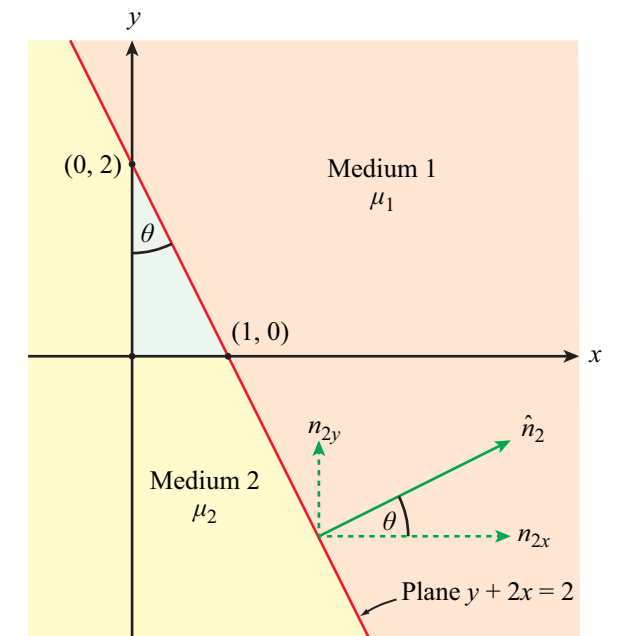


图 5-25 由平面 $y + 2x = 2$ 分隔的磁性介质（示例 5-7）。

Solution: To apply the boundary conditions at the plane between the two media, we first need to obtain an expression for the unit vector normal to the boundary. To apply the boundary condition given by Eq. (5.84), we need an expression for $\hat{\mathbf{n}}_2$, which is the unit vector pointing away from medium 2 (**Fig. 5-25**).

From the geometry of the small triangle in **Fig. 5-25**,

$$\theta = \tan^{-1} \left(\frac{1}{2} \right) = 26.57^\circ.$$

Hence,

$$\hat{\mathbf{n}}_2 = \hat{\mathbf{x}} \cos \theta + \hat{\mathbf{y}} \sin \theta = \hat{\mathbf{x}}0.89 + \hat{\mathbf{y}}0.45.$$

Magnetic field $\mathbf{B}_1$ has a normal component $\mathbf{B}_{1n}$ along $\hat{\mathbf{n}}_2$ and a tangential component $\mathbf{B}_{1t}$:

$$\mathbf{B}_1 = \mathbf{B}_{1n} + \mathbf{B}_{1t}$$

with

$$\mathbf{B}_{1n} = B_{1n} \hat{\mathbf{n}}_2$$

and

$$\begin{aligned} B_{1n} &= \hat{\mathbf{n}}_2 \cdot \mathbf{B}_1 \\ &= (\hat{\mathbf{x}} \cos \theta + \hat{\mathbf{y}} \sin \theta) \cdot (\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3) \\ &= 2 \cos \theta + 3 \sin \theta. \end{aligned}$$

The tangential component of $\mathbf{B}_1$ is

$$\begin{aligned} \mathbf{B}_{1t} &= \mathbf{B}_1 - \mathbf{B}_{1n} \\ &= \mathbf{B}_1 - B_{1n} \hat{\mathbf{n}}_2 \\ &= (\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3) - (2 \cos \theta + 3 \sin \theta)(\hat{\mathbf{x}} \cos \theta + \hat{\mathbf{y}} \sin \theta) \\ &= \hat{\mathbf{x}}(2 - 2 \cos^2 \theta - 3 \sin \theta \cos \theta) \\ &\quad + \hat{\mathbf{y}}(3 - 2 \cos \theta \sin \theta - 3 \sin^2 \theta). \end{aligned}$$

Boundary conditions require that

$$\mathbf{B}_{1n} = \mathbf{B}_{2n}$$

and

$$\frac{\mathbf{B}_{1t}}{\mu_1} = \frac{\mathbf{B}_{2t}}{\mu_2},$$

which leads to

$$\begin{aligned} \mathbf{B}_2 &= \mathbf{B}_{2n} + \mathbf{B}_{2t} = \mathbf{B}_{1n} + \frac{\mu_2}{\mu_1} \mathbf{B}_{1t} \\ &= B_{1n} \hat{\mathbf{n}}_2 + \frac{\mu_2}{\mu_1} \mathbf{B}_{1t} \\ &= (2 \cos \theta + 3 \sin \theta)(\hat{\mathbf{x}} \cos \theta + \hat{\mathbf{y}} \sin \theta) \\ &\quad + \frac{\mu_2}{\mu_1} [\hat{\mathbf{x}}(2 - 2 \cos^2 \theta - 3 \sin \theta \cos \theta) \\ &\quad + \hat{\mathbf{y}}(3 - 2 \cos \theta \sin \theta - 3 \sin^2 \theta)] \\ &= \hat{\mathbf{x}} \left[2 \cos^2 \theta + 3 \sin \theta \cos \theta \right. \\ &\quad \left. + \frac{\mu_2}{\mu_1} (2 - 2 \cos^2 \theta - 3 \sin \theta \cos \theta) \right] \\ &\quad + \hat{\mathbf{y}} \left[2 \cos \theta \sin \theta + 3 \sin^2 \theta \right. \\ &\quad \left. + \frac{\mu_2}{\mu_1} (3 - 2 \cos \theta \sin \theta - 3 \sin^2 \theta) \right]. \end{aligned}$$

For $\theta = 26.57^\circ$ and $\mu_2 = 2\mu_1$,

$$\mathbf{B}_2 = \hat{\mathbf{x}}1.2 + \hat{\mathbf{y}}4.6 \quad (\text{T}).$$

Exercise 5-13: With reference to **Fig. 5-24**, determine the angle between $\mathbf{H}_1$ and $\hat{\mathbf{n}}_2 = \hat{\mathbf{z}}$ if $\mathbf{H}_2 = (\hat{\mathbf{x}}3 + \hat{\mathbf{z}}2)$ (A/m), $\mu_{r1} = 2$, $\mu_{r2} = 8$, and $\mathbf{J}_s = 0$.

Answer: $\theta = 20.6^\circ$. (See )

5-7 Inductance

An inductor is the magnetic analogue of an electric capacitor. Just as a capacitor can store energy in the electric field in the medium between its conducting surfaces, an inductor can store energy in the magnetic field near its current-carrying conductors. A typical inductor consists of multiple turns of wire helically coiled around a cylindrical core (**Fig. 5-26(a)**). Such a structure is called a **solenoid**. Its core may be air filled or may contain a magnetic material with magnetic permeability μ . If the wire carries a current I and the turns are closely spaced, the solenoid will produce a relatively uniform magnetic field within its interior with magnetic field lines resembling those of the permanent magnet (**Fig. 5-26(b)**).

5-7.1 Magnetic Field in a Solenoid

As a prelude to our discussion of inductance, we derive an expression for the magnetic flux density $\mathbf{B}$ in the interior

解决方案：为了在两种介质之间的平面上应用边界条件，我们首先需要获得垂直于边界的单位向量的表达式。应用方程给出的边界条件。(5.84)，我们需要一个 $\hat{\mathbf{n}}_2$ 的表达式，它是指向远离介质 2 的单位向量（图 5-25）。

由图5-25中小三角形的几何形状可知，

$$\theta = \tan^{-1} \left(\frac{1}{2} \right) = 26.57^\circ.$$

因此，

$$\hat{\mathbf{n}}_2 = \hat{\mathbf{x}} \cos \theta + \hat{\mathbf{y}} \sin \theta = \hat{\mathbf{x}}0.89 + \hat{\mathbf{y}}0.45.$$

磁场 $\mathbf{B}_1$ 具有沿 $\hat{\mathbf{n}}_2$ 的法向分量 $\mathbf{B}_{1n}$ 和切向分量 $\mathbf{B}_{1t}$ ：

$$\mathbf{B}_1 = \mathbf{B}_{1n} + \mathbf{B}_{1t}$$

和

$$\mathbf{B}_{1n} = B_{1n} \hat{\mathbf{n}}_2$$

和

$$\begin{aligned} B_{1n} &= \hat{\mathbf{n}}_2 \cdot \mathbf{B}_1 \\ &= (\hat{\mathbf{x}} \cos \theta + \hat{\mathbf{y}} \sin \theta) \cdot (\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3) \\ &= 2 \cos \theta + 3 \sin \theta. \end{aligned}$$

$\mathbf{B}_1$ 的切向分量为

$$\begin{aligned} \mathbf{B}_{1t} &= \mathbf{B}_1 - \mathbf{B}_{1n} \\ &= \mathbf{B}_1 - B_{1n} \hat{\mathbf{n}}_2 \\ &= (\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3) - (2 \cos \theta + 3 \sin \theta)(\hat{\mathbf{x}} \cos \theta + \hat{\mathbf{y}} \sin \theta) \\ &= \hat{\mathbf{x}}(2 - 2 \cos^2 \theta - 3 \sin \theta \cos \theta) \\ &\quad + \hat{\mathbf{y}}(3 - 2 \cos \theta \sin \theta - 3 \sin^2 \theta). \end{aligned}$$

边界条件要求

$$\mathbf{B}_{1n} = \mathbf{B}_{2n}$$

和

$$\frac{\mathbf{B}_{1t}}{\mu_1} = \frac{\mathbf{B}_{2t}}{\mu_2},$$

这导致

$$\begin{aligned} \mathbf{B}_2 &= \mathbf{B}_{2n} + \mathbf{B}_{2t} = \mathbf{B}_{1n} + \frac{\mu_2}{\mu_1} \mathbf{B}_{1t} \\ &= B_{1n} \hat{\mathbf{n}}_2 + \frac{\mu_2}{\mu_1} \mathbf{B}_{1t} \\ &= (2 \cos \theta + 3 \sin \theta)(\hat{\mathbf{x}} \cos \theta + \hat{\mathbf{y}} \sin \theta) \\ &\quad + \frac{\mu_2}{\mu_1} [\hat{\mathbf{x}}(2 - 2 \cos^2 \theta - 3 \sin \theta \cos \theta) \\ &\quad + \hat{\mathbf{y}}(3 - 2 \cos \theta \sin \theta - 3 \sin^2 \theta)] \\ &= \hat{\mathbf{x}} \left[2 \cos^2 \theta + 3 \sin \theta \cos \theta \right. \\ &\quad \left. + \frac{\mu_2}{\mu_1} (2 - 2 \cos^2 \theta - 3 \sin \theta \cos \theta) \right] \\ &\quad + \hat{\mathbf{y}} \left[2 \cos \theta \sin \theta + 3 \sin^2 \theta \right. \\ &\quad \left. + \frac{\mu_2}{\mu_1} (3 - 2 \cos \theta \sin \theta - 3 \sin^2 \theta) \right]. \end{aligned}$$

对于 $\theta = 26.57^\circ$ 和 $\mu_2 = 2\mu_1$,

$$\mathbf{B}_2 = \hat{\mathbf{x}}1.2 + \hat{\mathbf{y}}4.6 \quad (\text{T}).$$

练习 5-13：参考图 5-24，如果 $\mathbf{H}_2 = (\hat{\mathbf{x}}3 + \hat{\mathbf{z}}2)$ (A/m)、 $\mu_{r1} = 2$ 、 $\mu_{r2} = 8$ 和 $\mathbf{J}_s = 0$ ，求 $\mathbf{H}_1$ 和 $\hat{\mathbf{n}}_2 = \hat{\mathbf{z}}$ 之间的角度。

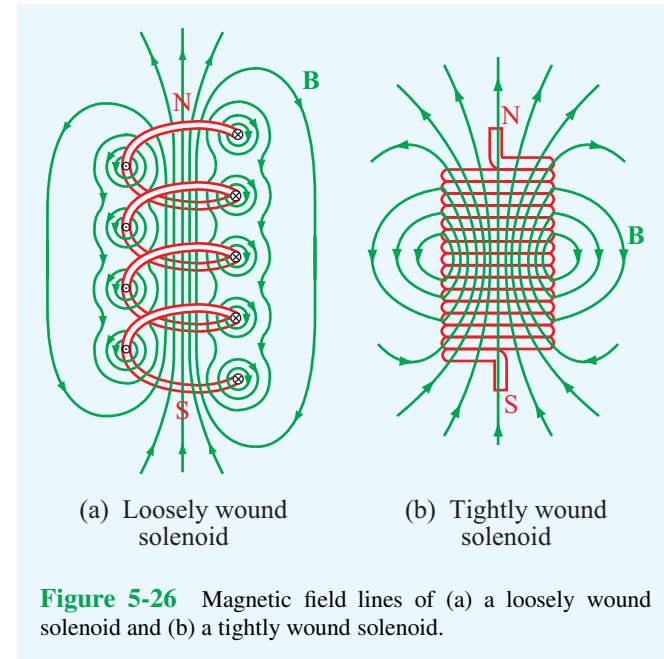
Answer: $\theta = 20.6^\circ$. (See )

5-7 电感

电感器是电容器的磁性类似物。正如电容器可以在其导电表面之间的介质中的电场中存储能量一样，电感器可以在其载流导体附近的磁场中存储能量。典型的电感器由多匝螺旋状缠绕在圆柱形磁芯上的导线组成（图 5-26(a)）。这种结构称为螺线管。其磁芯可以是空气填充的，也可以包含具有磁导率 μ 的磁性材料。如果导线中流过电流 I 并且匝数间隔很近，螺线管将在其内部产生相对均匀的磁场，其磁场线类似于永磁体的磁场线（图 5-26 (b)）。

5-7.1 螺线管中的磁场

作为讨论电感的前奏，我们推导了内部磁通密度 $\mathbf{B}$ 的表达式



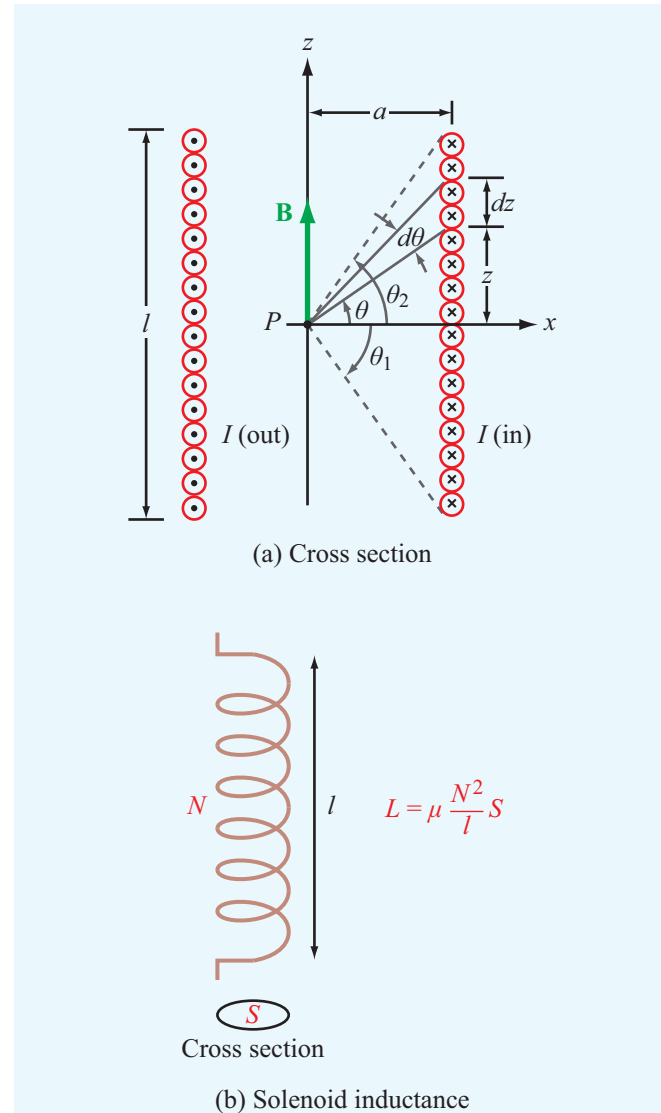
region of a tightly wound solenoid. The solenoid is of length l and radius a and comprises N turns carrying current I . The number of turns per unit length is $n = N/l$, and the fact that the turns are tightly wound implies that the pitch of a single turn is small compared with the solenoid's radius. Even though the turns are slightly helical in shape, we can treat them as circular loops (**Fig. 5-27(a)**). Let us start by considering the magnetic flux density $\mathbf{B}$ at point P on the axis of the solenoid. In Example 5-3, we derived the following expression for the magnetic field $\mathbf{H}$ along the axis of a circular loop of radius a , which is a distance z away from its center:

$$\mathbf{H} = \hat{\mathbf{z}} \frac{I' a^2}{2(a^2 + z^2)^{3/2}}, \quad (5.86)$$

where I' is the current carried by the loop. If we treat an incremental length dz of the solenoid as an equivalent loop composed of $n dz$ turns carrying a current $I' = In dz$, then the induced field at point P is

$$d\mathbf{B} = \mu d\mathbf{H} = \hat{\mathbf{z}} \frac{\mu n I a^2}{2(a^2 + z^2)^{3/2}} dz. \quad (5.87)$$

The total field $\mathbf{B}$ at P is obtained by integrating the contributions from the entire length of the solenoid. This is facilitated



by expressing the variable z in terms of the angle θ , as seen from P to a point on the solenoid rim. That is,

$$z = a \tan \theta, \quad (5.88a)$$

$$a^2 + z^2 = a^2 + a^2 \tan^2 \theta = a^2 \sec^2 \theta, \quad (5.88b)$$

$$dz = a \sec^2 \theta d\theta. \quad (5.88c)$$

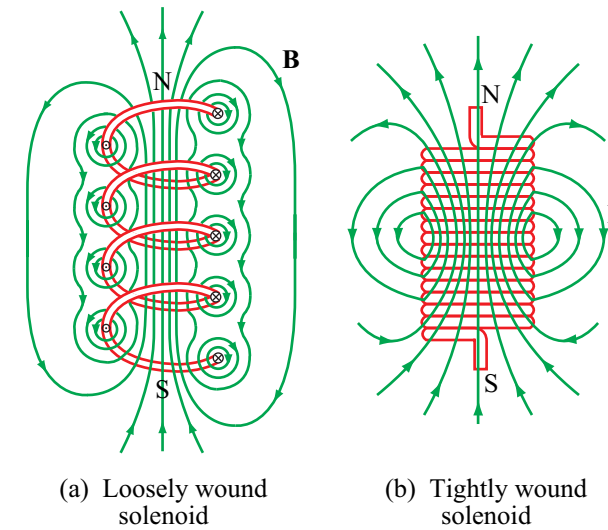


图 5-26 (a) 松散缠绕的螺线管和 (b) 紧密缠绕的螺线管的磁场线。

紧密缠绕的螺线管的区域。螺线管的长度为 l ，半径为 a ，包括承载电流 I 的 N 匝。每单位长度的匝数为 $n = N/l$ ，并且匝紧密缠绕的事实意味着单匝的节距与螺线管的半径相比较小。尽管匝的形状略呈螺旋状，但我们可以将它们视为圆形环（图 5-27 (a)）。让我们首先考虑螺线管轴上 P 点的磁通密度 $\mathbf{B}$ 。在例 5-3 中，我们推导出了沿半径为 a 的圆环轴的磁场 $\mathbf{H}$ 的以下表达式，圆环距圆心的距离为 z ：

$$\mathbf{H} = \hat{\mathbf{z}} \frac{I' a^2}{2(a^2 + z^2)^{3/2}}, \quad (5.86)$$

其中 I' 是环路承载的电流。如果我们将螺线管的增量长度 dz 视为由 $n dz$ 匝组成的等效回路，承载电流 $I' = In dz$ ，则 P 点处的感应场为

$$d\mathbf{B} = \mu d\mathbf{H} = \hat{\mathbf{z}} \frac{\mu n I a^2}{2(a^2 + z^2)^{3/2}} dz. \quad (5.87)$$

P 处的总磁场 $\mathbf{B}$ 是通过将螺线管整个长度的贡献进行积分而获得的。这有利于

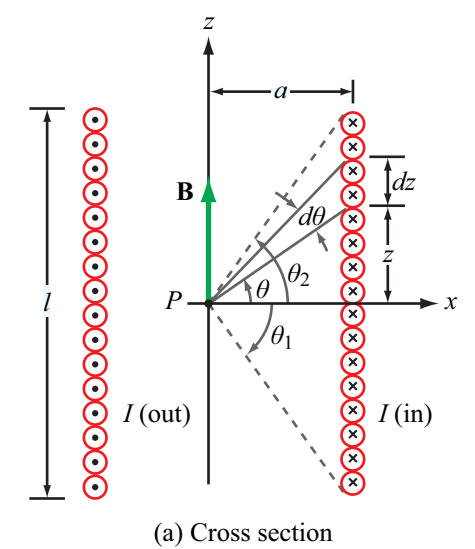


图 5-27 (a) 螺线管横截面显示了用于计算螺线管轴上点 P 处的 $\mathbf{H}$ 的几何形状，(b) 螺线管电感。

通过用角度 θ 表达变量 z ，如从 P 到螺线管边缘上的点所见。那是，

$$z = a \tan \theta, \quad (5.88a)$$

$$a^2 + z^2 = a^2 + a^2 \tan^2 \theta = a^2 \sec^2 \theta, \quad (5.88b)$$

$$dz = a \sec^2 \theta d\theta. \quad (5.88c)$$

Upon substituting the last two expressions in Eq. (5.87) and integrating from θ_1 to θ_2 , we have

$$\mathbf{B} = \hat{\mathbf{z}} \frac{\mu n I a^2}{2} \int_{\theta_1}^{\theta_2} \frac{a \sec^2 \theta \, d\theta}{a^3 \sec^3 \theta} = \hat{\mathbf{z}} \frac{\mu n I}{2} (\sin \theta_2 - \sin \theta_1). \quad (5.89)$$

If the solenoid length l is much larger than its radius a , then for points P away from the solenoid’s ends $\theta_1 \approx -90^\circ$ and $\theta_2 \approx 90^\circ$, in which case Eq. (5.89) reduces to

$$\mathbf{B} \approx \hat{\mathbf{z}} \mu n I = \frac{\hat{\mathbf{z}} \mu N I}{l} \quad (\text{long solenoid with } l/a \gg 1). \quad (5.90)$$

Even though Eq. (5.90) was derived for the field $\mathbf{B}$ at the midpoint of the solenoid, it is approximately valid everywhere in the solenoid’s interior, except near the ends.

We now return to a discussion of inductance, which includes the notion of **self-inductance**, representing the magnetic flux linkage of a coil or circuit with itself, and **mutual inductance**, which involves the magnetic flux linkage in a circuit due to the magnetic field generated by a current in another one. Usually, when the term **inductance** is used, the intended reference is to self-inductance.

Exercise 5-14: Use Eq. (5.89) to obtain an expression for $\mathbf{B}$ at a point on the axis of a very long solenoid but situated at its end points. How does $\mathbf{B}$ at the end points compare to $\mathbf{B}$ at the midpoint of the solenoid?

Answer: $\mathbf{B} = \hat{\mathbf{z}}(\mu N I / 2l)$ at the end points, which is half as large as $\mathbf{B}$ at the midpoint. (See [Ex. 5-14](#).)

5-7.2 Self-Inductance of a Solenoid

From Eq. (5.66), the magnetic flux Φ linking a surface S is

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{Wb}). \quad (5.91)$$

In a solenoid characterized by an approximately uniform magnetic field throughout its cross-section and given by Eq. (5.90), the flux linking a single loop is

$$\Phi = \int_S \hat{\mathbf{z}} \left(\mu \frac{N}{l} I \right) \cdot \hat{\mathbf{z}} \, ds = \mu \frac{N}{l} I S, \quad (5.92)$$

where S is the cross-sectional area of the loop ([Fig. 5-27\(b\)](#)). **Magnetic flux linkage** Λ is defined as the total magnetic flux linking a given circuit or conducting structure. If the structure

consists of a single conductor with multiple loops, as in the case of the solenoid, Λ equals the flux linking all loops of the structure. For a solenoid with N turns,

$$\Lambda = N\Phi = \mu \frac{N^2}{l} I S \quad (\text{Wb}). \quad (5.93)$$

The **self-inductance** of any conducting structure is defined as the ratio of the magnetic flux linkage Λ to the current I flowing through the structure:

$$L = \frac{\Lambda}{I} \quad (\text{H}). \quad (5.94)$$

The SI unit for inductance is the henry (H), which is equivalent to webers per ampere (Wb/A). For a solenoid, use of Eq. (5.93) gives

$$L = \mu \frac{N^2}{l} S \quad (\text{solenoid}). \quad (5.95)$$

5-7.3 Self-Inductance of Other Conductors

The solenoid consists of a single inductor of cross section S . If, on the other hand, the structure consists of two *separate* conductors, as in the case of the parallel-wire and coaxial transmission lines shown in [Fig. 5-28](#), the flux linkage Λ associated with a length l of either line refers to the flux Φ through a closed surface between the two conductors, such as the shaded areas in [Fig. 5-28](#). In reality, there is also some magnetic flux that passes through the conductors themselves, but it may be ignored by assuming that currents flow only on the surfaces of the conductors, in which case the magnetic field inside the conductors vanishes. This assumption is justified by the fact that our interest in calculating Λ is for the purpose of determining the inductance of a given structure, and inductance is of interest primarily in the ac case (i.e., time-varying currents, voltages, and fields). As we will see later in Section 7-5, the current flowing in a conductor under ac conditions is concentrated within a very thin layer on the skin of the conductor.

► For the parallel-wire transmission line, ac currents flow on the outer surfaces of the wires, and for the coaxial line, the current flows on the outer surface of the inner conductor and on the inner surface of the outer one (the current-carrying surfaces are those adjacent to the electric and magnetic fields present in the region between the conductors). ◀

代入等式中的最后两个表达式后: (5.87) 并从 θ_1 积分到 θ_2 , 我们有

$$\mathbf{B} = \hat{\mathbf{z}} \frac{\mu n I a^2}{2} \int_{\theta_1}^{\theta_2} \frac{a \sec^2 \theta \, d\theta}{a^3 \sec^3 \theta} = \hat{\mathbf{z}} \frac{\mu n I}{2} (\sin \theta_2 - \sin \theta_1). \quad (5.89)$$

如果螺线管长度 l 远大于其半径 a , 则对于远离螺线管末端的点 P $\theta_1 \approx -90^\circ$ 和 $\theta_2 \approx 90^\circ$, 在这种情况下(5.89) 减少为

$$\mathbf{B} \approx \hat{\mathbf{z}} \mu n I = \frac{\hat{\mathbf{z}} \mu N I}{l} \quad (\text{long solenoid with } l/a \gg 1). \quad (5.90)$$

即使方程。(5.90) 是针对螺线管中点处的 B 场导出的, 它在螺线管内部的任何地方都大致有效, 除了靠近端部的地方。

现在我们回到对电感的讨论, 其中包括自感和互感的概念, 自感代表线圈或电路与其自身的磁通链, 互感涉及一个电路中由于另一个电路中的电流产生的磁场而产生的磁通链。通常, 当使用术语 “电感” 时, 意指自感。

练习 5-14: 使用方程式(5.89) 以获得一个很长螺线管轴上但位于其端点的点处的 B 表达式。电磁阀端点处的 B 与中点处的 B 相比如何?

答案: 端点处的 $B = \hat{\mathbf{z}}(\mu N I / 2l)$ 是中点处 B 的一半。(参见^{Ex.}5-14)

5-7.2 螺线管的自感

从方程。(5.66), 连接表面 S 的磁通量 Φ 为

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{Wb}). \quad (5.91)$$

在螺线管中, 其整个横截面具有近似均匀的磁场, 由方程式给出。(5.90), 连接单个环路的通量为

$$\Phi = \int_S \hat{\mathbf{z}} \left(\mu \frac{N}{l} I \right) \cdot \hat{\mathbf{z}} \, ds = \mu \frac{N}{l} I S, \quad (5.92)$$

其中 S 是环路的横截面积 (图5-27 (b)) 。磁通量链 Λ 定义为给定电路或导电结构的总磁通量。如果结构

由具有多个环路的单个导体组成, 如螺线管的情况, Λ 等于连接该结构的所有环路的通量。对于 N 匝螺线管,

$$\Lambda = N\Phi = \mu \frac{N^2}{l} I S \quad (\text{Wb}). \quad (5.93)$$

任何导电结构的自感定义为磁通链 Λ 与流过该结构的电流 I 之比:

$$L = \frac{\Lambda}{I} \quad (\text{H}). \quad (5.94)$$

电感的 SI 单位是亨利 (H), 相当于韦伯每安培 (Wb/A)。

对于螺线管, 使用方程式。(5.93) 给出

$$L = \mu \frac{N^2}{l} S \quad (\text{solenoid}). \quad (5.95)$$

5-7.3 其他导体的自感

螺线管由横截面为 S 的单个电感器组成。另一方面, 如果该结构由两个独立的导体组成, 如图 5-28 中所示的平行线和同轴传输线的情况, 则与任一线的长度 l 相关的磁通链 Λ 指的是通过两个导体之间的封闭表面的磁通 Φ , 如图 5-28 中的阴影区域。实际上, 还有一些磁通量穿过导体本身, 但如果假设电流仅在导体表面流动, 则可以忽略它, 在这种情况下, 导体内部的磁场消失。这一假设是合理的, 因为我们计算 Λ 的目的是为了确定给定结构的电感, 而电感主要在交流情况下 (即随时间变化的电流、电压和场) 而感兴趣。正如我们稍后将在第 7-5 节中看到的, 交流条件下导体中流动的电流集中在导体表皮上的一个非常薄的层内。

► 对于平行线传输线, 交流电流在电线的外表面上流动, 对于同轴线, 电流在内导体的外表面上流动, 在外导体的内表面上流动 (载流表面是与导体之间区域中存在的电场和磁场相邻的表面) 。 ◀

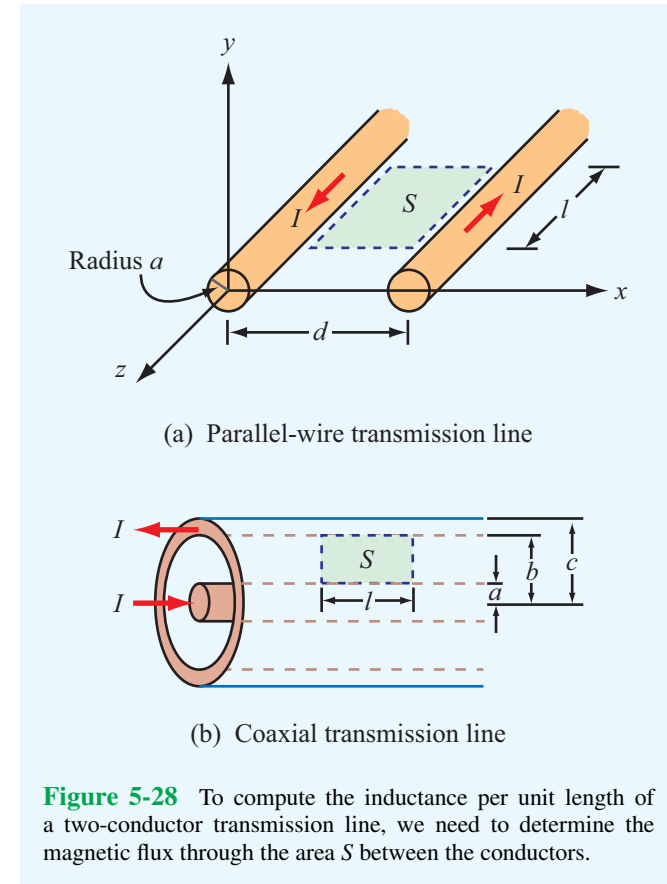


Figure 5-28 To compute the inductance per unit length of a two-conductor transmission line, we need to determine the magnetic flux through the area S between the conductors.

For a two-conductor configuration similar to those of **Fig. 5-28**, the inductance is given by

$$L = \frac{\Lambda}{I} = \frac{\Phi}{I} = \frac{1}{I} \int_S \mathbf{B} \cdot d\mathbf{s}. \quad (5.96)$$

Example 5-8: Inductance of a Coaxial Transmission Line

Develop an expression for the inductance per unit length of a coaxial transmission line with inner and outer conductors of radii a and b (**Fig. 5-29**) and an insulating material of permeability μ .

Solution: The current I in the inner conductor generates a magnetic field $\mathbf{B}$ throughout the region between the two

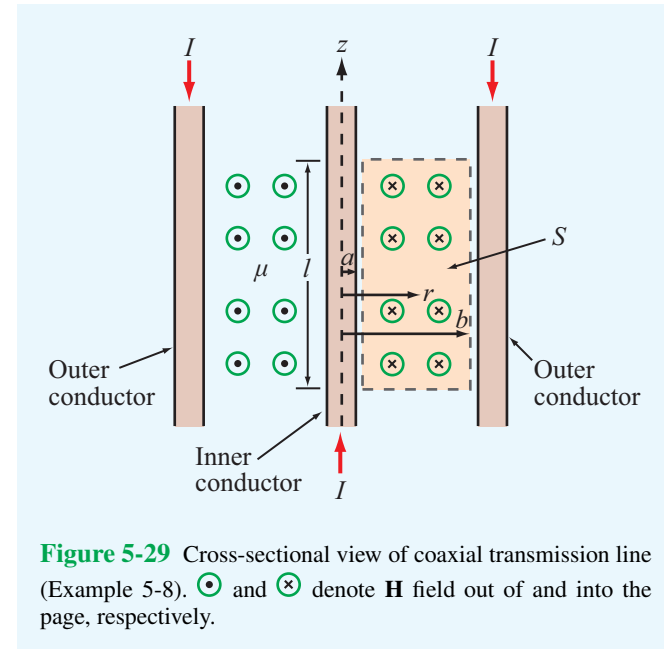


Figure 5-29 Cross-sectional view of coaxial transmission line (Example 5-8). $\odot$ and $\otimes$ denote $\mathbf{H}$ field out of and into the page, respectively.

conductors. It is given by Eq. (5.30) as

$$\mathbf{B} = \hat{\phi} \frac{\mu I}{2\pi r}, \quad (5.97)$$

where r is the radial distance from the axis of the coaxial line. Consider a transmission-line segment of length l as shown in **Fig. 5-29**. Because $\mathbf{B}$ is perpendicular to the planar surface S between the conductors, the flux through S is

$$\Phi = l \int_a^b B dr = l \int_a^b \frac{\mu I}{2\pi r} dr = \frac{\mu I l}{2\pi} \ln \left(\frac{b}{a} \right). \quad (5.98)$$

Using Eq. (5.96), the inductance per unit length of the coaxial transmission line is

$$L' = \frac{L}{l} = \frac{\Phi}{I l} = \frac{\mu}{2\pi} \ln \left(\frac{b}{a} \right). \quad (5.99)$$

5-7.4 Mutual Inductance

Magnetic coupling between two different conducting structures is described in terms of the mutual inductance between them. For simplicity, consider the case of two multiturn closed loops with surfaces S_1 and S_2 . Current I_1 flows through the first loop (**Fig. 5-30**), and no current flows through the second one. The magnetic field $\mathbf{B}_1$ generated by I_1 results in a flux Φ_{12}

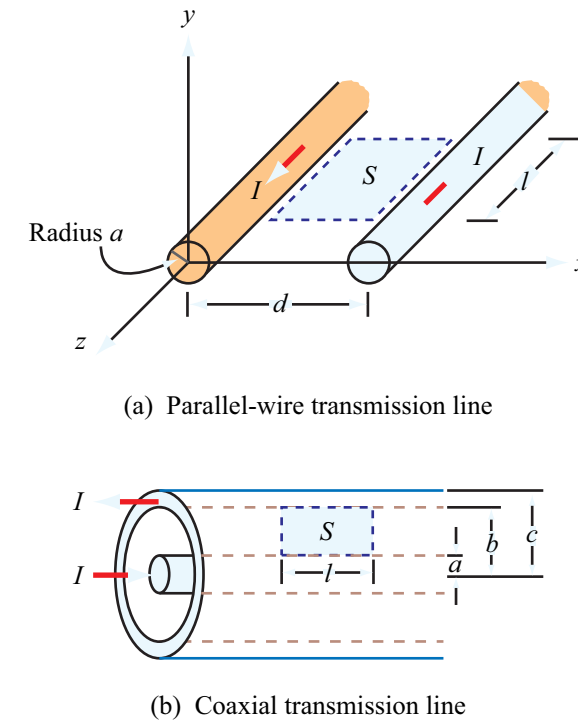


图 5-28 为了计算双导体传输线单位长度的电感，我们需要确定通过导体之间面积 S 的磁通量。

对于类似于图 5-28 的双导体配置，电感由下式给出

$$L = \frac{\Lambda}{I} = \frac{\Phi}{I} = \frac{1}{I} \int_S \mathbf{B} \cdot d\mathbf{s}. \quad (5.96)$$

例 5-8：同轴传输线的电感

建立同轴传输线每单位长度电感的表达式，该同轴传输线具有半径为 a 和 b 的内导体和外导体（图 5-29）以及磁导率为 μ 的绝缘材料。

解：内导体中的电流 I 在两者之间的区域产生磁场 $\mathbf{B}$

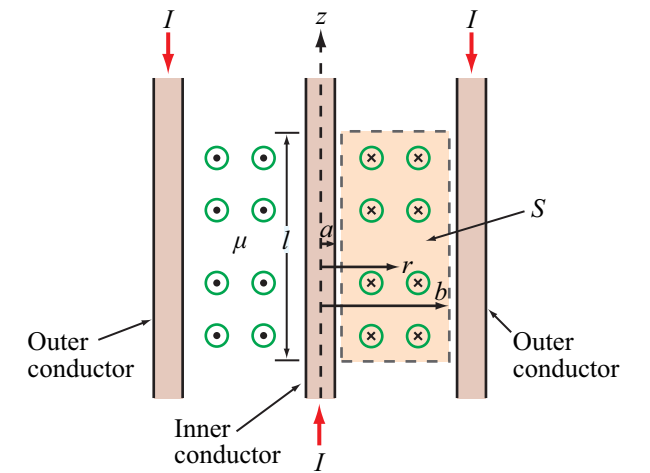


图5-29 同轴传输线的横截面图（示例5-8）。和 分别表示页面外的 $\odot$ 场和页面内的 $\otimes$ 场。

导体。它由等式给出。 (5.30) 作为

$$\mathbf{B} = \hat{\phi} \frac{\mu I}{2\pi r}, \quad (5.97)$$

其中 r 是距同轴线的径向距离。考虑长度为 l 的传输线段，如图 5-29 所示。因为 $\mathbf{B}$ 垂直于导体之间的平面 S ，所以通过 S 的通量为

$$\Phi = l \int_a^b B dr = l \int_a^b \frac{\mu I}{2\pi r} dr = \frac{\mu I l}{2\pi} \ln \left(\frac{b}{a} \right). \quad (5.98)$$

使用方程式(5.96)，同轴传输线单位长度的电感为

$$L' = \frac{L}{l} = \frac{\Phi}{I l} = \frac{\mu}{2\pi} \ln \left(\frac{b}{a} \right). \quad (5.99)$$

5-7.4 互感

两个不同导电结构之间的磁耦合用它们之间的互感来描述。为简单起见，考虑具有曲面 S_1 和 S_2 的两个多匝闭环的情况。电流 I_1 流过第一个环路（图 5-30），第二个环路没有电流流过。 I_1 产生的磁场 $\mathbf{B}_1$ 产生磁通 Φ_{12}

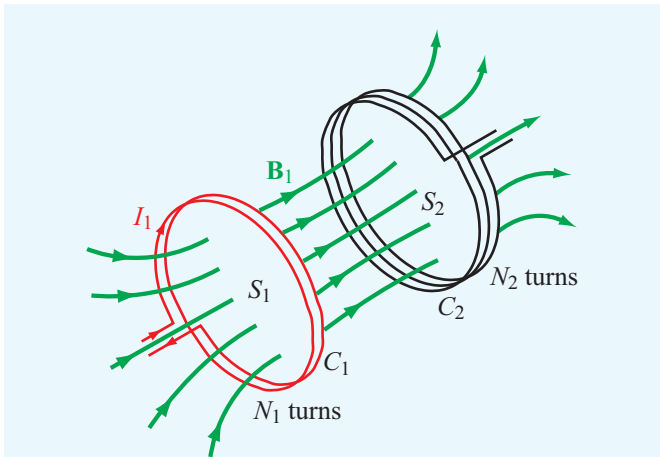


Figure 5-30 Magnetic field lines generated by current I_1 in loop 1 linking surface S_2 of loop 2.

through loop 2 given by

$$\Phi_{12} = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}, \quad (5.100)$$

and if loop 2 consists of N_2 turns all coupled by $\mathbf{B}_1$ in exactly the same way, then the total magnetic flux linkage through loop 2 is

$$\Lambda_{12} = N_2 \Phi_{12} = N_2 \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}. \quad (5.101)$$

The mutual inductance associated with this magnetic coupling is given by

$$L_{12} = \frac{\Lambda_{12}}{I_1} = \frac{N_2}{I_1} \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s} \quad (\text{H}). \quad (5.102)$$

Mutual inductance is important in transformers (as discussed in Chapter 6) wherein the windings of two or more circuits share a common magnetic core, as illustrated by the toroidal arrangement shown in **Fig. 5-31**.

Example 5-9: Mutual Inductance

The rectangular conducting loop shown in **Fig. 5-32** is coplanar with the long, straight wire carrying current I . Determine the mutual inductance between the wire and the loop.

Solution: According to Eq. (5.102), the mutual inductance is given by

$$L_{12} = \frac{N_2}{I_1} \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}.$$

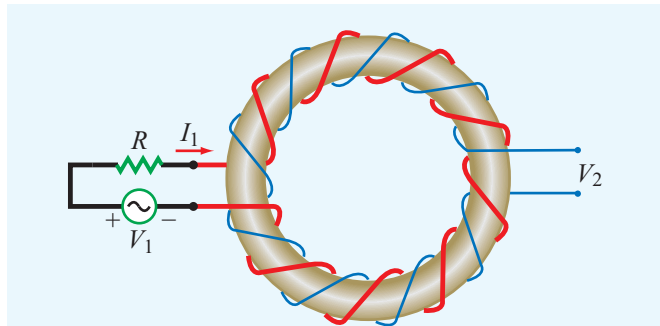


Figure 5-31 Toroidal coil with two windings used as a transformer.

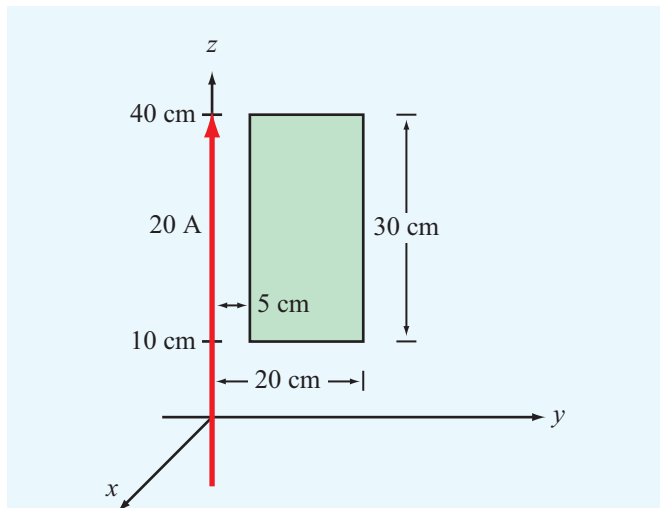


Figure 5-32 Mutual inductance between two conductors (Example 5-9).

In the present case, $I_1 = I$ is the current carried by the first conductor (straight wire), S_2 is the cross-sectional area of the second conductor (rectangular loop), and $\mathbf{B}_1$ is the magnetic field due to the first conductor (wire). Also, $N_2 = 1$ because the loop has a single turn. In the right-hand side of the y - z plane, the magnetic field due to current I is into the page ($-\hat{\mathbf{x}}$ direction) and given by

$$\mathbf{B} = -\hat{\mathbf{x}} \frac{\mu_0 I}{2\pi y},$$

where y is the distance away from the wire. The expression is identical with that given by Eq. (5.97), after replacing $\hat{\phi}$ with $-\hat{\mathbf{x}}$ and r with y . To compute the magnetic flux passing through the loop (into the page), we need to define the differential area

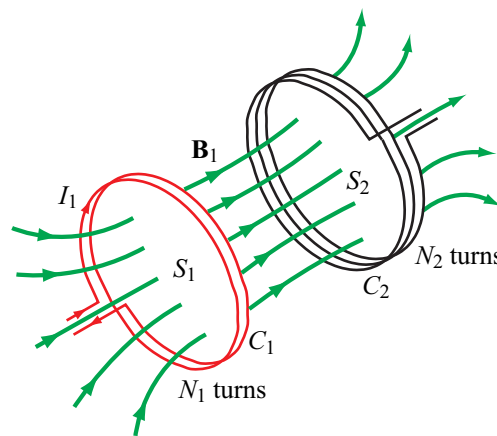


图 5-30 环路 1 中的电流 I_1 连接环路 2 的表面 S_2 时产生的磁场线。

通过由下式给出的循环 2

$$\Phi_{12} = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}, \quad (5.100)$$

如果环路 2 由 N_2 匝组成，所有匝数均以完全相同的方式通过 $\mathbf{B}_1$ 耦合，则通过环路 2 的总磁通链为

$$\Lambda_{12} = N_2 \Phi_{12} = N_2 \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}. \quad (5.101)$$

与该磁耦合相关的互感由下式给出

$$L_{12} = \frac{\Lambda_{12}}{I_1} = \frac{N_2}{I_1} \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s} \quad (\text{H}). \quad (5.102)$$

互感在变压器中很重要（如第 6 章所述），其中两个或多个电路的绕组共享一个公共磁芯，如图 5-31 所示的环形布置所示。

例 5-9：互感

图 5-32 所示的矩形导电环与承载电流 I 的长直导线共面。确定导线与环之间的互感。

解：根据方程(5.102)，互感由下式给出

$$L_{12} = \frac{N_2}{I_1} \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}.$$

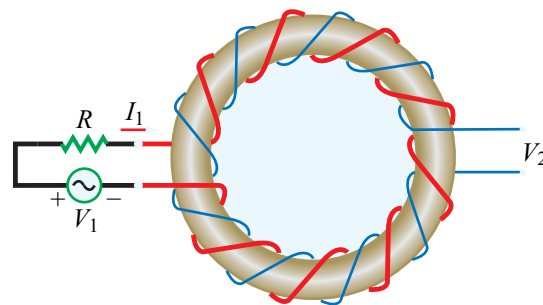


图 5-31 用作变压器的具有两个绕组的环形线圈。

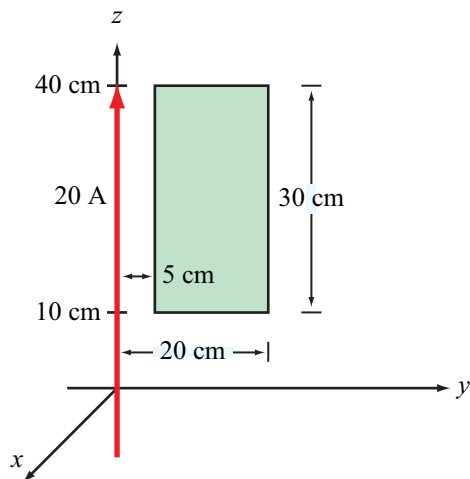


图 5-32 两个导体之间的互感（示例 5-9）。

在本例中， $I_1 = I$ 是第一导体（直线）承载的电流， S_2 是第二导体（矩形环）的横截面积， $\mathbf{B}_1$ 是第一导体（电线）产生的磁场。另外， $N_2 = 1$ 因为循环只有一匝。在 y - z 平面的右侧，电流 I 产生的磁场进入页面（ $-\hat{\mathbf{x}}$ 方向），由下式给出

$$\mathbf{B} = -\hat{\mathbf{x}} \frac{\mu_0 I}{2\pi y},$$

其中 y 是距电线的距离。该表达式与式 (1) 给出的表达式相同。(5.97)，将 $\hat{\phi}$ 替换为 $-\hat{\mathbf{x}}$ 并将 r 替换为 y 后。为了计算穿过线圈（进入页面）的磁通量，我们需要定义微分面积

as $ds = -\hat{\mathbf{x}} \, dy \, dz$. Hence,

$$\begin{aligned} L_{12} &= \frac{1}{I} \int_{y=0.05}^{0.20} \int_{z=0.1}^{0.4} \left(-\hat{\mathbf{x}} \frac{\mu_0 I}{2\pi y} \right) \cdot (-\hat{\mathbf{x}} \, dy \, dz) \\ &= \frac{\mu_0}{2\pi} \left(\int_{0.05}^{0.20} \frac{dy}{y} \right) \left(\int_{0.1}^{0.4} dz \right) \\ &= \frac{0.3\mu_0}{2\pi} (\ln y)|_{0.05}^{0.20} = \frac{0.3\mu_0}{2\pi} \ln \left(\frac{0.2}{0.05} \right) = 83 \text{ nH}. \end{aligned}$$

Concept Question 5-15: What is the magnetic field like in the interior of a long solenoid?

Concept Question 5-16: What is the difference between self-inductance and mutual inductance?

Concept Question 5-17: How is the inductance of a solenoid related to its number of turns N ?

5-8 Magnetic Energy

When we introduced electrostatic energy in Section 4-10, we did so by examining what happens to the energy expended in charging up a capacitor from zero voltage to some final voltage V . We introduce the concept of magnetic energy by considering an inductor with inductance L connected to a current source. Suppose that we were to increase the current i flowing through the inductor from zero to a final value I . From circuit theory, we know that the instantaneous voltage v across the inductor is given by $v = L \, di/dt$. We will derive this relationship from Maxwell's equations in Chapter 6, thereby justifying the use of the i – v relationship for the inductor. Power p equals the product of v and i , and the time integral of power is work, or energy. Hence, the total energy in joules (J) expended in building up a current I in the inductor is

$$W_m = \int p \, dt = \int i v \, dt = L \int_0^I i \, di = \frac{1}{2} L I^2 \quad (\text{J}). \quad (5.103)$$

We call this the **magnetic energy** stored in the inductor.

To justify this association, consider the solenoid inductor. Its inductance is given by Eq. (5.95) as $L = \mu N^2 S/l$, and the magnitude of the magnetic flux density in its interior is given by Eq. (5.90) as $B = \mu NI/l$, implying that $I = Bl/(\mu N)$. Using these expressions for L and I in Eq. (5.103), we obtain

$$\begin{aligned} W_m &= \frac{1}{2} L I^2 = \frac{1}{2} \left(\mu \frac{N^2}{l} S \right) \left(\frac{Bl}{\mu N} \right)^2 \\ &= \frac{1}{2} \frac{B^2}{\mu} (lS) = \frac{1}{2} \mu H^2 \mathcal{V}, \end{aligned} \quad (5.104)$$

where $\mathcal{V} = lS$ is the volume of the interior of the solenoid and $H = B/\mu$. The expression for W_m suggests that the energy expended in building up the current in the inductor is stored in the magnetic field with **magnetic energy density** w_m , defined as the magnetic energy W_m per unit volume,

$$w_m = \frac{W_m}{\mathcal{V}} = \frac{1}{2} \mu H^2 \quad (\text{J/m}^3). \quad (5.105)$$

► Even though this expression was derived for a solenoid, it remains valid for any medium with a magnetic field $\mathbf{H}$. ◀

Furthermore, for any volume $\mathcal{V}$ containing a material with permeability μ (including free space with permeability μ_0), the total magnetic energy stored in a magnetic field $\mathbf{H}$ is

$$W_m = \frac{1}{2} \int_{\mathcal{V}} \mu H^2 \, d\mathcal{V} \quad (\text{J}). \quad (5.106)$$

Example 5-10: Magnetic Energy in a Coaxial Cable

Derive an expression for the magnetic energy stored in a coaxial cable of length l and inner and outer radii a and b . The current flowing through the cable is I and its insulation material has permeability μ .

Solution: From Eq. (5.97), the magnitude of the magnetic field in the insulating material is

$$H = \frac{B}{\mu} = \frac{I}{2\pi r},$$

where r is the radial distance from the center of the inner conductor (**Fig. 5-29**). The magnetic energy stored in the coaxial cable therefore is

$$W_m = \frac{1}{2} \int_{\mathcal{V}} \mu H^2 \, d\mathcal{V} = \frac{\mu I^2}{8\pi^2} \int_{\mathcal{V}} \frac{1}{r^2} \, d\mathcal{V}.$$

Since H is a function of r only, we choose $d\mathcal{V}$ to be a cylindrical shell of length l , radius r , and thickness dr along the radial direction. Thus, $d\mathcal{V} = 2\pi r l \, dr$ and

$$W_m = \frac{\mu I^2}{8\pi^2} \int_a^b \frac{1}{r^2} \cdot 2\pi r l \, dr = \frac{\mu I^2 l}{4\pi} \ln \left(\frac{b}{a} \right) = \frac{1}{2} L I^2 \quad (\text{J})$$

with L given by Eq. (5.99).

作为 $ds = -\hat{\mathbf{x}} \, dy \, dz$ 。因此,

$$\begin{aligned} L_{12} &= \frac{1}{I} \int_{y=0.05}^{0.20} \int_{z=0.1}^{0.4} \left(-\hat{\mathbf{x}} \frac{\mu_0 I}{2\pi y} \right) \cdot (-\hat{\mathbf{x}} \, dy \, dz) \\ &= \frac{\mu_0}{2\pi} \left(\int_{0.05}^{0.20} \frac{dy}{y} \right) \left(\int_{0.1}^{0.4} dz \right) \\ &= \frac{0.3\mu_0}{2\pi} (\ln y)|_{0.05}^{0.20} = \frac{0.3\mu_0}{2\pi} \ln \left(\frac{0.2}{0.05} \right) = 83 \text{ nH}. \end{aligned}$$

概念问题 5-15: 长螺线管内部的磁场是什么样的?

概念问题 5-16: 自感和互感有什么区别?

概念问题 5-17: 螺线管的电感与其匝数 N 有何关系?

5-8 磁能

当我们在第 4-10 节中引入静电能时, 我们通过检查将电容器从零电压充电到某个最终电压 V 所消耗的能量发生了什么来做到这一点。我们通过考虑连接到电流源的电感 L 的电感器来引入磁能的概念。假设我们要将流过电感器的电流 i 从零增加到最终值 I 。根据电路理论, 我们知道电感器上的瞬时电压 v 由 $v = L \, di/dt$ 给出。我们将从第 6 章中的麦克斯韦方程组中推导出这个关系, 从而证明对电感器使用 i – v 关系是合理的。功率 p 等于 v 和 i 的乘积, 功率的时间积分就是功, 或者说能量。因此, 在电感器中建立电流 I 所消耗的总能量 (以焦耳 (J) 为单位) 为

$$W_m = \int p \, dt = \int i v \, dt = L \int_0^I i \, di = \frac{1}{2} L I^2 \quad (\text{J}). \quad (5.103)$$

我们称之为存储在电感器中的磁能。

为了证明这种关联的合理性, 请考虑螺线管电感器。其电感由方程式给出。(5.95) 为 $L = \mu N^2 S/l$, 其内部磁通密度的大小由方程 (5.95) 给出。(5.90) 为 $B = \mu NI/l$, 意味着 $I = Bl/(\mu N)$ 。在等式中使用这些表达式表示 L 和 I 。(5.103), 我们得到

$$\begin{aligned} W_m &= \frac{1}{2} L I^2 = \frac{1}{2} \left(\mu \frac{N^2}{l} S \right) \left(\frac{Bl}{\mu N} \right)^2 \\ &= \frac{1}{2} \frac{B^2}{\mu} (lS) = \frac{1}{2} \mu H^2 \mathcal{V}, \end{aligned} \quad (5.104)$$

其中, $\mathcal{V} = lS$ 是螺线管内部的体积, $H = B/\mu$ 。 W_m 的表达式表明, 在电感器中建立电流所消耗的能量存储在磁能密度为 w_m 的磁场中, 定义为每单位体积的磁能 w_m ,

$$w_m = \frac{W_m}{\mathcal{V}} = \frac{1}{2} \mu H^2 \quad (\text{J/m}^3). \quad (5.105)$$

► 尽管这个表达式是针对螺线管导出的, 但它对于任何具有磁场 H 的介质仍然有效。◀

此外, 对于包含具有磁导率 μ (的材料 (包括具有磁导率 μ_0) 的自由空间) 的任何体积 $\mathcal{V}$, 存储在磁场 H 中的总磁能为

$$W_m = \frac{1}{2} \int_{\mathcal{V}} \mu H^2 \, d\mathcal{V} \quad (\text{J}). \quad (5.106)$$

例 5-10: 同轴电缆中的磁能

导出长度为 l 、内半径和外半径为 a 和 b 的同轴电缆中存储的磁能的表达式。流过电缆的电流为 I , 其绝缘材料具有磁导率 μ 。

解决方案: 从方程。(5.97), 绝缘材料中的磁场强度为

$$H = \frac{B}{\mu} = \frac{I}{2\pi r},$$

其中 r 是距内导体中心的径向距离 (图 5-29)。因此, 同轴电缆中存储的磁能为

$$W_m = \frac{1}{2} \int_{\mathcal{V}} \mu H^2 \, d\mathcal{V} = \frac{\mu I^2}{8\pi^2} \int_{\mathcal{V}} \frac{1}{r^2} \, d\mathcal{V}.$$

由于 H 仅是 r 的函数, 因此我们选择 $d\mathcal{V}$ 作为沿径向长度为 l 、半径为 r 、厚度为 dr 的圆柱壳。因此, $d\mathcal{V} = 2\pi r l \, dr$ 和

$$W_m = \frac{\mu I^2}{8\pi^2} \int_a^b \frac{1}{r^2} \cdot 2\pi r l \, dr = \frac{\mu I^2 l}{4\pi} \ln \left(\frac{b}{a} \right) = \frac{1}{2} L I^2 \quad (\text{J})$$

L 由方程给出。(5.99)。

Technology Brief 11: Inductive Sensors

Magnetic coupling between different coils forms the basis of several different types of inductive sensors. Applications include the measurement of position and displacement (with submillimeter resolution) in device fabrication processes, **proximity detection** of conductive objects, and other related applications.

Linear Variable Differential Transformer (LVDT)

▶ An LVDT comprises a primary coil connected to an ac source (typically a sine wave at a frequency in the 1–10 kHz range) and a pair of secondary coils, all sharing a common **ferromagnetic core** (Fig. TF11-1). ◀

The magnetic core serves to couple the magnetic flux generated by the primary coil into the two secondaries, thereby inducing an output voltage across each of them. The secondary coils are connected in opposition, so when the core is positioned at the magnetic center of the LVDT, the individual output signals of the secondaries cancel each other out, producing a null output voltage. The core is connected to the outside world via a non-magnetic push rod. When the rod moves the core away from the magnetic center, the magnetic fluxes induced

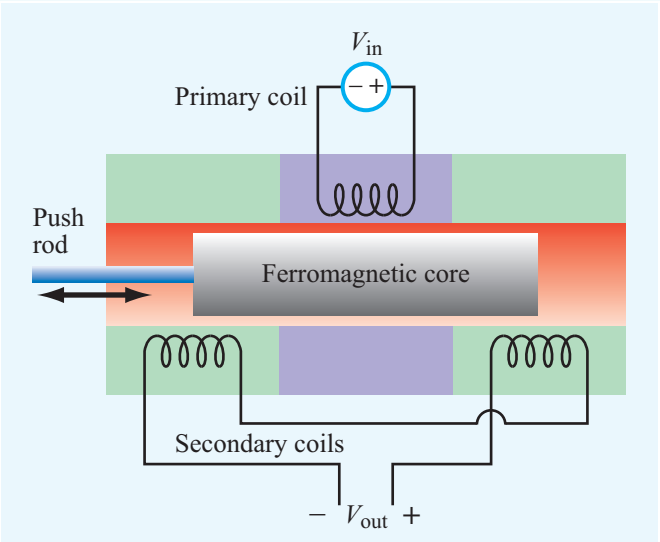


Figure TF11-1 Linear variable differential transformer (LVDT) circuit.

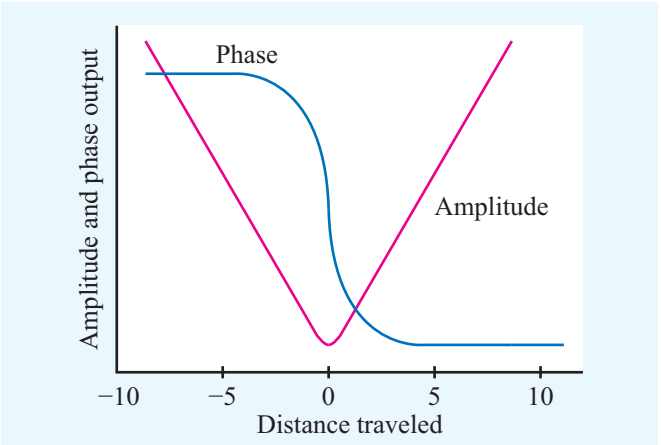


Figure TF11-2 Amplitude and phase responses as functions of the distance by which the magnetic core is moved away from the center position.

in the secondary coils are no longer equal, resulting in a nonzero output voltage. The LVDT is called a “linear” transformer because the amplitude of the output voltage is a linear function of displacement over a wide operating range (Fig. TF11-2).

The cutaway view of the LVDT model in Fig. TF11-3 depicts a configuration in which all three coils—with the primary straddled by the secondaries—are wound around a glass tube that contains the magnetic core and attached rod. Sample applications are illustrated in Fig. TF11-4.

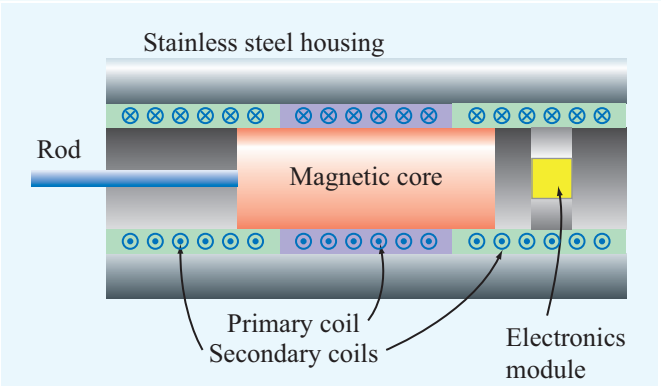


Figure TF11-3 Cutaway view of LVDT.

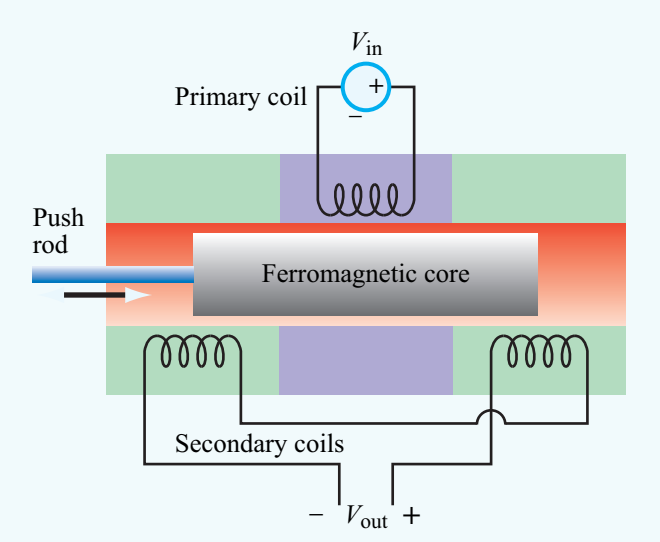
技术简介 11：电感式传感器

不同线圈之间的磁耦合构成了几种不同类型的电感式传感器的基础。应用包括设备制造过程中的位置和位移测量（亚毫米分辨率）、导电物体的接近检测以及其他相关应用。

线性可变差动变压器 (LVDT)

▶ LVDT 包括一个连接到交流电源（通常是频率在 1 -10 kHz 范围内的正弦波）的初级线圈和一对次级线圈，所有线圈共享一个公共铁磁芯（图 TF11-1）。 ◀

磁芯用于将初级线圈产生的磁通量耦合到两个次级线圈中，从而在每个次级线圈上感应出输出电压。次级线圈反向连接，因此当磁芯位于 LVDT 的磁中心时，次级线圈的各个输出信号相互抵消，产生零输出电压。核心通过非磁性推杆与外界相连。当棒将磁芯移离磁中心时，会产生感应磁通量



图TF11-1 线性可变差动变压器（LVDT）电路。

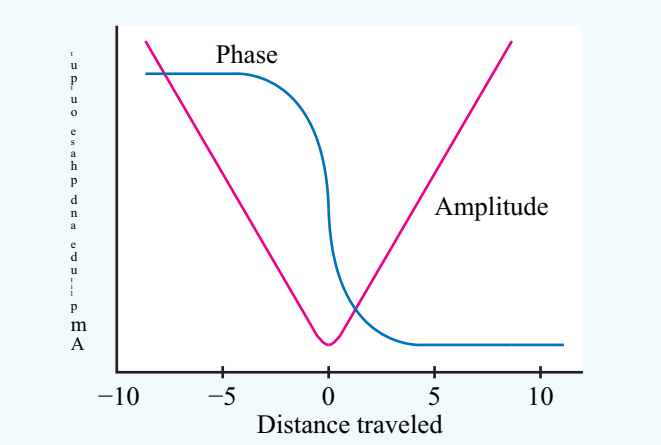


图 TF11-2 幅度和相位响应作为磁芯远离中心位置的距离的函数。

次级线圈中的电压不再相等，导致输出电压非零。LVDT 被称为“线性”变压器，因为输出电压的幅度是宽工作范围内位移的线性函数（图 TF11-2）。

图 TF11-3 中 LVDT 模型的剖面图描绘了一种配置，其中所有三个线圈（初级线圈跨在次级线圈上）都缠绕在包含磁芯和附加杆的玻璃管上。示例应用如图 TF11-4 所示。

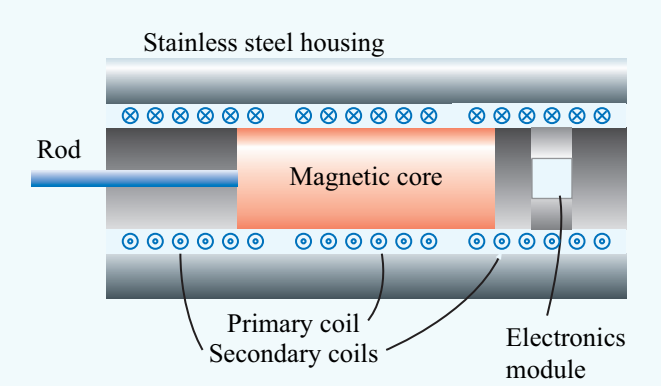


图 TF11-3 LVDT 剖面图。

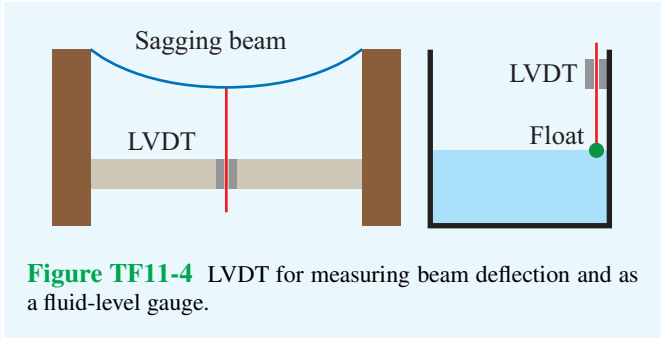


Figure TF11-4 LVDT for measuring beam deflection and as a fluid-level gauge.

Eddy-Current Proximity Sensor

The transformer principle can be applied to build a proximity sensor in which the output voltage of the secondary coil becomes a sensitive indicator of the presence of a conductive object in its immediate vicinity (Fig. TF11-5).

► When an object is placed in front of the secondary coil, the magnetic field of the coil induces eddy (circular) currents in the object that generate magnetic fields of their own having a direction that opposes the magnetic field of the secondary coil. ◄

The reduction in magnetic flux causes a drop in output voltage with the magnitude of the change being dependent on the conductive properties of the object and its distance from the sensor.

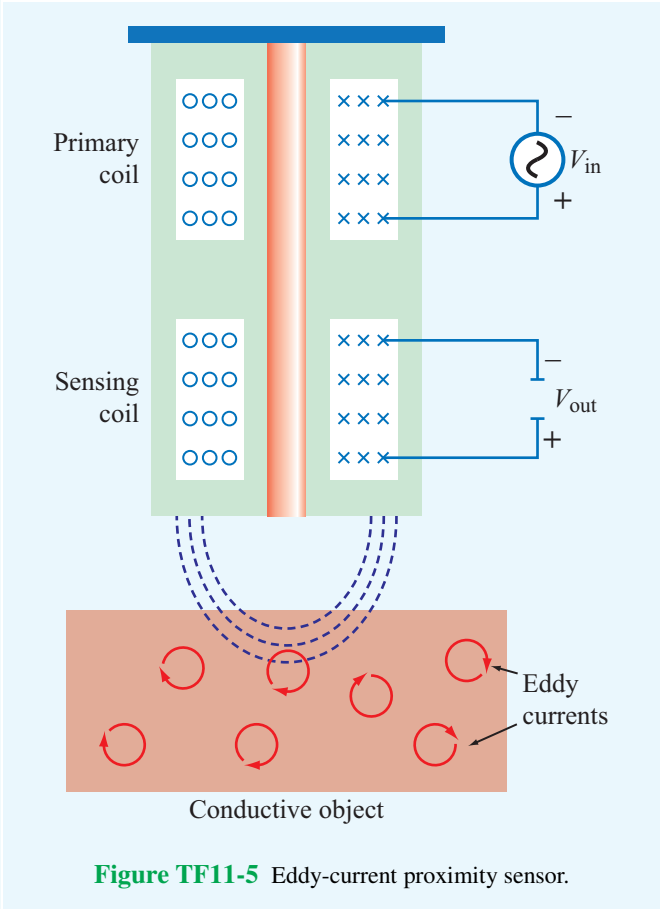


Figure TF11-5 Eddy-current proximity sensor.

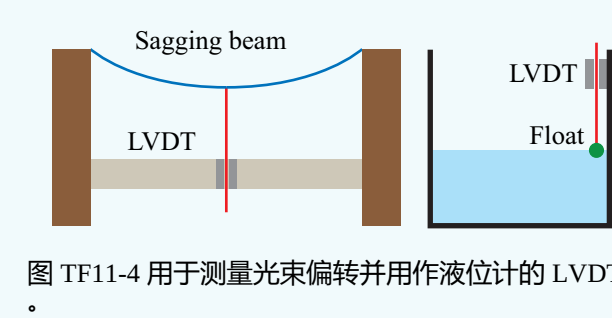


图 TF11-4 用于测量光束偏转并用作液位计的 LVDT。

电涡流接近传感器

变压器原理可用于构建接近传感器，其中次级线圈的输出电压成为其附近是否存在导电物体的敏感指示器（图TF11-5）。

► 当一个物体放置在次级线圈前面时，线圈的磁场会在物体中感应出涡流（环形）电流，该涡流产生自己的磁场，其方向与次级线圈的磁场相反。 ◄

磁通量的减少会导致输出电压下降，变化的幅度取决于物体的导电特性及其与传感器的距离。

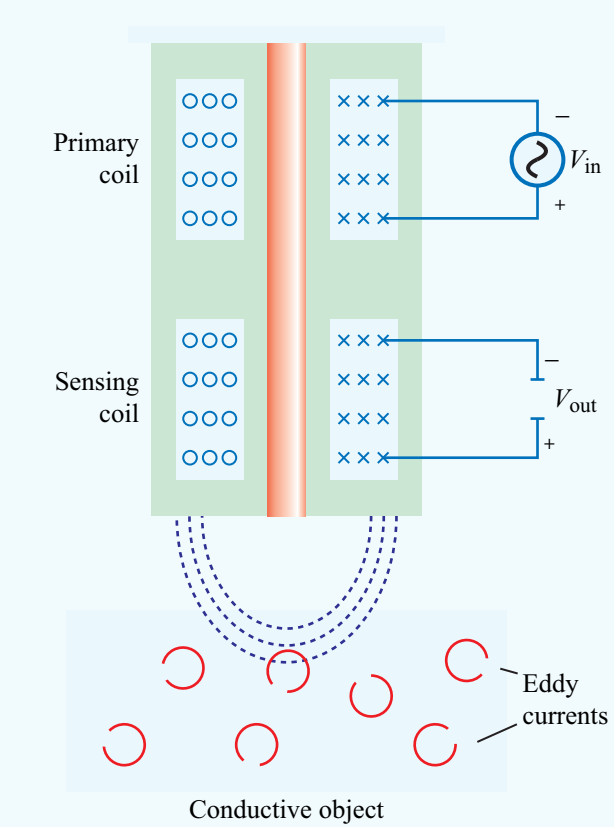


图 TF11-5 电涡流接近传感器。

Chapter 5 Summary

Concepts

- The magnetic force acting on a charged particle q moving with a velocity $\mathbf{u}$ in a region containing a magnetic flux density $\mathbf{B}$ is $\mathbf{F}_m = q\mathbf{u} \times \mathbf{B}$.
- The total electromagnetic force, known as the Lorentz force, acting on a moving charge in the presence of both electric and magnetic fields is $\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$.
- Magnetic forces acting on current loops can generate magnetic torques.
- The magnetic field intensity induced by a current element is defined by the Biot–Savart law.
- Gauss’s law for magnetism states that the net magnetic flux flowing out of any closed surface is zero.
- Ampère’s law states that the line integral of $\mathbf{H}$ over a closed contour is equal to the net current crossing the surface bounded by the contour.
- The vector magnetic potential $\mathbf{A}$ is related to $\mathbf{B}$ by $\mathbf{B} = \nabla \times \mathbf{A}$.
- Materials are classified as diamagnetic, paramagnetic,

- or ferromagnetic, depending on their crystalline structure and the behavior under the influence of an external magnetic field.
- Diamagnetic and paramagnetic materials exhibit a linear behavior between $\mathbf{B}$ and $\mathbf{H}$ with $\mu \approx \mu_0$ for both.
- Ferromagnetic materials exhibit a nonlinear hysteretic behavior between $\mathbf{B}$ and $\mathbf{H}$, and for some, μ may be as large as $10^5\mu_0$.
- At the boundary between two different media, the normal component of $\mathbf{B}$ is continuous, and the tangential components of $\mathbf{H}$ are related by $H_{2t} - H_{1t} = J_s$, where J_s is the surface current density flowing in a direction orthogonal to H_{1t} and H_{2t} .
- The inductance of a circuit is defined as the ratio of magnetic flux linking the circuit to the current flowing through it.
- Magnetic energy density is given by $w_m = \frac{1}{2}\mu H^2$.

Mathematical and Physical Models

Maxwell’s Magnetostatics Equations

Gauss’s Law for Magnetism

$$\nabla \cdot \mathbf{B} = 0 \quad \longleftrightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{s} = 0$$

Ampère’s Law

$$\nabla \times \mathbf{H} = \mathbf{J} \quad \longleftrightarrow \quad \oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I$$

Lorentz Force on Charge q

$$\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

Magnetic Force on Wire

$$\mathbf{F}_m = I \oint_C d\mathbf{l} \times \mathbf{B} \quad (\text{N})$$

Magnetic Torque on Loop

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} \quad (\text{N}\cdot\text{m})$$

$$\mathbf{m} = \hat{\mathbf{n}}NIA \quad (\text{A}\cdot\text{m}^2)$$

Biot–Savart Law

$$\mathbf{H} = \frac{I}{4\pi} \int_l \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} \quad (\text{A/m})$$

Magnetic Field

$$\text{Infinitely Long Wire} \quad \mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r} \quad (\text{Wb/m}^2)$$

$$\text{Circular Loop} \quad \mathbf{H} = \hat{\mathbf{z}} \frac{Ia^2}{2(a^2 + z^2)^{3/2}} \quad (\text{A/m})$$

$$\text{Solenoid} \quad \mathbf{B} \approx \hat{\mathbf{z}}\mu nI = \frac{\hat{\mathbf{z}}\mu NI}{l} \quad (\text{Wb/m}^2)$$

Vector Magnetic Potential

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (\text{Wb/m}^2)$$

Vector Poisson’s Equation

$$\nabla^2 \mathbf{A} = -\mu \mathbf{J}$$

Inductance

$$L = \frac{\Lambda}{I} = \frac{\Phi}{I} = \frac{1}{I} \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{H})$$

Magnetic Energy Density

$$w_m = \frac{1}{2} \mu H^2 \quad (\text{J/m}^3)$$

第五章总结

概念

- 在磁通密度为 $\mathbf{B}$ 的区域中，以速度 $\mathbf{u}$ 运动的带电粒子 q 受到的磁力为 $\mathbf{F}_m = q\mathbf{u} \times \mathbf{B}$ 。
- 在存在电场和磁场的情况下作用于移动电荷的总电磁力（称为洛伦兹力）为 $\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$ 。
- 作用在电流环上的磁力可以产生磁扭矩。
- 电流元件感应的磁场强度由毕奥-萨伐尔定律定义。
- 高斯磁定律指出，从任何闭合表面流出的净磁通量为零。
- 安培环路定律指出， $\mathbf{H}$ 在闭合轮廓上的线积分等于穿过由轮廓界定的表面的净电流。

- 矢量磁势 $\mathbf{A}$ 与 $\mathbf{B}$ 的关系为 $\mathbf{B} = \nabla \times \mathbf{A}$ 。

- 材料可分为抗磁性、顺磁性、

或铁磁性，取决于它们的晶体结构和在外部磁场影响下的行为。

- 抗磁性和顺磁性材料在 $\mathbf{B}$ 和 $\mathbf{H}$ 之间表现出线性行为，两者均具有 $\mu \approx \mu_0$ 。
- 铁磁材料在 $\mathbf{B}$ 和 $\mathbf{H}$ 之间表现出非线性磁滞行为，对于某些材料， μ 可能大至 $10^5\mu_0$ 。

- $\mathbf{B}$ 的法向分量连续；切向磁场满足 $\mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{J}_s$ ，其中 $\mathbf{n}$ 是从介质1指向介质2的单位法向量。

- 电感 L 为总磁链 Λ 与电流 I 之比： $L = \Lambda/I$ ；总磁链须计入全部匝数。

- 磁能密度由 $w_m = \frac{1}{2}\mu H^2$ 给出。

数学与物理模型

静磁场的麦克斯韦方程组

磁场的高斯定律

$$\nabla \cdot \mathbf{B} = 0 \quad \longleftrightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{s} = 0$$

安培环路定律

$$\nabla \times \mathbf{H} = \mathbf{J} \quad \longleftrightarrow \quad \oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I$$

电荷 q 所受洛伦兹力

$$\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

载流导线所受磁力

$$\mathbf{F}_m = I \oint_C d\mathbf{l} \times \mathbf{B} \quad (\text{N})$$

电流回路所受磁力矩

$$\mathbf{T} = \mathbf{m} \times \mathbf{B} \quad (\text{N}\cdot\text{m})$$

$$\mathbf{m} = \hat{\mathbf{n}}NIA \quad (\text{A}\cdot\text{m}^2)$$

毕奥-萨伐尔定律

$$\mathbf{H} = \frac{I}{4\pi} \int_l \frac{d\mathbf{l} \times \hat{\mathbf{R}}}{R^2} \quad (\text{A/m})$$

磁场

$$\text{无限长直导线} \quad \mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r} \quad (\text{Wb/m}^2)$$

$$\text{圆形电流回路} \quad \mathbf{H} = \hat{\mathbf{z}} \frac{Ia^2}{2(a^2 + z^2)^{3/2}} \quad (\text{A/m})$$

$$\text{螺线管} \quad \mathbf{B} \approx \hat{\mathbf{z}}\mu nI = \frac{\hat{\mathbf{z}}\mu NI}{l} \quad (\text{Wb/m}^2)$$

磁矢势

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (\text{Wb/m}^2)$$

矢量泊松方程

$$\nabla^2 \mathbf{A} = -\mu \mathbf{J}$$

单匝回路电感

$$L = \frac{\Lambda}{I} = \frac{\Phi}{I} = \frac{1}{I} \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{H})$$

磁能密度

$$w_m = \frac{1}{2} \mu H^2 \quad (\text{J/m}^3)$$

使用条件：圆形回路的磁场式用于其轴线上；螺线管式用于足够长的螺线管内部。电感一般定义为 $L = \Lambda/I$ ，其中 Λ 为总磁链；只有单匝时 $\Lambda = \Phi$ ，多匝且每匝磁通相同才有 $\Lambda = N\Phi$ 。

Important Terms

Provide definitions or explain the meaning of the following terms:

Ampère's law	Lorentz force $\mathbf{F}$	magnetic moment $\mathbf{m}$	solenoid
Ampèrian contour	magnetic dipole	magnetic potential A	surface current
Biot–Savart law	magnetic energy W_m	magnetic susceptibility χ_m	density $\mathbf{J}_s$
current density (volume) $\mathbf{J}$	magnetic energy	magnetization curve	toroid
diamagnetic	density w_m	magnetization vector $\mathbf{M}$	toroidal coil
ferromagnetic	magnetic flux Φ	magnetized domains	torque $\mathbf{T}$
Gauss's law for magnetism	magnetic flux density $\mathbf{B}$	moment arm $\mathbf{d}$	vector Poisson's
hard and soft ferromagnetic	magnetic flux linkage Λ	orbital and spin magnetic	equation
materials	magnetic force $\mathbf{F}_m$	moments	
inductance (self- and mutual)	magnetic hysteresis	paramagnetic	

PROBLEMS

Section 5-1: Magnetic Forces and Torques

*5.1 An electron with a speed of 8×10^6 m/s is projected along the positive x direction into a medium containing a uniform magnetic flux density $\mathbf{B} = (\hat{x}4 - \hat{z}3)$ T. Given that $e = 1.6 \times 10^{-19}$ C and the mass of an electron is $m_e = 9.1 \times 10^{-31}$ kg, determine the initial acceleration vector of the electron (at the moment it is projected into the medium).

5.2 When a particle with charge q and mass m is introduced into a medium with a uniform field $\mathbf{B}$ such that the initial velocity of the particle $\mathbf{u}$ is perpendicular to $\mathbf{B}$ (Fig. P5.2), the magnetic force exerted on the particle causes it to move in a circle of radius a . By equating $\mathbf{F}_m$ to the centripetal force on the particle, determine a in terms of q , m , u , and $\mathbf{B}$.

5.3 The circuit shown in Fig. P5.3 uses two identical springs to support a 10 cm long horizontal wire with a mass of 20 g. In the absence of a magnetic field, the weight of the wire causes the springs to stretch a distance of 0.2 cm each. When a uniform magnetic field is turned on in the region containing the horizontal wire, the springs are observed to stretch an additional 0.5 cm each. What is the intensity of the magnetic flux density $\mathbf{B}$? The force equation for a spring is $F = kd$, where k is the spring constant and d is the distance it has been stretched.

* Answer(s) available in Appendix E.

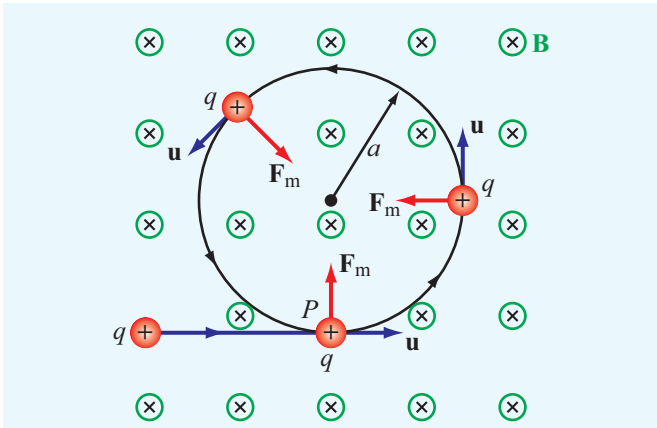


Figure P5.2 Particle of charge q projected with velocity $\mathbf{u}$ into a medium with a uniform field $\mathbf{B}$ perpendicular to $\mathbf{u}$ (Problem 5.2).

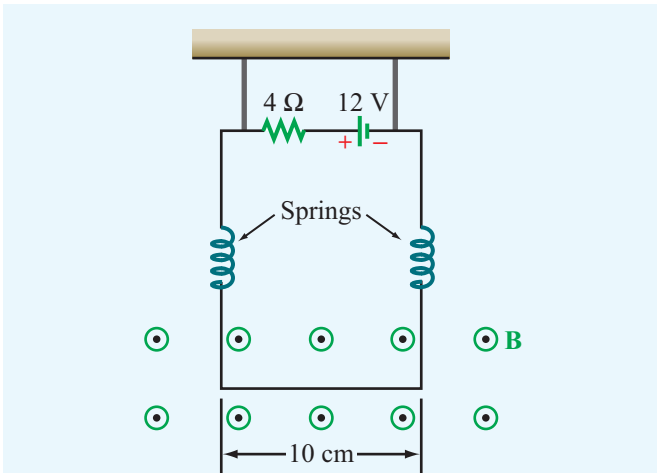


Figure P5.3 Configuration of Problem 5.3.

重要术语 提供以下术语的定义或解释其含义：

安培环路定律 安培环路 毕	洛伦兹力 $\mathbf{F}$ 磁偶极子
奥-萨伐尔定律 电流密度 (	磁能 W_m 磁能密度 w_m
体积) $\mathbf{J}$ 抗磁铁磁 高斯磁	磁通 Φ 磁通密度 $\mathbf{B}$ 磁
性定律 硬磁和软磁材料 电	通链 Λ 磁力 $\mathbf{F}_m$ 磁滞
感 (自电感和互电感)	

磁矩 $\mathbf{m}$ 磁势 A 磁化率	螺线管表面电
χ_m 磁化曲线 磁化矢量	流密度 $\mathbf{J}_s$ 环形
M 磁畴 力矩臂 $\mathbf{d}$ 轨道和	线圈 环形线圈
自旋磁矩 顺磁	扭矩 $\mathbf{T}$ 矢量 泊
	松方程

问题

第 5-1 节：磁力和扭矩

*5.1 电子速度为 8×10^6 m/s 沿正 x 方向投射到含有均匀磁通密度 $\mathbf{B} = (\hat{x}4 - \hat{z}3)$ T 的介质中。假设 $e = 1.6 \times 10^{-19}$ C 且电子质量为 $m_e = 9.1 \times 10^{-31}$ kg，确定电子的初始加速度矢量（在其投射到介质中的那一刻）。

5.2 当带电荷 q 、质量 m 的粒子被引入均匀场 $\mathbf{B}$ 的介质中，使得粒子的初速度 $\mathbf{u}$ 垂直于 $\mathbf{B}$ 时（图 P5.2），施加在粒子上的磁力使其沿半径为 a 的圆运动。通过将 $\mathbf{F}_m$ 等同于粒子上的向心力，根据 q 、 m 、 u 和 $\mathbf{B}$ 确定 a 。

5.3 图P5.3所示电路采用两个相同的弹簧支撑一根10厘米长、质量为20克的水平导线。在没有磁场的情况下，金属丝的重量会导致每个弹簧拉伸0.2厘米的距离。当在包含水平线的区域中打开均匀磁场时，可以观察到每个弹簧额外拉伸0.5厘米。磁通密度 $\mathbf{B}$ 的强度是多少？弹簧的力方程为 $F = kd$ ，其中 k 是弹簧常数， d 是弹簧拉伸的距离。

* Answer(s) available in Appendix E.

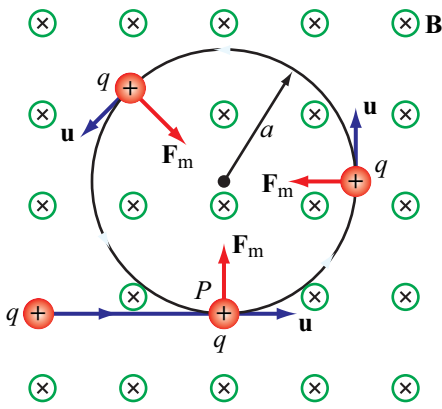


图 P5.2 电荷 q 的粒子以速度 $\mathbf{u}$ 投射到具有垂直于 $\mathbf{u}$ 的均匀场 $\mathbf{B}$ 的介质中（问题 5.2）。

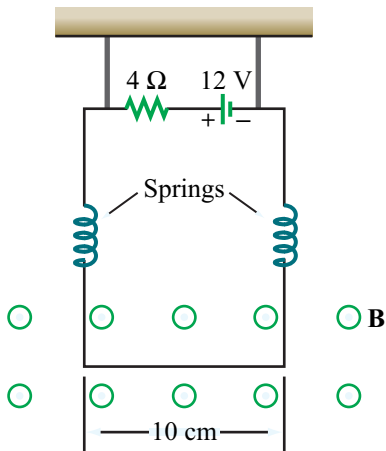


图 P5.3 问题 5.3 的配置。